Advisory Board to the Global Innovation Index

In 2011, an Advisory Board was set up to advise on the research underlying the Global Innovation Index (GII), generate synergies at its development stages, and assist with the dissemination of its messages and results.

The Advisory Board is a select group of leading international practitioners and experts with unique knowledge and skills in the realm of innovation. Its members, while coming from diverse geographical and institutional backgrounds (international organizations, the public sector, non-governmental organizations, business, and academia), participate in their personal capacity.

We are grateful for the time and support provided by the Advisory Board members.

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Rankings

Global Innovation Index rankings

Country/Economy	Score (0–100)	Rank	Income	Rank	Region	Rank	GII PAST YEARS 2010	2009
Switzerland	63.82	1	HI	1	ECS	1	4	7
Sweden	62.12	2	HI	2	ECS	2	2	3
Singapore	59.64	3	HI	3	EAS	1	7	5
Hong Kong (SAR), China	58.80	4	HI	4	EAS	2	3	12
Finland	57.50	5	HI	5	ECS	3	6	13
Denmark	56.96	6	HI	6	ECS	4	5	8
United States of America	56.57	7	HI	7	NAC	1	11	1
Canada	56.33	8	HI	8	NAC	2	12	11
Netherlands	56.31	9	HI	9	ECS	5	8	10
United Kingdom	55.96	10	HI	10	ECS	6	14	4
Iceland	55.10	11	HI	11	ECS	7	1	20
Germany	54.89	12	HI	12	ECS	8	16	2
Ireland	54.10	13	HI	13	ECS	9	19	21
Israel	54.03	14	HI	14	MEA	1	23	23
New Zealand	53.79	15	HI	15	EAS	3	9	27
Korea, Rep.	53.68	16	HI	16	EAS	4	20	6
Luxembourg	52.65	17	HI	17	ECS	10	15	17
Norway	52.60	18	HI	18	ECS	11	10	14
Austria	50.75	19	HI	19	ECS	12	21	15
Japan	50.32	20	HI	20	EAS	5	13	9
Australia	49.85	21	HI	21	EAS	6	18	22
France	49.25	22	HI	22	ECS	13	22	19
Estonia	49.18	23	HI	23	ECS	14	29	29
Belgium	49.05	24	HI	24	ECS	15	17	18
Hungary	48.12	25	HI	25	ECS	16	36	47
Qatar	47.74	26	HI	26	MEA	2	35	24
Czech Republic	47.30	27	HI	27	ECS	17	27	33
Cyprus	46.45	28	HI	28	ECS	18	32	45
China	46.43	29	LM	1	EAS	7	43	37
Slovenia	45.07	30	HI	29	ECS	19	26	36
Malaysia	44.05	31	UM	1	EAS	8	28	25
Spain	43.81	32	HI	30	ECS	20	30	28
Portugal	42.40	33	HI	31	ECS	21	34	40
United Arab Emirates	41.99	34	HI	32	MEA	3	24	26
Italy	40.69	35	HI	33	ECS	22	38	31
Latvia	39.80	36	HI	34	ECS	23	44	60
Slovak Republic	39.05	37	HI	35	ECS	24	37	35
Chile	38.84	38	UM	2	LCN	1	42	39
Moldova, Rep.	38.66	39	LM	2	ECS	25	n/a	116
Lithuania	38.49	40	UM	3	ECS	26	39	42
Jordan	38.43	41	LM	3	MEA	4	58	55
Bulgaria	38.42	42	UM	4	ECS	27	49	74
Poland	38.02	43	HI	36	ECS	28	47	56
Croatia	37.98	44	HI	37	ECS	29	45	62
Costa Rica	37.91	45	UM	5	LCN	2	41	48
Bahrain	37.80	46	HI	38	MEA	5	40	34
Brazil	37.75	47	UM	6	LCN	3	68	50
Thailand	37.63	48	LM	4	EAS	9	60	44
Lebanon	37.11	49	UM	7	MEA	6	n/a	n/a
Romania	36.83	50	UM	8	ECS	30	52	69
Viet Nam	36.71	51	LM	5	EAS	10	71	64
Kuwait	36.64	52	HI	39	MEA	7	33	30
Mauritius	36.47	53	UM	9	SSF	1	73	66
Saudi Arabia	36.44	54	HI	40	MEA	8	54	32
Serbia	36.31	55	UM	10	ECS	31	101	92
Russian Federation	35.85	56	UM	11	ECS	32	64	68
Oman	35.51	57	HI	41	MEA	9	65	52
Argentina	35.36	58	UM	12	LCN	4	75	84
South Africa	35.22	59	UM	13	SSF	2	51	43
Ukraine	35.01	60	LM	6	ECS	33	61	79
Guyana	34.83	61	LM	7	LCN	5	113	103
India	34.52	62	LM	8	SAS	1	56	41
Greece	34.18	63	HI	42	ECS	34	46	54

Global Innovation Index rankings *(continued)*

Country/Economy	Score (0–100)	Rank	Income	Rank	Region	Rank	GII PAST YEARS 2010	2009
Uruguay	34.18	64	UM	14	LCN	6	53	80
Turkey	34.11	65	UM	15	ECS	35	67	51
Tunisia	33.89	66	LM	9	MEA	10	62	46
Macedonia	33.47	67	UM	16	ECS	36	77	89
Mongolia	33.40	68	LM	10	EAS	11	87	105
Armenia	33.00	69	LM	11	ECS	37	82	104
Ghana	32.48	70	LI	1	SSF	3	105	n/a
Colombia	32.32	71	UM	17	LCN	7	90	75
Trinidad and Tobago	32.17	72	HI	43	LCN	8	55	65
Georgia	31.87	73	LM	12	ECS	38	84	98
Paraguay	31.17	74	LM	13	LCN	9	127	118
Brunei Darussalam	30.93	75	HI	44	EAS	12	48	n/a
Bosnia & Herzegovina	30.84	76	UM	18	ECS	39	116	n/a
Panama	30.77	77	UM	19	LCN	10	66	67
Namibia	30.74	78	UM	20	SSF	4	92	95
Botswana	30.51	79	UM	21	SSF	5	86	77
Albania	30.45	80	UM	22	ECS	40	81	121
Mexico	30.45	81	UM	23	LCN	11	69	61
Sri Lanka	30.36	82	LM	14	SAS	2	79	58
Peru	30.34	83	UM	24	LCN	12	88	85
Kazakhstan	30.32	84	UM	25	ECS	41	63	72
Kyrgyzstan	29.79	85	LI	2	ECS	42	104	122
Guatemala	29.33	86	LM	15	LCN	13	95	81
Egypt	29.21	87	LM	16	MEA	11	74	76
Azerbaijan	29.17	88	UM	26	ECS	43	57	57
Kenya	29.15	89	LI	3	SSF	6	83	78
El Salvador	29.14	90	LM	17	LCN	14	91	88
Philippines	28.98	91	LM	18	EAS	13	76	63
Jamaica	28.88	92	UM	27	LCN	15	70	73
Ecuador	28.75	93	LM	19	LCN	16	126	109
Morocco	28.73	94	LM	20	MEA	12	94	82
Iran	28.41	95	UM	28	MEA	13	n/a	n/a
Nigeria	28.15	96	LM	21	SSF	7	96	70
Bangladesh	28.05	97	LI	4	SAS	3	120	111
Honduras	27.81	98	LM	22	LCN	17	112	83
Indonesia	27.78	99	LM	23	EAS	14	72	49
Senegal	27.56	100	LM	24	SSF	8	106	90
Swaziland	27.52	101	LM	25	SSF	9	n/a	n/a
Venezuela	27.41	102	UM	29	LCN	18	124	101
Cameroon	26.95	103	LM	26	SSF	10	119	106
Tanzania	26.88	104	LI	5	SSF	11	98	86
Pakistan	26.75	105	LM	27	SAS	4	103	93
Uganda	26.37	106	LI	6	SSF	12	108	100
Mali	26.35	107	LI	7	SSF	13	107	97
Malawi	25.96	108	LI	8	SSF	14	97	n/a
Rwanda	25.86	109	LI	9	SSF	15	n/a	n/a
Nicaragua	25.78	110	LM	28	LCN	19	117	114
Cambodia	25.46	111	LI	10	EAS	15	102	117
Bolivia	25.44	112	LM	29	LCN	20	129	123
Madagascar	25.41	113	LI	11	SSF	16	125	113
Zambia	25.27	114	LI	12	SSF	17	111	96
Syrian Arab Republic	24.82	115	LM	30	MEA	14	132	94
Tajikistan	24.50	116	LI	13	ECS	44	115	112
Côte d'Ivoire	24.08	117	LM	31	SSF	18	89	n/a
Benin	23.81	118	LI	14	SSF	19	118	99
Zimbabwe	23.54	119	LI	15	SSF	20	131	126
Burkina Faso	23.14	120	LI	16	SSF	21	122	115
Ethiopia	22.88	121	LI	17	SSF	22	123	120
Niger	21.41	122	LI	18	SSF	23	n/a	n/a
Yemen	20.72	123	LM	32	MEA	15	n/a	n/a
Sudan	20.36	124	LM	33	SSF	24	n/a	n/a
Algeria	19.79	125	UM	30	MEA	16	121	108

Note: World Bank Income Group Classification (January 2011): LI = low income; LM = lower-middle income; UM = upper-middle income; and HI = high income; World Bank Regional Classification (January 2011): ECS = Europe & Central Asia; MEA = Middle East & North Africa; SSF = Sub-Saharan Africa; EAS = East Asia & Pacific; SAS = South Asia; NAC = North America; and LCN = Latin America & Caribbean.

Chapters

Measuring Innovation Potential and Results: The Best Performing Economies

Soumitra Dutta and **Daniela Benavente**, INSEAD

The Global Innovation Index (GII) project was launched by INSEAD in 2007 with the simple goal of determining how to find metrics and approaches to better capture the richness of innovation in society and go beyond such traditional measures of innovation as the number of PhDs, the number of research articles produced, the research centres created, the patents issued, and research and development (R&D) expenditures.

There were several motivations for setting this goal. First, innovation is important for driving economic progress and competitiveness—both for developed and developing economies. Many governments are putting innovation at the centre of their growth strategies. Second, there is awareness that the definition of innovation has broadened—it is no longer restricted to R&D laboratories and to published scientific papers. Innovation could be and is more general and horizontal in nature, and includes social innovations and business model innovations as well. Last but not least, recognizing and celebrating innovation in emerging markets is seen as critical for inspiring people—especially the next generation of entrepreneurs and innovators.

However, reaching this goal has not been simple. A serious body of literature (see the next section) has attempted to outline metrics for innovation over the last several years. The GII builds on these prior approaches and attempts to incorporate new perspectives on both traditional and emerging views of innovation. Many aspects of innovation, such as those in the informal economy, remain hard to identify and harder to measure with objective metrics. The GII innovation model, described in further detail in this chapter, takes several important steps in this direction, but feedback from experts and practitioners allows the model to continue to evolve.

An ambition of the GII has been to maximize the number of economies evaluated in the study. This continues to be a challenge because obtaining timely and relevant metrics on a global basis is often not possible. All available official data from international organizations such as the World Bank, the United Nations Educational, Scientific and Cultural Organization (UNESCO), and the International Telecommunications Union (ITU) were considered, although many critical measures of innovation are not covered in the efforts of these organizations. Finally, combining various metrics into a simple measure of innovation for an economy is fraught with statistical and other complexities,[1] especially when considering economies that are often vastly different in size, population, and stage of economic development.

As a sign of the increasing validation and importance of the GII project, four key Knowledge Partners have contributed to the project this year: Alcatel-Lucent, Booz & Company, the Confederation of Indian Industry (CII),[2] and the World Intellectual Property Organization (WIPO, a specialized agency of the United Nations). Each of these partners shares a common vision of the importance of a broader notion of innovation in our world today. The GII project has benefited from the knowledge and input of these partners, and contributions from other public- and private-sector leaders who are interested in understanding and improving innovation in their economies will continue to provide valuable input.

This chapter presents selected findings from a review of innovation literature that has allowed us to refine the theoretical underpinnings of the GII model and guide the revision of pillars and sub-pillars and the selection of indicators. The chapter also includes details on the innovation rankings emerging from the GII project in 2011.

Box 1: Low-tech innovations

Although innovations with low technological content have always existed, the fact that innovations do not necessarily entail a technological component emerges as an important theme in the recent literature on innovation, in sharp contrast with the past (OECD/EC, 2005, p. 17).

According to the OECD classification of innovativeness based on R&D intensity, low-tech industries are those that have an R&D intensity that ranges between 0 and 0.9% (this intensity is greater than 5% for high-tech industries). Since this classification applies exclusively to manufactures (textiles, wood, pulp, etc.), a different taxonomy was proposed by Pavitt in 1984, which included four groups: (1) supplier-dominated firms, (2) scale-intensive firms, (3) specialized suppliers, and (4) science-based firms. Low-tech sectors mainly fall into the first group.

This distinction is crucial, as high-tech industries represent a small proportion of total manufacturing industries, including in developed economies. High-tech industries have represented around 6% and 10% of the value-added of the manufacturing sector in the European Union at 15 countries (1979–2003) and the United States of America, respectively (1979–2004).

Innovation in low-tech industries has particular characteristics:

1. It is more 'market pulled' than 'technology pushed'; demand factors, niche markets, product differentiation, and mature brands are crucial to innovation in low-tech industries.
2. Product innovations are not intensive in research and development (R&D), although process innovations have more technological content (cf. investment in equipment and machinery).
3. New technologies often spill over (through acquisition) from other industries, so that low-tech firms provide a demand pull for high-tech firms.
4. For this absorption of innovation to be effective, a skilled workforce and learning capabilities are required.

Source

Based on Joint Research Centre of the European Commission, 2009.

Prior research in innovation: Selected perspectives

Different definitions of innovation have been proposed in the literature. In this report, we embrace a broad definition that has the advantage of being both short and well suited to capture global innovation:

> An innovation is the implementation of a new or significantly improved product (good or service), a new process, a new marketing method, or a new organizational method in business practices, workplace organization, or external relations.[3]

The modern evolution of the concept of innovation can be sketched by referring to the different versions of the *Oslo Manual*, which guides statisticians in their recent attempts to measure innovation.[4] In its first editions in 1992 and 1997, the *Manual* focused exclusively on technological innovations and covered only products and processes. The breakthrough made after 1997 was to expand the sectoral coverage from manufacturing industries to services. The 2005 edition incorporated three crucial developments. First, the 'technological' qualifier was eliminated (see Box 1). Second, innovations in methods were added to the list. Third, for the first time, innovation in the public sector was mentioned as an area deserving further attention.[5] In 2010, the Ministerial Report on the Organisation for Economic Co-operation and Development (OECD) Innovation Strategy added that 'consideration [was] being given to extending the methodology to public sector innovation and social innovation so as to correspond to the reality of innovation today'.[6]

An innovation can be new to the world, or new to a sector or market, or new to an agent. It can also be a disruptive innovation, where the focus is on impact, rather than on novelty.[7] Most studies agree that the commercial introduction of innovations developed elsewhere constitutes an innovation:

> Innovation also occurs when a firm introduces a product or process to a country for the first time. It occurs when other firms imitate this pioneering firm. Moreover, it occurs when the initial or follower firms make minor improvements and adaptations to improve a product or production process, leading to productivity improvements. In short, innovation occurs through 'creative imitation'.[8]

Innovations are not restricted to the firm, they originate in all agents of society: at the level of the firm, or of an industry; in governmental services or in the public sector; in academia; and in society in general.[9] Innovation activities by these agents are usually complementary: Prototypes might be developed in a university research lab and the final product introduced in the market by a firm, for instance.

A distinction made by Gibbons et al. (1994) has been highly influential in the literature on innovation. These authors label traditional 19th and early 20th century research as 'mode 1': Characterized by a cleavage between academia and society, this type of research is academic, autonomous, self-sustained, investigator-initiated, peer-reviewed, and discipline-based in nature. By contrast, 'mode 2' refers to more recent forms of knowledge production, which is centred mostly around the firm where research is increasingly context-driven, problem-focused, application-oriented, and interdisciplinary—task-force teams and tailored processes are created to work on specific projects.[10] Other theorists, such as those of the 'Triple Helix of Innovation', have stressed the historical continuities and linkages in the relationship among academia, industry, and government.[11]

The different legal, institutional, organizational, and governance regimes surrounding innovative activities are of special relevance to their success. These include the political environment, openness to credit, the treatment of investment and trade, the presence of competition laws, the protection of intellectual property rights, tax laws, and the transportation and telecommunications infrastructure.

With technological catch-up and market and business sophistication, innovation acquires a strong regional or sectoral component. Sub-national systems of innovation might develop, for example, around local public research institutions, large dynamic firms, or industry clusters. In addition, good infrastructure, venture capital, and a strong entrepreneurial environment can influence the innovative performance of regions.[12]

International market linkages foster the development of technological capabilities in developing and least-developed countries because they integrate global value chains through exports, the import of machinery and equipment, transfers of technology, the spill-over effects of foreign direct investment (FDI), and licensing. A recent study

Box 2: Innovation in emerging markets

Terms such as 'reverse innovation',[1] 'Gandhian innovation',[2] 'frugal innovation', 'inclusive innovation', [3] 'constraint-based innovation', or even 'poor people's knowledge'[4] have all been recently coined to describe the type of innovation by which technological products are customized at low prices and high volumes in and for emerging markets. Examples abound: Tata's Nano car, solar-powered cellular phones, micro-spinning in the textiles industry, and the hand-held electrocardiogram Mac 400 are just a few.

Chapter 3, 'Innovation in India: Affordable Innovations' analyses this and other phenomena in India today. Big multinational corporations such as Microsoft, PepsiCo, IBM, Cisco, Nokia, GE, and Xerox as well as Indian major players such as Tata, Godrej, and Mahindras are shifting their focus towards the rapidly expanding middle-income group of customers by coming up with frugal innovations, keeping in mind the price sensitivity of Indian consumers. A trend of 'reverse innovation' has set in, where an innovation is developed and/or adopted first in the developing world and then deployed in mature markets.

In a recent publication, the World Bank notes the technological divide in developing countries: 'Slow diffusion within countries reflects a nonlinear process. . . .The surprisingly low level of overall technological achievement in countries such as China and India contrasts with popular perceptions, which are based on the relative technological sophistication of some of the two countries' major cities and trading centers. . . . [t]he same kind of technological diversity observed across countries is visible within countries as well.'[5]

How to account 'fairly' for these islands of progress is a real conundrum. The same report gives one clue: 'The rise in China's index of diffusion of new technologies is almost double that of India, in part because the more technologically backward regions in China have made progress in closing the gap with the more technologically advanced regions on the coast'.[6]

Notes

1 http://www.vijaygovindarajan.com/2009/10/what_is_reverse_innovation.htm.
2 Mashelkar and Prahalad, 2010.
3 Mashelkar and Prahalad, 2010.
4 Finger, 2004.
5 World Bank, 2008, p. 90.
6 World Bank, 2008, p. 91.

Box 3: Innovation surveys

The Organisation for Economic Co-operation and Development (OECD) and the European Commission have been guiding the collection and interpretation of data on innovation since the first edition of the *Oslo Manual* in 1992, which is now in its third edition (2005).

Innovation surveys started with the European Community Innovation Survey in the early 1990s and are now being conducted in about 50–60 countries worldwide (mostly the European countries but also a number of Latin American, Asian, African, and other countries).

Firm-level innovation surveys seek to identify the characteristics of innovative enterprise activities. After asking firms to answer certain basic questions (industry affiliation, turnover, R&D spending), firms are asked to identify whether they are an 'innovator' and, if yes, they are asked to respond to a number of questions regarding their innovation characteristics. Firms are also being asked about factors that hamper their ability to innovate. Finally, these surveys aim to assess the effect of innovation on sales, productivity, employment, and other factors.

These innovation surveys are a rich data source for analytical work on innovation. However, a number of problems exist: (1) the questionnaires are given only to firms, so that innovation outside the business sector is not captured; (2) the quality of responses varies greatly, as one cannot control who is replying to the questionnaire and as respondents have a tendency to overrate their innovative activity; (3) the country coverage is still very limited, because most developing countries—but also some large developed countries—do not conduct these surveys; and (4) survey results across studies are not always comparable.

on the readiness, by 2020, of a set of countries to apply recently developed technologies in bio- and nano-technology, materials, information, and so on shows that a limited technological adaptive capacity may reduce the diffusion of future technologies.[13]

The type of innovation taking place in emerging markets presents its own peculiarities, which are difficult to capture with traditional metrics (see Box 2). The challenge for the research team behind the Global Innovation Index was to find statistics that would gauge the developments and trends of innovation in low-tech industries, in emerging markets, and in business models, while at the same time covering the traditional sectors of innovation and enabling environments.

Composite indicators for innovation

The previous section surveyed important developments in the conceptualization of innovation and a series of recent issues and trends in the realm of innovation. The literature review allowed us to refine the theoretical underpinnings of the GII and guided the revision of pillars, sub-pillars, and indicators. One general preliminary conclusion in light of this survey was that the GII conceptual framework developed in previous versions was well suited to uncover innovation as it occurs today—it did not require a major overhaul. Some aspects, however, needed to be strengthened.

A key inspiration behind the GII comes from the literature on total quality management (TQM), which has a long history in benchmarking and data analysis. The first TQM award, the Deming Award, was given in Japan in 1951—this award initially focused on product and process quality. Subsequent versions have evolved into a broader notion of business excellence that looks at the whole business, including enablers and results.

The focus in TQM expanded from a narrow technical on to a much broader concept. Innovation today is expanding its focus in a similar way. The same distinction between enablers and results has been incorporated into the GII, providing the theoretical underpinnings of the conceptual framework. The GII also draws on other composite indicators in its design, although it differs in many respects from a host of other indices on innovation. Some key pieces from prior research are mentioned below.

The Boston Consulting Group/National Association of Manufacturers Index

The BCG/NAM International Innovation Index was built in 2009 to establish a ranking among US states and among countries. The BCG/NAM Index is built on a model comprising two major blocks: Innovation Inputs and Innovation Performance. Innovation Inputs are measured by three aspects: fiscal policies, other policies, and innovation environment. Innovation Performance is measured by R&D results, business performance, and public impact of innovation. The focus of the BCG/NAM Index is on business performance specifically in

the manufacturing sector, and most of the data used were generated through surveys and interviews. Only one edition (2009) of the BCG/NAM Innovation Index has been published thus far.[14]

The European Innovation Scoreboard/ Innovation Union Scoreboard

The European Innovation Scoreboard, renamed the Innovation Union Scoreboard in 2010, has been in existence for nearly a decade. Until 2007, it lacked an underlying model of innovation and focused primarily on the technological sector. Since 2008, it has been modified to include an underlying model comprising three blocks: Enablers (human resources, research systems, and finance/support); Firm Activities (firm investments, linkages/ entrepreneurship, and intellectual assets); and Outputs (innovators and economic effects) and has included a broader set of 25 indicators to measure the above blocks. The focus of the Innovation Union Scoreboard is on the European Union (EU) member states with selective comparisons to a few international reference countries such as the United States of America (US), China, and India.[15]

The Global Innovation Index of the Economist Intelligence Unit

The Economist Intelligence Unit index ranks 41 countries along a model consisting of Innovation Inputs and Innovation Outputs. Innovation Inputs are measured by direct innovation inputs (six measures, such as education of the workforce) and innovation environment (nine indicators, such as foreign trade and exchange controls). Innovation Outputs are measured by a single indicator: the sum of patents granted by the European, Japanese, and US Patent Offices (EPO, JPO, and USPTO, respectively).[16]

The Global Competitiveness Index (GCI) of the World Economic Forum

The Global Competitiveness Index (GCI) of the World Economic Forum (WEF), while dealing with the theme of competitiveness, includes 12 pillars that overlap on some enabling factors for innovation in the GII. Innovation is a separate pillar within the GCI that includes metrics traditionally attributed to innovation related to R&D, intellectual property protection, and patenting.[17]

Statistics on innovation

All efforts at capturing innovation confront the same challenge: Direct official measures that would quantify innovation outputs are frequently not available across many countries. This is particularly true if one considers our broadening notion of innovation, which encapsulates non-technological, softer or local types of innovation (including those in developing countries). Most existing measures also struggle to appropriately capture innovation outputs of a wider spectrum of innovation actors, as mentioned above (e.g., the services sector, public entities, etc.).

In recent years, the generation of data from firm-level innovation surveys (see Box 3) has improved the data situation somewhat. However, there are several unresolved issues with these data. They are generally not available and comparable for more than about 50 countries, and they still do not provide a good count on innovation outputs per country. Moreover, they target only innovations at the firm level—broader public-sector and social innovations are not included.

Science and technology indicators are not all available internationally, so they provide, at best, information on innovation inputs / throughputs (such as R&D expenditures and the number of scientists in a country), intermediate innovation outputs (such as numbers of scientific publications or patents), or certain forms of technology-related commercial activity (such as data on high-technology exports). Rarely do they provide data on the aforementioned innovation itself, and they are often specific to technological and product innovations of research organizations and firms.

A trade-off between precision and country coverage was often made in selecting the indicators to be included in the GII model. The balance was struck in favour of selecting a combination of three to six indicators that would capture the latent dimension within each sub-pillar in the best possible way, with an overall coherence within pillars. The Joint Research Centre, which assisted in the assessment of the conceptual and statistical coherence of the overall structure of the GII, confirmed the soundness of this approach (details in the appendix to this chapter).

A development particularly relevant to the fine-tuning of this year's GII was the release of the OECD Innovation Strategy Report in 2010 along with its accompanying compendium of close to 100 indicators on innovation (see Box 4).

The OECD Innovation Strategy metrics confirmed the continued relevance of traditional

Box 4: OECD Innovation Strategy (2010)

The Organisation for Economic Co-operation and Development (OECD) issued its *Innovation Strategy* report in 2010. This report includes the findings of a three-year multi-disciplinary effort aimed at proposing new perspectives and prospective metrics on innovation, broadly defined as 'the introduction of a new or significantly improved product, process or method'.[1]

The report is accompanied by a measurement compendium, entitled *Measuring Innovation: A New Perspective*, which is comprised of 100 innovation indicators grouped by six different topics. Indicators traditionally used to monitor innovation are complemented by indicators from other domains that describe the broader context in which innovation occurs. It includes some experimental indicators that provide insight into new areas of policy interest. An important objective of the report is to highlight measurement gaps and propose ways to advance the measurement agenda. Although these metrics represent state-of-the-art innovation statistics, most are developed on a prospective basis and/or are provided for only a handful of countries.[2] For example, the data on investment in fixed and intangible assets as a share of GDP is available only for 16 countries for 2006.[3]

Source

OECD, 2010a.

Notes

1 OECD, 2010a, p. 9.

2 Online measurement guide available at http://www.oecd.org/document/22/0,3746,en_41462537_41454856_44979734_1_1_1_1,00.html.

3 Includes Machinery and equipment, Software and databases, R&D and other intellectual property products, and Brand equity, firm-specific human capital, and organizational capital.

variables included in earlier editions of the GII. Examples are in the areas of education and R&D, patents, scientific and technical journal publications, and labour productivity.

The OECD compendium was also an inspiration for the inclusion of new variables. The statistics that were incorporated only this year to the GII and that were inspired, to a large extent, by the OECD measurement exercise and other expert publications include: graduate inbound and outbound mobility and gross enrolment ratios, the OECD PISA scores on performance in reading, mathematics and science in elementary education, school life expectancy, percentages of graduates in science and engineering, venture capital, joint venture and strategic alliance deals, total tax rate, software spending, R&D performed and financed by business, and the share of renewables in energy use, to name a few.

In an effort to capture a broader vision of innovation that goes beyond science and technology indicators to accommodate, whenever possible, the type of innovation found in emerging markets, and in view of its holistic approach to innovation, the GII also includes statistics on the enabling environment for innovation (pillars 1, 3, and 4 on institutions, infrastructure, and market conditions for credit, investment, and trade); knowledge absorption and technological catch-up (sub-pillar 5.3); and creative outputs (pillar 7). Chapter 6 describes in detail the particular challenges involved in the measurement of creative industries and copyright-related industries.

The Global Innovation Index

The GII is an evolving project, which builds upon previous editions of the Index while incorporating the latest research on the measurement of innovation. This section looks at the GII 2011 framework and considers the indicators that comprise each pillar and sub-pillar.

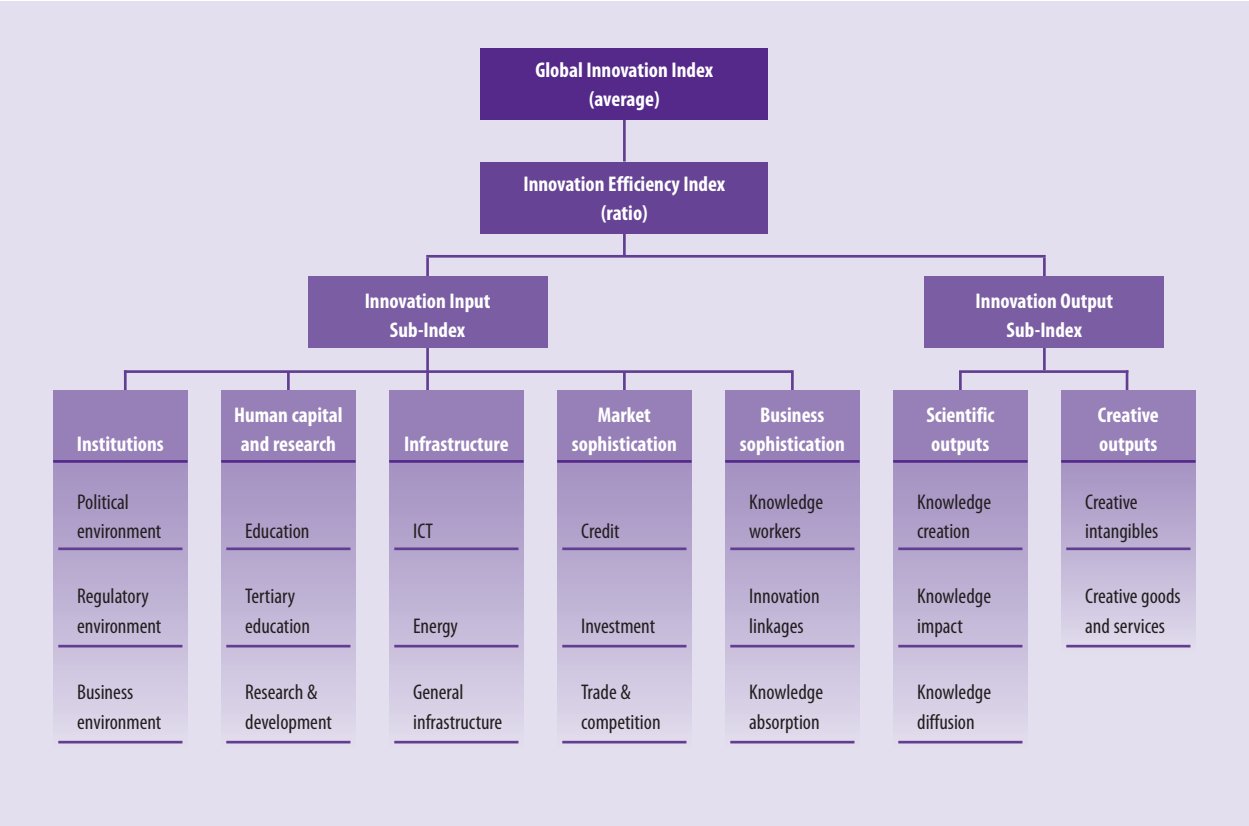
Conceptual framework

The Global Innovation Index (GII) relies on two sub-indices, the Innovation Input Sub-Index and the Innovation Output Sub-Index, each built around pillars. Five input pillars capture elements of the national economy that enable innovative activities: (1) Institutions, (2) Human capital and research, (3) Infrastructure, (4) Market sophistication, and (5) Business sophistication. Two output pillars capture actual evidence of innovation outputs: (6) Scientific outputs and (7) Creative outputs. Each pillar is divided into sub-pillars and each sub-pillar is composed of individual indicators.

Sub-pillar scores are calculated as the weighted average of individual indicators; pillar scores are calculated as the simple average of the sub-pillar scores (refer to Appendix IV Technical Notes). Four measures are then calculated (see Figure 1):

1. The Innovation Input Sub-Index is the simple average of the first five pillar scores.

Figure 1: Framework of the Global Innovation Index 2011



2. The Innovation Output Sub-Index is the simple average of the last two pillar scores.

3. The overall GII is the simple average of the Input and Output Sub-Indices.

4. The Innovation Efficiency Index is the ratio of the Output Sub-Index over the Input Sub-Index.

Innovation Input Sub-Index

The GII has five enabler pillars: Institutions, Human capital and research, Infrastructure, Market sophistication, and Business sophistication. Enabler pillars define aspects of the environment conducive to innovation within an economy. What follows is a description of each pillar (refer to Appendix III, Sources and Definitions, for further details).

A table is included for each pillar that provides a list of its indicators; their type (composite indicators are identified with an asterisk '*', survey questions with a dagger '†', and the remaining indicators are hard data); their weight (indicators with half weight are identified with the letter 'a'); and the direction of their effect (indicators for which higher values imply worse outcomes are identified with the letter 'b'). The table then provides for each indicator the average values (in their respective units) per income group (World Bank classification) and for the whole sample of 125 economies retained in the final computation.

Institutions

Nurturing an institutional framework that attracts business and fosters growth by providing good governance and the correct levels of protection and incentives is essential to innovation. The Institutions pillar captures the institutional framework of a country through three sub-pillars (Table 1a).

The political environment sub-pillar includes three indices that reflect perceptions of the likelihood that a government might be destabilized; the quality of public and civil services, policy formulation, and

Table 1a: Institutions pillar

	Indicator	Average value by income group: High income	Upper-middle income	Lower-middle income	Low income	Mean
1	**Institutions**					
1.1	*Political environment*					
1.1.1	Political stability*	72.0	42.3	25.8	29.5	46.5
1.1.2	Government effectiveness*	84.0	53.9	36.8	30.1	56.5
1.1.3	Press freedom*[b]	13.7	29.2	44.9	31.5	28.2
1.2	*Regulatory environment*					
1.2.1	Regulatory quality*	84.6	54.1	39.7	34.6	58.2
1.2.2	Rule of law*	83.0	46.6	32.7	29.5	53.3
1.2.3	Rigidity of employment*[b]	23.6	28.8	31.7	30.8	28.0
1.3	*Business environment*					
1.3.1	Time to start a business, days[b]	15.9	30.1	26.9	26.4	23.7
1.3.2	Cost to start a business, % income/cap[b]	5.2	12.8	37.2	63.4	23.9
1.3.3	Tota l tax rate, % profits[b]	37.3	42.2	43.4	42.0	40.8

Note: (*) index, (†) survey question, (a) half weight, (b) higher values indicate worse outcomes.

Table 1b: Human capital and research pillar

	Indicator	Average value by income group: High income	Upper-middle income	Lower-middle income	Low income	Mean
2	**Human capital & research**					
2.1	*Education*					
2.1.1	Education expenditure, % GNI[a]	4.8	4.2	3.7	3.6	4.2
2.1.2	Public expenditure/pupil, % GDP/cap[a]	21.6	16.5	19.9	19.3	19.7
2.1.3	School life expectancy, years	15.6	13.5	11.1	9.4	13.0
2.1.4	PISA scales in reading, maths, & science[a]	496.5	415.4	435.6	324.9	465.3
2.1.5	Pupil-teacher ratio, secondary[b]	11.4	15.0	19.4	25.5	16.5
2.2	*Tertiary education*					
2.2.1	Tertiary enrolment, % gross	57.6	41.7	24.8	7.9	37.8
2.2.2	Graduates in science, %	9.5	7.2	8.1	8.9	8.6
2.2.3	Graduates in engineering, %	12.3	12.6	11.0	8.4	11.7
2.2.4	Tertiary inbound mobility, %[a]	9.4	3.2	1.9	3.5	6.0
2.2.5	Tertiary outbound mobility, %[a]	14.8	8.0	5.7	9.2	10.0
2.2.6	Gross tertiary outbound enrolment, %[a]	4.0	2.0	1.0	0.4	2.1
2.3	*Research & development (R&D)*					
2.3.1	Researchers headcount/million pop.	4,754.0	1,071.5	667.9	99.8	2,192.6
2.3.2	Gross expenditure on R&D, % GDP	1.7	0.5	0.3	0.1	0.9
2.3.3	Quality research institutions†	4.8	3.6	3.2	3.2	3.9

Note: (*) index, (†) survey question, (a) half weight, (b) higher values indicate worse outcomes.

implementation; and perceptions on violations to press freedom.[18]

The regulatory environment sub-pillar draws on three World Bank indices aimed at capturing perceptions on the ability of the government to formulate and implement cohesive policies that promote the development of the private sector; at evaluating the extent to which the rule of law prevails (in aspects such as contract enforcement, property rights, the police, and the courts); and at measuring the level of rigidity of employment regulations.

The business environment sub-pillar expands on three aspects that directly affect private entrepreneurial endeavours: time and cost constraints to starting a business and the total amount of taxes and mandatory contributions borne by businesses (profit or corporate income tax, social contributions and labour taxes paid by the employer, property taxes, etc.).[19]

Human capital and research

The level and standard of education and research activity in a country are the prime determinants of the innovation capacity of a nation. This pillar tries to gauge the human capital of countries through three sub-pillars (Table 1b).

The first sub-pillar includes a mix of indicators aimed at capturing achievements at the elementary and secondary education levels. Education expenditure and school life expectancy are good proxies for coverage. Public expenditure per pupil gives a sense of the level of priority given to education by the state. The quality of education is measured through the results to the OECD Programme for International Student Assessment (PISA), which examines 15-year-old students' performances in reading, mathematics, and science, and the pupil-teacher ratio.

Higher education is crucial for economies to move up the value chain beyond simple production processes and products. The sub-pillar on tertiary education aims at capturing coverage (tertiary enrolment); the priority given to the sectors traditionally associated with innovation (science and engineering); and the inbound and outbound mobility of tertiary students, which plays a crucial role in the exchange of ideas and skills necessary to innovation.

The last sub-pillar, on R&D, measures the level and quality of R&D activities, with indicators on expenditure, researchers (headcounts), and perceptions of the

quality of scientific and research institutions (a survey question).

Infrastructure

Information and communication technologies (ICT), energy supply, and infrastructure are respectively the nervous system, the circulatory system, and the backbone of any economy (Table 1c). They facilitate the production and exchange of ideas, services, and goods and feed into the innovation system through increased productivity and efficiency, lower transaction costs, and better access to markets.

In the past 50 years, ICT has revolutionized the way people interact, businesses transact, and governments serve. The ICT sub-pillar includes four indices developed by international organizations on ICT access, ICT use, online service by governments, and online participation of citizens (see Box 5).[20]

The sub-pillar on energy regroups three indicators related to energy supply, efficiency in energy use, and sustainability.[21] Energy supply is essential to the proper functioning of any economy; however, energy policy is an area that typically goes beyond energy supply to address a series of elements such as the risk of supply-and-demand mismatch, environmental issues, alternative sources of energy, cost-efficiency, and so on.

The sub-pillar on infrastructure includes a composite indicator on the quality of trade- and transport-related infrastructure (e.g., ports, railroads, roads, and information technology).[22] The second variable of this sub-pillar, gross capital formation, consists of outlays on additions to the fixed assets and net inventories of the economy, including land improvements (fences, ditches, drains); plant, machinery, and equipment purchases; and the construction of roads, railways, and the like, including schools, offices, hospitals, private residential dwellings, and commercial and industrial buildings. Ecological concerns are also addressed through the inclusion of a measure on the ecological biocapacity and footprint reserve or deficit of countries.

Market sophistication

The recent global financial crisis has underscored how crucial the availability of credit, investment funds, and access to international markets are for businesses to prosper. This

Table 1c: Infrastructure pillar

	Indicator	Average value by income group: High income	Upper-middle income	Lower-middle income	Low income	Mean
3	**Infrastructure**					
3.1	*Info & comm. technologies (ICT)*					
3.1.1	ICT access*	7.1	4.1	2.9	1.6	4.5
3.1.2	ICT use*	4.2	1.4	0.6	0.2	2.0
3.1.3	Government's Online Service*[a]	0.5	0.4	0.3	0.2	0.4
3.1.4	E-Participation*[a]	0.4	0.2	0.2	0.1	0.3
3.2	*Energy*					
3.2.1	Electricity output, kWh/cap[a]	9,995.2	3,033.9	1,374.9	640.5	5,021.2
3.2.2	Electricity consumption, kWh/capita[a]	9,581.7	2,736.2	995.8	553.7	4,677.4
3.2.3	GDP/unit of energy use, PPP$/kg oil eq.	6.2	6.4	5.7	4.3	6.0
3.2.4	Share of renewables in energy use, %	11.6	14.8	36.0	68.2	24.4
3.3	*General infrastructure*					
3.3.1	Quality of trade & transport infrastructure*	3.6	2.6	2.4	2.1	2.8
3.3.2	Gross capital formation, % GDP	21.0	23.7	24.4	22.0	22.7
3.3.3	Ecological footprint & biocapacity, ha/cap	(2.1)	(0.0)	0.7	(0.1)	(0.5)

Note: (*) index, (†) survey question, (a) half weight, (b) higher values indicate worse outcomes.

Table 1d: Market sophistication pillar

	Indicator	Average value by income group: High income	Upper-middle income	Lower-middle income	Low income	Mean
4	**Market sophistication**					
4.1	*Credit*					
4.1.1	Strength of legal rights for credit*[a]	6.8	5.7	4.7	6.1	5.9
4.1.2	Depth of credit information*[a]	4.4	4.7	3.7	1.7	3.9
4.1.3	Domestic credit to private sector, % GDP	107.3	49.7	43.5	19.4	59.8
4.1.4	Microfinance gross loans, % GDP[a]	0.0	1.0	2.0	2.0	1.5
4.2	*Investment*					
4.2.1	Strength of investor protection*	6.0	5.7	4.7	5.0	5.4
4.2.2	Market capitalization, % GDP	78.7	47.4	34.5	22.3	55.3
4.2.3	Total value of stocks traded, % GDP	74.1	13.8	20.0	3.5	39.6
4.2.4	Venture capital deals/tr GDP PPP$[a]	104.7	6.1	22.8	30.6	48.8
4.3	*Trade & competition*					
4.3.1	Applied tariff rate weighted mean, %[b]	1.8	4.9	6.4	9.2	4.8
4.3.2	Market access trade restrictiveness*, %[a,b]	7.2	8.5	13.1	16.6	11.1
4.3.3	Imports of goods & services, % GDP	52.4	39.8	45.0	39.9	45.6
4.3.4	Exports of goods & services, % GDP	59.6	36.9	36.0	25.8	43.1
4.3.5	Intensity of local competition†	5.4	4.6	4.7	4.4	4.9

Note: (*) index, (†) survey question, (a) half weight, (b) higher values indicate worse outcomes.

Box 5: Composition of sub-pillar 3.1 on information and communication technologies (ICT)

ICT access indicator 3.1.1

Indicator 3.1.1 is a composite indicator developed by the International Telecommunication Union (ITU). The 'ICT access sub-index', is one component in ITU's *ICT Development Index* (IDI), together with the 'ICT use sub-index' and the 'ICT skills sub-index' (*Measuring the Information Society,* ITU, 2010). It is composed of five ICT indicators (20% each):

1. fixed telephone lines per 100 inhabitants,
2. mobile cellular telephone subscriptions per 100 inhabitants,
3. international Internet bandwidth (bit/s) per Internet user,
4. proportion of households with a computer, and
5. proportion of households with Internet access at home

ICT use indicator 3.1.2

Indicator 3.1.2 is the second component in ITU's IDI. The 'ICT use sub-index' is composed of three ICT indicators (33% each):

1. Internet users per 100 inhabitants,
2. fixed broadband Internet subscribers per 100 inhabitants, and
3. mobile broadband subscriptions per 100 inhabitants.

Government's online service indicator 3.1.3

Indicator 3.1.3 is a composite indicator developed by the United Nations Public Administration Network (UNPAN). The 'Government's online service index' is a component in UNPAN's *E-Government Development Index* together with two indices on telecommunication infrastructure and human capital.

It is constructed on the basis of the United Nations (UN) e-Government Survey (*United Nations E-Government Survey 2010 Leveraging e-Government at a Time of Financial and Economic Crisis,* UN Department of Economic and Social Affairs, 2010). The survey covers four stages of government's online service development with points assigned for emerging information services, enhanced information services, transaction services, and a connected approach. In addition, research teams assess each country's national website as well as the websites of the ministries of education, labour, social services, health, and finance for a minimal level of content accessibility (with a citizen-centric approach).

E-participation indicator 3.1.4

Indicator 3.1.4 is a composite indicator, the 'E-participation index', developed by UNPAN on the basis of the UN e-Government Survey. The survey was expanded with questions emphasizing quality in the connected presence stage of e-government. These questions focus on the use of the Internet to facilitate provision of information by governments to citizens ('e-information sharing'), the interaction with stakeholders ('e-consultation'), and the engagement in decision-making processes ('e-decision making').

pillar has three sub-pillars structured around market conditions and the total level of transactions (Table 1d).

The credit sub-pillar includes two indices aimed at measuring the degree to which collateral and bankruptcy laws facilitate lending by protecting the rights of borrowers and lenders, as well as the rules and practices affecting the coverage, scope, and accessibility of credit information. Transactions are given by the total value of domestic credit and, in an attempt to make the model more applicable to emerging markets, a measure on the level of development of microfinance institutions.[23]

The investment sub-pillar includes an index measuring the extent of disclosure and of director liability and the ease of shareholder suits. To show whether market size is matched by market dynamism, stock market capitalization is complemented by the value of shares traded. This year, for the first time, hard data on venture capital deals have been included, taking into account a total of 7,937 deals in 81 countries for this variable.[24]

The last sub-pillar tackles trade and competition. The market conditions for trade are given by two indicators: the average tariff rate weighted by import shares and a measure capturing market access conditions to foreign markets. The sub-pillar then includes the total value of exports and imports as a percentage of GDP. The last indicator is a survey question that reflects on the intensity of competition in the local markets. Efforts made at finding hard data on competition proved unsuccessful.

Business sophistication

The last enabler pillar tries to capture the level of business sophistication to assess how conducive firms are to innovation activity (Table 1e). The Human capital and research pillar (pillar 2) made the case that the accumulation of human capital through education, and particularly higher education and the prioritization of R&D activities, is an indispensable condition for innovation to take place. That logic is taken one step further here with the assertion that businesses foster their productivity, competitiveness, and innovation potential with the employment of highly qualified professionals and technicians.

The first sub-pillar includes four quantitative indicators on knowledge workers: employment in knowledge-intensive services; the availability of formal training at the level of the firm; and the percentage of total gross expenditure of R&D that is either financed or performed by business enterprise.

Innovation linkages and public/private/academic partnerships are essential to innovation. In emerging markets, pockets of wealth have developed around industrial or technological clusters and networks in sharp contrast with the poverty that may prevail in the rest of the territory. The sub-pillar draws on both qualitative and quantitative data regarding business/university collaboration on R&D, the prevalence of well-developed and deep clusters, collaboration in inventive activities, and the level of gross R&D expenditure financed by abroad. For the first time this year, a measure on the number of deals on joint ventures and strategic alliances is included. It covers a total of 920 joint ventures and 327 strategic alliances announced in 2010, with firms headquartered in 95 participating economies.[25] In addition, the share of published patent applications filed by residents through the Patent Cooperation Treaty with at least one foreign inventor is included to proxy for international linkages.

In broad terms, pillar 4 on market sophistication makes the case that well-functioning markets contribute to the innovation environment through competitive pressure, efficiency gains, and economies of transaction and by allowing supply to meet demand. Open markets to

Table 1e: Business sophistication pillar

	Indicator	Average value by income group: High income	Upper-middle income	Lower-middle income	Low income	Mean
5	**Business sophistication**					
5.1	*Knowledge workers*					
5.1.1	Knowledge-intensive employment, %	36.4	23.5	16.8	7.0	26.1
5.1.2	Firms offering formal training, % firms	42.6	39.3	34.4	31.7	36.8
5.1.3	R&D performed by business, %[a]	55.7	30.8	26.9	15.5	41.1
5.1.4	R&D financed by business, %[a]	49.3	31.1	24.9	15.4	38.5
5.2	*Innovation linkages*					
5.2.1	University/industry collaboration†	4.5	3.5	3.2	3.2	3.8
5.2.2	State of cluster development†	4.1	3.3	3.3	3.1	3.6
5.2.3	R&D financed by abroad, %[a]	8.2	8.1	9.3	19.9	9.4
5.2.4	JV/strategic alliance deals/tr GDP PPP$[a]	26.8	6.2	6.8	9.9	14.1
5.2.5	PCT patent filings with foreign inventor, %	37.6	22.0	13.2	19.0	25.2
5.3	*Knowledge absorption*					
5.3.1	Royalty & license fees payments, % GDP	1.0	0.2	0.4	0.0	0.5
5.3.2	High-tech imports less re-imports, %	13.2	11.0	8.9	7.6	10.9
5.3.3	Computer & comm. service imports, %	39.3	32.5	25.4	21.4	31.4
5.3.4	FDI net inflows, % GDP	12.0	3.9	3.6	3.3	6.5

Note: (*) index, (†) survey question, (a) half weight, (b) higher values indicate worse outcomes.

Table 1f: Scientific outputs pillar

	Indicator	Average value by income group: High income	Upper-middle income	Lower-middle income	Low income	Mean
6	**Scientific outputs**					
6.1	*Knowledge creation*					
6.1.1	Domestic resident patent ap/bn GDP PPP$	9.8	2.9	4.3	1.6	5.9
6.1.2	PCT resident patent ap/bn GDP PPP$	3.6	0.3	0.2	0.0	1.5
6.1.3	Domestic res utility model ap/bn GDP PPP$[a]	2.5	1.1	10.6	2.5	4.0
6.1.4	Scientific & technical articles/bn GDP PPP$	15.3	4.4	3.3	3.7	7.8
6.2	*Knowledge impact*					
6.2.1	Growth rate of GDP PPP$/worker, %	0.6	2.1	3.1	3.5	2.0
6.2.2	New businesses/1,000 pop. 15–64 yrs	5.5	2.5	0.7	0.5	3.0
6.2.3	Computer software spending, % GDP[a]	0.6	0.2	0.2	0.1	0.4
6.3	*Knowledge diffusion*					
6.3.1	Royalty & license fees receipts, % GDP	0.3	0.0	0.3	0.0	0.2
6.3.2	High-tech exports less re-exports, %	10.7	4.9	3.3	0.9	6.2
6.3.3	Computer & comm service exports, %	38.3	26.5	30.2	22.9	31.1
6.3.4	FDI net outflows, % GDP	13.5	0.8	0.3	0.1	5.1

Note: (*) index, (†) survey question, (a) half weight, (b) higher values indicate worse outcomes.

Table 1g: Creative outputs pillar

	Indicator	Average value by income group: High income	Upper-middle income	Lower-middle income	Low income	Mean
7	**Creative outputs**					
7.1	*Creative intangibles*					
7.1.1	Domestic res trademark ap/bn GDP PPP$	31.5	47.3	63.2	15.7	40.6
7.1.2	Madrid resident trademark ap/bn GDP PPP$[a]	13.9	6.3	4.2	0.1	9.2
7.1.3	ICT & business models[†]	5.1	4.4	4.3	4.1	4.6
7.1.4	ICT & organizational models[†]	4.7	4.1	3.9	3.5	4.2
7.2	*Creative goods & services*					
7.2.1	Recreation & culture consumption, %[a]	7.8	4.1	2.4	2.0	5.7
7.2.2	National feature films/mn pop.[a]	4.0	0.6	0.9	1.4	2.3
7.2.3	Daily newspapers/1,000 literate pop.[a]	304.3	89.1	79.1	6.8	164.6
7.2.4	Creative goods exports, %	2.1	1.5	1.8	0.9	1.7
7.2.5	Creative services exports, %	6.2	4.7	1.8	1.9	4.1

Note: (*) index, (†) survey question, (a) half weight, (b) higher values indicate worse outcomes.

foreign trade and investment have the additional effect of exposing domestic firms to best practices around the globe, which is critical to innovation through knowledge absorption and diffusion. The rationale behind sub-pillars 5.3 on knowledge absorption (an enabler) and 6.3 on knowledge diffusion (a result), two sub-pillars designed to be mirror images of each other, is precisely that together they will reveal how good countries are at absorbing and diffusing knowledge.

Sub-pillar 5.3 includes four statistics all linked to sectors with high-tech content or that are key to innovation: royalty and license fees payments as a percentage of GDP; high-tech imports (net of re-imports) as a percentage of total imports; imports of computer, communications, and other services as a percentage of commercial service imports; and net inflows of FDI as a percentage of GDP.

Innovation Output Sub-Index

Innovation outputs are the results of innovative activities within the economy. Although the Output Sub-Index includes only two pillars, it is averaged with the Input Sub-Index in the overall GII. This asymmetry, which is only apparent, serves two purposes: to give innovation results the same weight as innovation enablers, and to redress an unbalance at the level of available statistics.

There are two output pillars: Scientific outputs and Creative outputs.

Scientific outputs

This pillar covers all those variables that are traditionally thought to be the fruits of innovation (Table 1f). The first sub-pillar refers to the creation of knowledge. It includes four indicators that are the result of inventive and innovation activities: patent applications filed by residents both at the national patent office and at the international level through the Patent Cooperation Treaty (see Box 6); utility model applications filed by residents at the national office; and scientific and technical published articles in peer-reviewed journals.

The second sub-pillar, on knowledge impact, includes three statistics representing the impact of innovation activities at the micro and macroeconomic level: increases in labour productivity, the entry density of new firms, and spending on software.

The third sub-pillar, on knowledge diffusion, is the mirror image of the Knowledge absorption sub-pillar under pillar 5. It includes four statistics all linked to sectors with high-tech content or that are key to innovation: royalty and license fees receipts as a percentage of GDP; high-tech exports (net of re-exports) as a percentage of total exports (net of re-exports); exports of computer, communications, and other services as a percentage of commercial service exports; and net outflows of FDI as a percentage of GDP.

Creative outputs

The last pillar, on creative outputs, has only two sub-pillars (Table 1g). This is essentially the result of the lack of reliable indicators across many countries on all copyrighted-related industries and creativity in general (see Chapter 6, 'Accounting for Creativity in Innovation'). As new and better statistics become available in the coming years, this pillar will be strengthened.

The first sub-pillar on creative intangibles includes statistics on trademark registrations by residents at the national office and under the Madrid System, as well as two survey questions regarding the use of ICT in business and organizational models, new areas that are increasingly linked to innovation in the literature.

The last sub-pillar, on creative goods and services, includes the share of household expenditure in recreation and culture as a proxy for creative activities in a given country. Two UNESCO series for which

Box 6: Capturing innovation: The patent system

This edition of the Global Innovation Index (GII) relies on two patent-based measures drawn from the WIPO Statistics Database (www.wipo.int/ipstats/en) described below. In addition, statistics on utility models and trademarks are used in different pillars.

The first measure is concerned with the number of patent applications filed by residents at their national patent office (resident applications).

When an inventor decides to protect an invention through the patent system, the first step is to file an application with a patent office. In most cases, applicants tend to file at their national patent office. Data on resident patent applications (2009 or latest available year) capture this patenting activity of residents in a given country. An application is filed with a patent office by an applicant residing in the country in which that office has jurisdiction. For example, a patent application filed with the Japan Patent Office (JPO) by a resident of Japan is considered a resident application for the JPO.

In contrast, patent indicators based on a specific office will introduce a home bias between resident (domestic) and non-resident (foreign) applications because the propensity to patent at the national patent office is considerably higher than the propensity to patent abroad. For example, only 4.4% of total Chinese patent applications in 2008 were filed abroad. Patents submitted to one single patent office are also likely to reflect the trade patterns of that particular country. Moreover, data of one single office will capture only a fraction of world innovation.

In addition, the use of statistics on patent applications—rather than data on patents granted—ensures that innovative performance is captured in a more timely and comprehensive manner. In contrast, data on patent grants reflect inventions that obtain patent protection and that are most likely several years old. This is the result of the lengthy (and increasing) processing and examination periods that are part of the patenting process.

The second patent-based measure presented in this report is the number of patents filed under the WIPO-administered Patent Cooperation Treaty (PCT). To complement national data, the second metric provides the number of PCT international applications by residents of a given country in 2010.

National patent office data are frequently criticized on the grounds that there is a lack of international comparability. The use of PCT data to some extent alleviates this criticism.

An inventor of a promising technology with international market potential will wish to protect his or her invention in more than one country. In addition to filing patents directly in other jurisdictions, inventors can file an 'international application' through the PCT, which facilitates the acquisition of patent rights in a large number of jurisdictions (142 contracting states) by reducing the requirement to file a separate application in each jurisdiction.

The use of PCT data sheds light on patents that might be most economically valuable, as these are the ones that inventors are likely to patent abroad and for which inventors are willing to incur the extra costs that the process of patenting abroad requires.

This combination of data on national IP filings and filings under the PCT system makes for a strong and timely indicator of inventive activity and innovation with very good country coverage. It better achieves the goal of capturing worldwide innovative activity, with a particular emphasis on inventions in medium- or lower-income economies and inventions that may have a strong international appeal.

Source

Khan and Wunsch-Vincent, 2011; WIPO, 2010.

data are available were included: national feature films produced in a given country, and daily newspapers. Other data series, notably on the publication of books, have been discontinued. Since statistics on creative and copyright-related industries' national output values are not available for a sufficient number of countries, trade figures were incorporated. These are based on the classifications used by UNCTAD in its *Creative Economy Report,* which includes data on creative goods and services exports.

Discussion of results: The world's top innovators

This year the GII model includes 125 countries that represent 93.2% of the world's population and 98.0% of the world's GDP (in current US dollars). This section provides a general discussion of the GII and the rankings of the related sub-indices. The emphasis, for each index, is on general trends and includes details of the results for the global leaders and the best performers within each income category (high, upper-middle, lower-middle, and low-income

groups).[26] A short discussion of the rankings at the regional level follows, along with some additional information on regional leaders by income group.

Tables 2 through 4 report on the overall GII and the Input and Output Sub-Indices, with regional and income group rankings. Tables 5a through 5g provide the rankings per pillar, with details on sub-pillar scores. Table 6 shows the top 10 Efficiency Index rankings, and Tables 7a through 7d report on the Efficiency Index by income group. These tables should be read in conjunction with the country/economy profiles (Appendix I) and the data tables (Appendix II). The profiles provide only normalized scores in the [0, 100] range, to facilitate the replicability of results and provide a sense of the scores with greater leverage for each country. The data tables provide both the raw indicator and the normalized score.[27]

This section includes a brief discussion of some general results, with detailed discussions provided for index and income group leaders (in bold). Each economy has its own strengths and weaknesses, the full details of which are available in the country/economy profiles (Appendix I).

The Global Innovation Index

The overall GII scores provide a composite picture of the state of each country's innovation performance.

Top 10

The top 10 countries in the GII 2001 edition are dominated by Europe, with six countries, and includes two Asian economies and two North American countries: Switzerland, Sweden, Singapore, Hong Kong (SAR, China), Finland, Denmark, the United States of America (US), Canada, the Netherlands, and the United Kingdom (UK). Leaders in their respective regions are Switzerland (1st), Singapore (3rd), the US (7th), Israel (14th), Chile (38th), Mauritius (53rd), and India (62nd).

Switzerland comes in at top place in the overall GII 2011 rankings (up from position 4th last year) on the basis of its strong position in both the Input and Output Sub-Indices (3rd and 2nd). Although the country does not top any individual pillar, it places within the top 5 in three Input pillars (Institutions at 5th, Market sophistication at 5th, and Business sophistication at 4th) and both Output pillars (Scientific outputs at 2nd and Creative outputs at 3rd). Its many strengths include its top 10 positions on 30 indicators as well as on the following sub-pillars: political (2nd) and regulatory environment (7th), R&D (9th), energy (7th), credit (10th), investment (6th), knowledge workers (2nd), innovation linkages (6th), knowledge creation (4th), knowledge diffusion (10th), and creative intangibles (9th). A high degree of Innovation Efficiency (3rd among high-income countries) allows Switzerland to translate these strengths into visible innovation outputs, with high marks on key output indicators: resident patent filings at the Patent Cooperation Treaty applications (1st), scientific and technical journal articles (2nd), computer software spending (1st), high-tech exports (6th), FDI net outflows (9th), and resident international trademark applications under the Madrid system (1st). It also exhibits high marks on creativity, at positions 7th, 4th, and 8th on national feature films, daily newspapers, and creative goods exports, respectively.

The runner-up, **Sweden,** is the only country in the top 10 on all four indices, and the only country in the top 10 in the GII to be among the 10 most efficient innovators (ranked 6th on the Efficiency Index). A knowledge-based economy, this outstanding performance is driven by 1st place on Scientific outputs, with positions in the top 10 on patent filings through the Patent Cooperation Treaty (3rd), scientific and technical journal articles (3rd), computer software spending (4th), royalty and license fees' receipts (3rd), and FDI net outflows (7th). Sweden is ranked 2nd on Creative outputs, placing among the top 10 on four indicators. Its position 5 on the Input side is also one of the best, with top 10 positions on seven of the 15 sub-pillars: Political environment (4th), Education (8th), R&D (4th), ICT (10th), Energy (5th), Investment (10th) and Knowledge workers (6th).

Singapore is at 3rd position overall, moving up significantly from position 7 last year. Singapore shows its strongest performance in the Input Sub-Index, coming in at 1st place on the basis of its strengths in the Institutions (9th), Human capital and research (1st), Infrastructure (9th), Market sophistication (2nd), and Business sophistication (1st) pillars. However, Singapore's Innovation Efficiency Index ranking is low (36th among high-income countries, 94th in the general rankings); this shows up in its relative weak performance in the Output Sub-Index, where it is ranked 17th overall (Scientific outputs at 15th place and Creative outputs at 30th). Singapore is ranked

40th in patent applications at the national office, 79th in national office trademark applications, and 79th in creative services exports.

Top performers by income group

By income group, from high- to low-income countries, the leaders are Switzerland (1st), Malaysia (31st), China (29th), and Ghana (70th). China, at position 29, is the only developing country to be among the top 30; Malaysia (31st), Chile (38th), Moldova (39th), and Lithuania (40th) make it to the top 40. Among high-income countries, three countries lag behind: Greece reached the median score (63rd), followed by Trinidad and Tobago (72nd) and Brunei Darussalam (75th).

Malaysia tops the overall ranking within the upper-middle-income group at 31st position overall, placing 27th on the Input Sub-Index (1st within its income group) and 35th on the Output one. On the Input side, Malaysia's strengths come from the Market and Business sophistication pillars, where it is ranked 1st on R&D performed and financed by business at 84.9% and 84.7%, respectively, and on the strength of legal rights for credit and the depth of credit information indicators—positions matched with the country's level of domestic credit to the private sector (20th, at 100.8% of GDP). Malaysia also ranks within the top 10 worldwide on the strength of investor protection and market capitalization, as well as on imports and exports of goods and services. Its performances on the other three input pillars are generally weak, except in two particular indicators where it does very well: graduates in science (8th) and engineering (2nd). On the Output side, Malaysia's leverage is provided by its 19th placement on sub-pillar 6.3, Knowledge diffusion, where it ranks 2nd—after Singapore—on high-tech exports (which represent an impressive 35.4% of total exports), 13th on FDI net outflows, and 23rd on royalty and license fees' receipts.

From the same region, at position 29, **China** tops the GII rankings among lower-middle-income countries, and is the only country from this income group in the top 30. China exhibits many strengths. It is 1st within its income group on all three main Indices (GII, Input, and Output). On the Input side, it ranks among the top 30 in Market (26th) and Business sophistication (29th). The country places among the top 10 on the Investment sub-pillar, with the 5th most dynamic stock market (with the total value of stocks traded reaching 179.7% of GDP) and among the top 20 on market capitalization (placing 16th with 100.5% of GDP). China ranks 1st on the percentage of firms offering formal training (at 84.8%), 4th on high-tech imports (at 26.8% of total imports), 6th and 10th on R&D financed and performed by business (at 70.4% and 72.3%, respectively), 7th on the state of cluster development (a survey question), 13th on domestic credit to private sector (127.3% of GDP), and 17th on the intensity of local competition (survey question). On a less positive note, however, China presents important weaknesses on Institutions (98th) and Human capital and research (56th), where the 1st position on the PISA assessment on reading, mathematics and science corresponds to Shanghai only. It achieves, however, relatively good rankings on R&D, placing 24th on gross expenditure on R&D and 36th on the quality of research institutions (a survey question). On the Output Sub-Index, its 14th position overall is well deserved: China gets its leverage from placing within the top 10 on a third of the indicators in pillars 6 and 7 for which data were available, including resident utility model (1st), patent (3rd), and trademark (9th) applications at the national office, growth in labour productivity (3rd position, with an impressive 8.4%), high-tech exports (3rd, with 29.9% of total exports), and creative goods exports (4th, with 5.9% of total exports).

Among low-income countries, **Ghana** prevails with an overall rank of 70. Ghana's relative strengths lie in its Institutions, Human capital and research, Scientific outputs, and Business sophistication, with scores in the top 40 on freedom of the press (25th), public expenditure per pupil (34th), and percentage of graduates in science (31st); its commendable share of renewables in energy use (with 72.5%, it is ranked 10th globally), and its strength of legal rights for credit (19th) and investor protection (34th). These strong points translate into relatively high scores on venture capital deals (41st), joint venture and strategic alliance deals (41st), and FDI net inflows (20th); they also provide a strong growth in labour productivity (with 4.4%, it is ranked 23rd on this indicator).

The Innovation Output Sub-Index

The Innovation Output Sub-Index variables provide information on elements that are the result of innovation within an economy.

Table 2: Global Innovation Index rankings

Country/Economy	Score (0–100)	Rank	Income	Rank	Region	Rank	GII PAST YEARS 2010	2009
Switzerland	63.82	1	HI	1	ECS	1	4	7
Sweden	62.12	2	HI	2	ECS	2	2	3
Singapore	59.64	3	HI	3	EAS	1	7	5
Hong Kong (SAR), China	58.80	4	HI	4	EAS	2	3	12
Finland	57.50	5	HI	5	ECS	3	6	13
Denmark	56.96	6	HI	6	ECS	4	5	8
United States of America	56.57	7	HI	7	NAC	1	11	1
Canada	56.33	8	HI	8	NAC	2	12	11
Netherlands	56.31	9	HI	9	ECS	5	8	10
United Kingdom	55.96	10	HI	10	ECS	6	14	4
Iceland	55.10	11	HI	11	ECS	7	1	20
Germany	54.89	12	HI	12	ECS	8	16	2
Ireland	54.10	13	HI	13	ECS	9	19	21
Israel	54.03	14	HI	14	MEA	1	23	23
New Zealand	53.79	15	HI	15	EAS	3	9	27
Korea, Rep.	53.68	16	HI	16	EAS	4	20	6
Luxembourg	52.65	17	HI	17	ECS	10	15	17
Norway	52.60	18	HI	18	ECS	11	10	14
Austria	50.75	19	HI	19	ECS	12	21	15
Japan	50.32	20	HI	20	EAS	5	13	9
Australia	49.85	21	HI	21	EAS	6	18	22
France	49.25	22	HI	22	ECS	13	22	19
Estonia	49.18	23	HI	23	ECS	14	29	29
Belgium	49.05	24	HI	24	ECS	15	17	18
Hungary	48.12	25	HI	25	ECS	16	36	47
Qatar	47.74	26	HI	26	MEA	2	35	24
Czech Republic	47.30	27	HI	27	ECS	17	27	33
Cyprus	46.45	28	HI	28	ECS	18	32	45
China	46.43	29	LM	1	EAS	7	43	37
Slovenia	45.07	30	HI	29	ECS	19	26	36
Malaysia	44.05	31	UM	1	EAS	8	28	25
Spain	43.81	32	HI	30	ECS	20	30	28
Portugal	42.40	33	HI	31	ECS	21	34	40
United Arab Emirates	41.99	34	HI	32	MEA	3	24	26
Italy	40.69	35	HI	33	ECS	22	38	31
Latvia	39.80	36	HI	34	ECS	23	44	60
Slovak Republic	39.05	37	HI	35	ECS	24	37	35
Chile	38.84	38	UM	2	LCN	1	42	39
Moldova, Rep.	38.66	39	LM	2	ECS	25	n/a	116
Lithuania	38.49	40	UM	3	ECS	26	39	42
Jordan	38.43	41	LM	3	MEA	4	58	55
Bulgaria	38.42	42	UM	4	ECS	27	49	74
Poland	38.02	43	HI	36	ECS	28	47	56
Croatia	37.98	44	HI	37	ECS	29	45	62
Costa Rica	37.91	45	UM	5	LCN	2	41	48
Bahrain	37.80	46	HI	38	MEA	5	40	34
Brazil	37.75	47	UM	6	LCN	3	68	50
Thailand	37.63	48	LM	4	EAS	9	60	44
Lebanon	37.11	49	UM	7	MEA	6	n/a	n/a
Romania	36.83	50	UM	8	ECS	30	52	69
Viet Nam	36.71	51	LM	5	EAS	10	71	64
Kuwait	36.64	52	HI	39	MEA	7	33	30
Mauritius	36.47	53	UM	9	SSF	1	73	66
Saudi Arabia	36.44	54	HI	40	MEA	8	54	32
Serbia	36.31	55	UM	10	ECS	31	101	92
Russian Federation	35.85	56	UM	11	ECS	32	64	68
Oman	35.51	57	HI	41	MEA	9	65	52
Argentina	35.36	58	UM	12	LCN	4	75	84
South Africa	35.22	59	UM	13	SSF	2	51	43
Ukraine	35.01	60	LM	6	ECS	33	61	79
Guyana	34.83	61	LM	7	LCN	5	113	103
India	34.52	62	LM	8	SAS	1	56	41
Greece	34.18	63	HI	42	ECS	34	46	54

Table 2: Global Innovation Index rankings *(continued)*

Country/Economy	Score (0–100)	Rank	Income	Rank	Region	Rank	GII PAST YEARS 2010	2009
Uruguay	34.18	64	UM	14	LCN	6	53	80
Turkey	34.11	65	UM	15	ECS	35	67	51
Tunisia	33.89	66	LM	9	MEA	10	62	46
Macedonia	33.47	67	UM	16	ECS	36	77	89
Mongolia	33.40	68	LM	10	EAS	11	87	105
Armenia	33.00	69	LM	11	ECS	37	82	104
Ghana	32.48	70	LI	**1**	SSF	3	105	n/a
Colombia	32.32	71	UM	17	LCN	7	90	75
Trinidad and Tobago	32.17	72	HI	43	LCN	8	55	65
Georgia	31.87	73	LM	12	ECS	38	84	98
Paraguay	31.17	74	LM	13	LCN	9	127	118
Brunei Darussalam	30.93	75	HI	44	EAS	12	48	n/a
Bosnia & Herzegovina	30.84	76	UM	18	ECS	39	116	n/a
Panama	30.77	77	UM	19	LCN	10	66	67
Namibia	30.74	78	UM	20	SSF	4	92	95
Botswana	30.51	79	UM	21	SSF	5	86	77
Albania	30.45	80	UM	22	ECS	40	81	121
Mexico	30.45	81	UM	23	LCN	11	69	61
Sri Lanka	30.36	82	LM	14	SAS	2	79	58
Peru	30.34	83	UM	24	LCN	12	88	85
Kazakhstan	30.32	84	UM	25	ECS	41	63	72
Kyrgyzstan	29.79	85	LI	2	ECS	42	104	122
Guatemala	29.33	86	LM	15	LCN	13	95	81
Egypt	29.21	87	LM	16	MEA	11	74	76
Azerbaijan	29.17	88	UM	26	ECS	43	57	57
Kenya	29.15	89	LI	3	SSF	6	83	78
El Salvador	29.14	90	LM	17	LCN	14	91	88
Philippines	28.98	91	LM	18	EAS	13	76	63
Jamaica	28.88	92	UM	27	LCN	15	70	73
Ecuador	28.75	93	LM	19	LCN	16	126	109
Morocco	28.73	94	LM	20	MEA	12	94	82
Iran	28.41	95	UM	28	MEA	13	n/a	n/a
Nigeria	28.15	96	LM	21	SSF	7	96	70
Bangladesh	28.05	97	LI	4	SAS	3	120	111
Honduras	27.81	98	LM	22	LCN	17	112	83
Indonesia	27.78	99	LM	23	EAS	14	72	49
Senegal	27.56	100	LM	24	SSF	8	106	90
Swaziland	27.52	101	LM	25	SSF	9	n/a	n/a
Venezuela	27.41	102	UM	29	LCN	18	124	101
Cameroon	26.95	103	LM	26	SSF	10	119	106
Tanzania	26.88	104	LI	5	SSF	11	98	86
Pakistan	26.75	105	LM	27	SAS	4	103	93
Uganda	26.37	106	LI	6	SSF	12	108	100
Mali	26.35	107	LI	7	SSF	13	107	97
Malawi	25.96	108	LI	8	SSF	14	97	n/a
Rwanda	25.86	109	LI	9	SSF	15	n/a	n/a
Nicaragua	25.78	110	LM	28	LCN	19	117	114
Cambodia	25.46	111	LI	10	EAS	15	102	117
Bolivia	25.44	112	LM	29	LCN	20	129	123
Madagascar	25.41	113	LI	11	SSF	16	125	113
Zambia	25.27	114	LI	12	SSF	17	111	96
Syrian Arab Republic	24.82	115	LM	30	MEA	14	132	94
Tajikistan	24.50	116	LI	13	ECS	44	115	112
Côte d'Ivoire	24.08	117	LM	31	SSF	18	89	n/a
Benin	23.81	118	LI	14	SSF	19	118	99
Zimbabwe	23.54	119	LI	15	SSF	20	131	126
Burkina Faso	23.14	120	LI	16	SSF	21	122	115
Ethiopia	22.88	121	LI	17	SSF	22	123	120
Niger	21.41	122	LI	18	SSF	23	n/a	n/a
Yemen	20.72	123	LM	32	MEA	15	n/a	n/a
Sudan	20.36	124	LM	33	SSF	24	n/a	n/a
Algeria	19.79	125	UM	30	MEA	16	121	108

Note: World Bank Income Group Classification (January 2011): LI = low income; LM = lower-middle income; UM = upper-middle income; and HI = high income; World Bank Regional Classification (January 2011): ECS = Europe & Central Asia; MEA = Middle East & North Africa; SSF = Sub-Saharan Africa; EAS = East Asia & Pacific; SAS = South Asia; NAC = North America; and LCN = Latin America & Caribbean.

Table 3: Innovation Input Sub-Index rankings

Country/Economy	Score (0–100)	Rank	Income	Rank	Region	Rank
Singapore	74.11	1	HI	1	EAS	1
Hong Kong (SAR), China	69.77	2	HI	2	EAS	2
Switzerland	66.07	3	HI	3	ECS	1
Ireland	65.53	4	HI	4	ECS	2
Sweden	64.85	5	HI	5	ECS	3
Finland	64.71	6	HI	6	ECS	4
Denmark	64.57	7	HI	7	ECS	5
Canada	64.41	8	HI	8	NAC	1
Luxembourg	63.93	9	HI	9	ECS	6
United Kingdom	63.66	10	HI	10	ECS	7
United States of America	62.84	11	HI	11	NAC	2
Australia	62.81	12	HI	12	EAS	3
Iceland	62.48	13	HI	13	ECS	8
Norway	61.15	14	HI	14	ECS	9
New Zealand	60.97	15	HI	15	EAS	4
Netherlands	60.42	16	HI	16	ECS	10
Korea, Rep.	59.43	17	HI	17	EAS	5
Japan	59.34	18	HI	18	EAS	6
Austria	59.28	19	HI	19	ECS	11
Israel	59.12	20	HI	20	MEA	1
Germany	59.04	21	HI	21	ECS	12
Belgium	58.44	22	HI	22	ECS	13
France	55.61	23	HI	23	ECS	14
Estonia	54.86	24	HI	24	ECS	15
United Arab Emirates	54.38	25	HI	25	MEA	2
Czech Republic	53.11	26	HI	26	ECS	16
Malaysia	52.94	27	UM	1	EAS	7
Bahrain	52.73	28	HI	27	MEA	3
Spain	52.43	29	HI	28	ECS	17
Cyprus	52.38	30	HI	29	ECS	18
Qatar	51.71	31	HI	30	MEA	4
Slovenia	51.29	32	HI	31	ECS	19
Hungary	51.04	33	HI	32	ECS	20
Portugal	50.32	34	HI	33	ECS	21
Slovak Republic	48.27	35	HI	34	ECS	22
Chile	48.09	36	UM	2	LCN	1
Italy	47.88	37	HI	35	ECS	23
Latvia	47.46	38	HI	36	ECS	24
Lithuania	47.46	39	UM	3	ECS	25
South Africa	46.37	40	UM	4	SSF	1
Poland	46.29	41	HI	37	ECS	26
Oman	46.23	42	HI	38	MEA	5
China	46.08	43	LM	1	EAS	8
Saudi Arabia	45.94	44	HI	39	MEA	6
Croatia	45.00	45	HI	40	ECS	27
Mauritius	44.79	46	UM	5	SSF	2
Bulgaria	44.20	47	UM	6	ECS	28
Thailand	43.33	48	LM	2	EAS	9
Namibia	43.01	49	UM	7	SSF	3
Greece	42.48	50	HI	41	ECS	29
Kuwait	42.44	51	HI	42	MEA	7
Mongolia	42.31	52	LM	3	EAS	10
Costa Rica	42.22	53	UM	8	LCN	2
Bosnia & Herzegovina	42.10	54	UM	9	ECS	30
Romania	41.80	55	UM	10	ECS	31
Jordan	41.34	56	LM	4	MEA	8
Lebanon	40.88	57	UM	11	MEA	9
Trinidad and Tobago	40.86	58	HI	43	LCN	3
Russian Federation	40.79	59	UM	12	ECS	32
Panama	40.73	60	UM	13	LCN	4
Macedonia	40.37	61	UM	14	ECS	33
Botswana	40.37	62	UM	15	SSF	4
Viet Nam	40.09	63	LM	5	EAS	11

Table 3: Innovation Input Sub-Index rankings *(continued)*

Country/Economy	Score (0–100)	Rank	Income	Rank	Region	Rank
Kazakhstan	39.86	64	UM	16	ECS	34
Ghana	39.84	65	LI	**1**	SSF	5
Uruguay	39.69	66	UM	17	LCN	5
Ukraine	39.59	67	LM	6	ECS	35
Brazil	39.47	68	UM	18	LCN	6
Kenya	39.24	69	LI	2	SSF	6
Brunei Darussalam	39.19	70	HI	44	EAS	12
Serbia	39.09	71	UM	19	ECS	36
Peru	39.06	72	UM	20	LCN	7
Jamaica	38.89	73	UM	21	LCN	8
Colombia	38.72	74	UM	22	LCN	9
Guyana	38.70	75	LM	7	LCN	10
Georgia	38.54	76	LM	8	ECS	37
Moldova, Rep.	38.40	77	LM	9	ECS	38
Albania	38.29	78	UM	23	ECS	39
Tunisia	38.21	79	LM	10	MEA	10
Turkey	37.96	80	UM	24	ECS	40
Mexico	37.47	81	UM	25	LCN	11
Argentina	37.29	82	UM	26	LCN	12
Azerbaijan	37.21	83	UM	27	ECS	41
Armenia	37.10	84	LM	11	ECS	42
Swaziland	36.93	85	LM	12	SSF	7
Morocco	36.65	86	LM	13	MEA	11
India	36.47	87	LM	14	SAS	**1**
Egypt	35.08	88	LM	15	MEA	12
Kyrgyzstan	34.93	89	LI	3	ECS	43
Rwanda	34.73	90	LI	4	SSF	8
El Salvador	34.60	91	LM	16	LCN	13
Paraguay	34.45	92	LM	17	LCN	14
Philippines	34.00	93	LM	18	EAS	13
Zambia	33.81	94	LI	5	SSF	9
Indonesia	33.57	95	LM	19	EAS	14
Sri Lanka	33.20	96	LM	20	SAS	2
Guatemala	33.18	97	LM	21	LCN	15
Honduras	33.08	98	LM	22	LCN	16
Malawi	32.82	99	LI	6	SSF	10
Ecuador	32.57	100	LM	23	LCN	17
Algeria	32.07	101	UM	28	MEA	13
Niger	31.44	102	LI	7	SSF	11
Cambodia	31.24	103	LI	8	EAS	15
Madagascar	31.20	104	LI	9	SSF	12
Nicaragua	31.13	105	LM	24	LCN	18
Iran	30.91	106	UM	29	MEA	14
Senegal	30.73	107	LM	25	SSF	13
Tanzania	30.45	108	LI	10	SSF	14
Bolivia	30.37	109	LM	26	LCN	19
Cameroon	30.12	110	LM	27	SSF	15
Syrian Arab Republic	30.03	111	LM	28	MEA	15
Uganda	29.86	112	LI	11	SSF	16
Mali	29.85	113	LI	12	SSF	17
Bangladesh	29.64	114	LI	13	SAS	3
Venezuela	29.48	115	UM	30	LCN	20
Ethiopia	29.29	116	LI	14	SSF	18
Burkina Faso	29.24	117	LI	15	SSF	19
Benin	28.26	118	LI	16	SSF	20
Nigeria	27.72	119	LM	29	SSF	21
Tajikistan	27.64	120	LI	17	ECS	44
Yemen	27.00	121	LM	30	MEA	16
Zimbabwe	26.82	122	LI	18	SSF	22
Pakistan	26.57	123	LM	31	SAS	4
Sudan	26.06	124	LM	32	SSF	23
Côte d'Ivoire	23.40	125	LM	33	SSF	24

Note: World Bank Income Group Classification (January 2011): LI = low income; LM = lower-middle income; UM = upper-middle income; and HI = high income; World Bank Regional Classification (January 2011): ECS = Europe & Central Asia; MEA = Middle East & North Africa; SSF = Sub-Saharan Africa; EAS = East Asia & Pacific; SAS = South Asia; NAC = North America; and LCN = Latin America & Caribbean.

Table 4: Innovation Output Sub-Index rankings

Country/Economy	Score (0–100)	Rank	Income	Rank	Region	Rank
Sweden	59.40	**1**	HI	**1**	ECS	**1**
Switzerland	58.20	2	HI	2	ECS	2
Netherlands	52.20	3	HI	3	ECS	3
Germany	50.74	4	HI	4	ECS	4
United States of America	50.30	5	HI	5	NAC	**1**
Finland	50.29	6	HI	6	ECS	5
Denmark	49.34	7	HI	7	ECS	6
Israel	48.94	8	HI	8	MEA	**1**
United Kingdom	48.27	9	HI	9	ECS	7
Canada	48.26	10	HI	10	NAC	2
Korea, Rep.	47.93	11	HI	11	EAS	**1**
Hong Kong (SAR), China	47.83	12	HI	12	EAS	2
Iceland	47.72	13	HI	13	ECS	8
China	46.77	14	LM	**1**	EAS	3
New Zealand	46.61	15	HI	14	EAS	4
Hungary	45.20	16	HI	15	ECS	9
Singapore	45.18	17	HI	16	EAS	5
Norway	44.04	18	HI	17	ECS	10
Qatar	43.77	19	HI	18	MEA	2
Estonia	43.50	20	HI	19	ECS	11
France	42.90	21	HI	20	ECS	12
Ireland	42.67	22	HI	21	ECS	13
Austria	42.21	23	HI	22	ECS	14
Czech Republic	41.49	24	HI	23	ECS	15
Luxembourg	41.37	25	HI	24	ECS	16
Japan	41.30	26	HI	25	EAS	6
Cyprus	40.52	27	HI	26	ECS	17
Belgium	39.66	28	HI	27	ECS	18
Moldova, Rep.	38.92	29	LM	2	ECS	19
Slovenia	38.86	30	HI	28	ECS	20
Australia	36.89	31	HI	29	EAS	7
Brazil	36.03	32	UM	**1**	LCN	**1**
Jordan	35.52	33	LM	3	MEA	3
Spain	35.19	34	HI	30	ECS	21
Malaysia	35.17	35	UM	2	EAS	8
Portugal	34.47	36	HI	31	ECS	22
Costa Rica	33.60	37	UM	3	LCN	2
Serbia	33.53	38	UM	4	ECS	23
Italy	33.49	39	HI	32	ECS	24
Argentina	33.44	40	UM	5	LCN	3
Lebanon	33.34	41	UM	6	MEA	4
Viet Nam	33.34	42	LM	4	EAS	9
Bulgaria	32.64	43	UM	7	ECS	25
India	32.56	44	LM	5	SAS	**1**
Latvia	32.14	45	HI	33	ECS	26
Thailand	31.93	46	LM	6	EAS	10
Romania	31.86	47	UM	8	ECS	27
Croatia	30.96	48	HI	34	ECS	28
Guyana	30.95	49	LM	7	LCN	4
Russian Federation	30.91	50	UM	9	ECS	29
Kuwait	30.85	51	HI	35	MEA	5
Ukraine	30.42	52	LM	8	ECS	30
Turkey	30.25	53	UM	10	ECS	31
Slovak Republic	29.83	54	HI	36	ECS	32
Poland	29.74	55	HI	37	ECS	33
United Arab Emirates	29.61	56	HI	38	MEA	6
Chile	29.60	57	UM	11	LCN	5
Tunisia	29.57	58	LM	9	MEA	7
Lithuania	29.52	59	UM	12	ECS	34
Armenia	28.90	60	LM	10	ECS	35
Uruguay	28.67	61	UM	13	LCN	6
Nigeria	28.58	62	LM	11	SSF	**1**
Mauritius	28.15	63	UM	14	SSF	2

Table 4: Innovation Output Sub-Index rankings *(continued)*

Country/Economy	Score (0–100)	Rank	Income	Rank	Region	Rank
Paraguay	27.90	64	LM	12	LCN	7
Sri Lanka	27.53	65	LM	13	SAS	2
Saudi Arabia	26.94	66	HI	39	MEA	8
Pakistan	26.94	67	LM	14	SAS	3
Macedonia	26.57	68	UM	15	ECS	36
Bangladesh	26.47	69	LI	1	SAS	4
Colombia	25.92	70	UM	16	LCN	8
Iran	25.91	71	UM	17	MEA	9
Greece	25.89	72	HI	40	ECS	37
Guatemala	25.49	73	LM	15	LCN	9
Venezuela	25.35	74	UM	18	LCN	10
Georgia	25.20	75	LM	16	ECS	38
Ghana	25.12	76	LI	2	SSF	3
Ecuador	24.94	77	LM	17	LCN	11
Oman	24.79	78	HI	41	MEA	10
Côte d'Ivoire	24.77	79	LM	18	SSF	4
Kyrgyzstan	24.65	80	LI	3	ECS	39
Mongolia	24.49	81	LM	19	EAS	11
Senegal	24.38	82	LM	20	SSF	5
South Africa	24.07	83	UM	19	SSF	6
Philippines	23.96	84	LM	21	EAS	12
Cameroon	23.79	85	LM	22	SSF	7
El Salvador	23.67	86	LM	23	LCN	12
Trinidad and Tobago	23.47	87	HI	42	LCN	13
Mexico	23.42	88	UM	20	LCN	14
Egypt	23.34	89	LM	24	MEA	11
Tanzania	23.30	90	LI	4	SSF	8
Uganda	22.87	91	LI	5	SSF	9
Bahrain	22.87	92	HI	43	MEA	12
Mali	22.85	93	LI	6	SSF	10
Brunei Darussalam	22.68	94	HI	44	EAS	13
Albania	22.62	95	UM	21	ECS	40
Honduras	22.53	96	LM	25	LCN	15
Indonesia	21.99	97	LM	26	EAS	14
Peru	21.63	98	UM	22	LCN	16
Tajikistan	21.36	99	LI	7	ECS	41
Azerbaijan	21.13	100	UM	23	ECS	42
Panama	20.82	101	UM	24	LCN	17
Morocco	20.81	102	LM	27	MEA	13
Kazakhstan	20.77	103	UM	25	ECS	43
Botswana	20.65	104	UM	26	SSF	11
Bolivia	20.51	105	LM	28	LCN	18
Nicaragua	20.44	106	LM	29	LCN	19
Zimbabwe	20.26	107	LI	8	SSF	12
Cambodia	19.68	108	LI	9	EAS	15
Madagascar	19.63	109	LI	10	SSF	13
Syrian Arab Republic	19.61	110	LM	30	MEA	14
Bosnia & Herzegovina	19.58	111	UM	27	ECS	44
Benin	19.35	112	LI	11	SSF	14
Malawi	19.11	113	LI	12	SSF	15
Kenya	19.05	114	LI	13	SSF	16
Jamaica	18.87	115	UM	28	LCN	20
Namibia	18.46	116	UM	29	SSF	17
Swaziland	18.11	117	LM	31	SSF	18
Burkina Faso	17.04	118	LI	14	SSF	19
Rwanda	17.00	119	LI	15	SSF	20
Zambia	16.73	120	LI	16	SSF	21
Ethiopia	16.47	121	LI	17	SSF	22
Sudan	14.65	122	LM	32	SSF	23
Yemen	14.44	123	LM	33	MEA	15
Niger	11.38	124	LI	18	SSF	24
Algeria	7.52	125	UM	30	MEA	16

Note: World Bank Income Group Classification (January 2011): LI = low income; LM = lower-middle income; UM = upper-middle income; and HI = high income; World Bank Regional Classification (January 2011): ECS = Europe & Central Asia; MEA = Middle East & North Africa; SSF = Sub-Saharan Africa; EAS = East Asia & Pacific; SAS = South Asia; NAC = North America; and LCN = Latin America & Caribbean.

Table 5a: Institutions pillar

	INSTITUTIONS		Political environment		Regulatory environment		Business environment	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank	Score	Rank
Denmark	94.2	1	94.2	7	96.7	4	91.8	12
New Zealand	94.0	2	93.6	9	96.9	3	91.5	14
Canada	93.3	3	91.5	11	96.3	6	92.0	11
Hong Kong (SAR), China	92.8	4	88.6	15	96.7	5	93.0	7
Switzerland	92.6	5	96.8	2	94.5	7	86.4	30
Ireland	91.2	6	90.1	12	93.2	9	90.3	16
Australia	91.0	7	88.7	14	97.8	1	86.5	28
Iceland	90.6	8	95.1	6	84.3	19	92.3	10
Singapore	90.4	9	80.0	28	97.5	2	93.8	5
Finland	89.2	10	98.3	1	85.5	15	83.7	48
Luxembourg	88.3	11	96.1	3	79.1	27	89.8	18
Norway	88.1	12	95.4	5	82.0	22	86.8	25
Netherlands	87.5	13	92.4	10	84.1	20	85.8	34
Sweden	87.3	14	95.6	4	86.1	14	80.1	63
United States of America	86.5	15	80.3	27	93.7	8	85.5	35
United Kingdom	86.4	16	79.8	29	92.7	10	86.6	27
Austria	85.7	17	94.1	8	88.4	11	74.5	82
Chile	85.2	18	81.3	22	87.8	12	86.3	32
Belgium	84.8	19	86.8	17	86.1	13	81.6	56
Japan	83.8	20	89.2	13	84.4	18	77.8	69
Germany	83.5	21	88.1	16	81.1	23	81.2	59
Cyprus	83.5	22	77.3	34	83.2	21	89.9	17
Qatar	83.5	23	77.4	33	79.5	25	93.4	6
Czech Republic	82.6	24	84.1	18	85.4	16	78.2	68
Mauritius	82.0	25	74.1	38	79.3	26	92.5	8
United Arab Emirates	81.8	26	77.6	32	75.4	32	92.3	9
Estonia	80.8	27	83.2	19	75.3	33	84.1	46
Portugal	80.4	28	82.2	21	73.7	39	85.3	36
Slovenia	80.4	29	82.5	20	69.0	47	89.7	20
Slovak Republic	79.6	30	81.1	23	76.1	31	81.5	58
Hungary	79.3	31	77.9	31	77.8	28	82.1	54
Oman	78.7	32	68.1	44	76.6	29	91.6	13
Lithuania	78.0	33	80.5	25	70.9	44	82.7	52
France	77.9	34	80.5	26	74.3	35	78.8	66
Korea, Rep.	77.4	35	73.9	40	73.3	40	85.2	39
Latvia	76.4	36	74.0	39	70.4	46	85.0	42
Poland	76.4	37	80.6	24	74.1	36	74.5	83
Botswana	76.1	38	77.1	35	74.3	34	76.9	75
Kuwait	75.3	39	65.2	50	73.9	37	86.8	26
Bulgaria	74.5	40	67.2	46	68.7	48	87.6	22
Bahrain	74.4	41	51.8	71	76.1	30	95.4	3
Croatia	73.6	42	73.1	41	59.5	65	88.1	21
Macedonia	73.0	43	56.1	62	64.4	55	98.4	1
Namibia	72.6	44	76.2	36	67.4	51	74.4	84
Georgia	72.4	45	50.1	75	71.0	43	96.1	2
Israel	72.1	46	55.7	63	79.7	24	80.8	61
Trinidad and Tobago	71.9	47	66.7	47	70.5	45	78.4	67
Brunei Darussalam	71.2	48	72.2	43	85.0	17	56.3	113
Italy	71.1	49	72.3	42	67.5	49	73.6	85
South Africa	71.0	50	66.4	48	61.8	60	84.6	43
Malaysia	70.5	51	57.5	60	71.7	42	82.2	53
Romania	69.8	52	64.2	52	60.7	62	84.5	44
Ghana	69.8	53	66.1	49	59.9	64	83.4	49
Uruguay	69.5	54	79.3	30	71.7	41	57.6	112
Jamaica	69.3	55	61.2	55	63.9	56	82.7	51
Spain	68.9	56	67.6	45	73.7	38	65.5	102
Armenia	67.9	57	58.7	58	60.8	61	84.2	45
Costa Rica	67.9	58	75.8	37	64.7	54	63.0	106
Greece	67.8	59	63.5	53	63.8	57	76.1	76
Saudi Arabia	67.5	60	39.8	92	67.4	50	95.2	4
Mongolia	66.3	61	52.5	67	55.6	73	90.8	15
Jordan	65.8	62	53.5	66	66.6	53	77.3	73
Albania	65.2	63	55.5	64	56.3	72	83.9	47

Table 5a: Institutions pillar *(continued)*

	INSTITUTIONS		Political environment		Regulatory environment		Business environment	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank	Score	Rank
Panama	64.8	64	62.9	54	50.4	89	81.1	60
Zambia	64.3	65	57.0	61	50.8	87	85.1	40
Serbia	63.2	66	51.2	72	52.5	81	85.9	33
El Salvador	63.1	67	60.8	56	54.0	75	74.6	80
Kazakhstan	62.6	68	48.5	76	52.2	82	87.2	23
Turkey	62.1	69	43.5	83	60.5	63	82.1	55
Egypt	61.7	70	41.0	88	58.8	67	85.3	37
Thailand	61.5	71	38.0	97	67.3	52	79.2	65
Moldova, Rep.	61.2	72	47.8	78	48.7	91	87.1	24
Tunisia	61.1	73	47.3	79	58.4	68	77.5	71
Burkina Faso	60.4	74	51.9	70	58.2	69	71.0	90
Rwanda	59.6	75	32.1	111	56.9	70	89.7	19
Mexico	58.6	76	44.1	81	51.3	86	80.4	62
Bosnia & Herzegovina	58.6	77	48.0	77	54.1	74	73.6	86
Guyana	58.5	78	52.1	69	47.4	94	75.9	77
Azerbaijan	57.9	79	35.3	106	52.0	83	86.4	31
Morocco	57.6	80	43.8	82	47.5	93	81.5	57
Peru	57.5	81	43.2	84	51.7	85	77.8	70
Tanzania	56.7	82	57.6	59	41.4	105	71.0	89
Colombia	55.7	83	36.3	101	62.3	59	68.7	96
Viet Nam	54.9	84	39.2	96	50.5	88	75.1	78
Senegal	54.7	85	51.1	73	44.0	98	69.1	94
Lebanon	54.3	86	39.3	94	52.5	80	71.0	88
Brazil	54.1	87	64.8	51	52.9	78	44.7	118
Uganda	54.0	88	40.6	89	62.4	58	59.0	110
Mali	54.0	89	50.2	74	49.1	90	62.6	107
Indonesia	53.4	90	44.3	80	45.8	95	70.1	92
Madagascar	53.3	91	39.9	91	34.9	114	85.2	38
Malawi	53.3	92	52.3	68	53.2	77	54.4	116
Sri Lanka	53.2	93	31.6	112	58.9	66	69.1	93
India	52.3	94	42.2	86	56.6	71	58.2	111
Kenya	51.9	95	41.0	87	48.4	92	66.2	100
Ethiopia	51.8	96	31.5	113	37.6	110	86.5	29
Russian Federation	51.8	97	37.9	98	40.3	107	77.3	72
China	51.7	98	32.8	108	53.5	76	68.8	95
Swaziland	51.4	99	37.7	100	51.8	84	64.6	105
Guatemala	51.2	100	43.0	85	45.5	96	65.1	104
Philippines	51.2	101	32.5	110	52.9	79	68.2	97
Argentina	51.1	102	55.2	65	43.2	100	55.0	115
Ukraine	51.0	103	36.2	103	42.3	104	74.6	81
Kyrgyzstan	49.5	104	25.6	118	42.9	103	80.1	64
Bangladesh	48.2	105	26.4	117	41.1	106	77.0	74
Honduras	47.7	106	36.0	104	36.5	112	70.5	91
Benin	46.9	107	59.9	57	42.9	102	38.0	122
Pakistan	46.7	108	20.0	121	36.6	111	83.4	50
Syrian Arab Republic	46.3	109	20.1	120	43.8	99	75.0	79
Paraguay	46.2	110	39.9	90	32.4	117	66.2	101
Cameroon	45.9	111	35.9	105	34.3	115	67.6	99
Algeria	45.4	112	32.5	109	35.5	113	68.2	98
Ecuador	42.8	113	37.9	99	25.2	121	65.2	103
Iran	42.7	114	11.6	123	31.4	118	85.1	41
Nigeria	41.6	115	19.5	122	43.0	101	62.2	108
Nicaragua	40.8	116	39.3	93	44.1	97	39.1	121
Niger	40.1	117	36.3	102	32.5	116	51.6	117
Tajikistan	37.3	118	30.6	114	25.1	122	56.3	114
Yemen	36.9	119	9.0	124	39.6	109	62.2	109
Cambodia	36.6	120	34.8	107	39.7	108	35.4	123
Sudan	34.4	121	6.1	125	25.9	120	71.1	87
Côte d'Ivoire	33.3	122	26.5	116	30.9	119	42.4	120
Bolivia	30.2	123	39.2	95	17.0	124	34.3	124
Venezuela	27.8	124	26.6	115	12.5	125	44.3	119
Zimbabwe	24.8	125	23.5	119	23.1	123	27.8	125

Table 5b: Human capital and research pillar

	HUMAN CAPITAL AND RESEARCH		Education		Tertiary education		Research and Development	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank	Score	Rank
Singapore	74.7	1	69.5	23	94.4	1	60.2	10
Israel	69.8	2	68.0	30	47.7	16	93.7	1
Finland	66.5	3	76.9	5	49.0	13	73.5	3
Iceland	65.7	4	78.8	4	44.9	18	73.5	2
Sweden	63.3	5	74.3	8	42.3	25	73.2	4
Denmark	60.2	6	79.4	1	37.6	40	63.6	8
Korea, Rep.	59.9	7	64.9	39	56.4	4	58.4	12
Austria	58.7	8	69.9	19	48.9	14	57.4	15
Australia	57.8	9	71.3	14	43.2	20	59.0	11
Ireland	57.8	10	79.3	2	49.3	10	44.7	26
Germany	57.5	11	72.5	12	42.4	24	57.8	14
Norway	57.5	12	76.7	7	40.2	31	55.5	17
United States of America	57.4	13	66.2	36	35.7	46	70.2	5
Luxembourg	56.6	14	61.1	48	63.9	3	44.7	25
New Zealand	56.4	15	76.8	6	42.4	22	50.1	20
United Kingdom	56.1	16	68.8	27	42.8	21	56.6	16
Switzerland	55.1	17	62.2	45	39.7	34	63.5	9
Bahrain	54.0	18	67.9	31	64.4	2	29.7	46
Canada	53.9	19	65.5	38	38.1	38	58.0	13
Japan	53.7	20	64.8	40	31.2	60	65.1	7
France	53.0	21	69.3	25	41.5	26	48.2	22
Belgium	52.9	22	73.9	10	33.2	52	51.7	19
Portugal	52.5	23	73.7	11	40.8	30	43.0	29
United Arab Emirates	52.5	24	56.8	65	48.3	15	52.3	18
Qatar	52.5	25	52.2	82	37.2	41	68.1	6
Slovenia	51.3	26	74.3	9	36.8	43	42.9	30
Estonia	50.5	27	69.8	20	39.1	36	42.6	31
Czech Republic	49.9	28	66.2	37	40.0	32	43.5	28
Cyprus	48.6	29	69.6	22	50.5	9	25.8	55
Hong Kong (SAR), China	48.4	30	59.4	55	53.6	5	32.1	42
Spain	48.2	31	67.3	33	38.9	37	38.4	36
Netherlands	47.6	32	70.6	16	28.4	73	43.9	27
Greece	47.4	33	66.3	35	51.7	7	24.1	62
Lithuania	47.0	34	67.7	32	39.9	33	33.4	40
Oman	45.9	35	52.9	80	36.6	44	48.0	23
Hungary	45.7	36	70.0	18	28.9	68	38.2	38
Venezuela	45.6	37	70.1	17	49.1	12	17.6	90
Russian Federation	45.1	38	62.0	46	43.3	19	30.0	44
Italy	44.5	39	69.4	24	35.1	49	29.1	47
Ukraine	44.3	40	70.7	15	37.8	39	24.5	59
Moldova, Rep.	43.6	41	78.9	3	35.5	47	16.3	94
Malaysia	43.5	42	55.0	69	49.2	11	26.4	54
Croatia	43.5	43	64.5	41	36.8	42	29.1	48
Kenya	43.3	44	46.7	96	34.3	50	48.9	21
Mauritius	43.2	45	53.5	76	53.3	6	22.8	68
Slovak Republic	42.8	46	61.3	47	42.4	23	24.8	56
Latvia	42.8	47	69.7	21	30.5	65	28.3	50
Poland	42.4	48	68.8	26	30.7	63	27.6	52
Bosnia & Herzegovina	42.0	49	71.6	13	41.4	27	12.9	108
Jordan	41.4	50	63.2	43	39.6	35	21.3	71
Lebanon	41.0	51	52.6	81	46.8	17	23.6	64
Zimbabwe	40.8	52	68.7	28	21.7	92	32.1	43
Saudi Arabia	40.4	53	68.6	29	33.9	51	18.9	79
Serbia	40.3	54	63.9	42	35.2	48	21.9	69
Syrian Arab Republic	40.2	55	55.9	67	n/a	n/a	24.5	58
China	39.9	56	59.9	51	17.3	102	42.3	32
Tunisia	39.8	57	62.4	44	24.6	82	32.4	41
Iran	39.5	58	44.1	104	51.0	8	23.2	66
Bulgaria	39.2	59	59.8	52	36.3	45	21.5	70
Ghana	39.0	60	51.8	83	19.2	97	46.1	24
Morocco	38.0	61	54.8	72	40.9	28	18.3	85
Namibia	37.9	62	53.9	74	19.4	96	40.2	34
Argentina	37.2	63	59.3	56	28.0	74	24.4	61

Table 5b: Human capital and research pillar *(continued)*

	HUMAN CAPITAL AND RESEARCH		Education		Tertiary education		Research and Development	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank	Score	Rank
Botswana	37.0	64	66.6	34	25.7	79	18.6	82
Romania	36.8	65	58.6	58	31.3	58	20.4	75
Trinidad and Tobago	36.7	66	53.5	75	40.9	29	15.6	97
Kuwait	35.8	67	61.1	49	31.8	56	14.5	101
Jamaica	35.5	68	58.0	61	25.7	80	23.0	67
Guyana	35.1	69	53.5	77	24.9	81	27.1	53
Uruguay	35.0	70	54.9	71	29.0	67	21.2	73
Chile	34.9	71	50.1	89	30.8	62	23.6	63
Macedonia	34.7	72	54.9	70	31.1	61	18.0	87
Mexico	34.7	73	53.4	78	31.5	57	19.0	78
Kazakhstan	34.6	74	58.2	59	32.3	53	13.2	106
Mongolia	34.1	75	57.0	63	31.9	55	13.4	105
Brazil	33.9	76	54.3	73	19.4	95	27.9	51
Brunei Darussalam	33.8	77	59.6	54	27.8	76	14.0	102
Swaziland	33.1	78	59.7	53	16.2	104	23.3	65
Costa Rica	33.1	79	58.2	60	16.5	103	24.5	57
Turkey	32.9	80	49.9	90	27.9	75	20.9	74
Algeria	32.8	81	57.0	64	28.5	72	12.9	107
Albania	32.7	82	49.8	91	23.8	86	24.4	60
Georgia	32.6	83	58.8	57	24.0	84	15.0	98
Panama	32.0	84	49.1	92	29.9	66	16.9	92
Viet Nam	31.7	85	45.1	100	32.1	54	17.8	89
Bolivia	31.2	86	56.7	66	26.2	78	10.8	116
Thailand	31.0	87	48.2	94	26.2	77	18.5	83
Armenia	30.7	88	55.1	68	21.3	94	15.7	96
Yemen	30.7	89	50.3	88	11.2	115	n/a	n/a
Azerbaijan	30.4	90	50.9	87	22.4	90	17.9	88
Colombia	30.0	91	44.9	101	30.5	64	14.6	100
South Africa	30.0	92	57.8	62	3.3	124	28.8	49
Rwanda	29.9	93	46.3	98	8.2	122	35.3	39
Malawi	29.7	94	30.4	118	18.1	101	40.8	33
Kyrgyzstan	29.7	95	51.6	85	28.5	71	8.9	118
Indonesia	29.6	96	46.1	99	24.6	83	18.1	86
Tanzania	29.4	97	28.2	122	21.7	93	38.3	37
Tajikistan	29.4	98	47.6	95	28.6	70	12.0	111
Cameroon	29.2	99	46.3	97	24.0	85	17.5	91
Niger	29.0	100	34.6	114	23.4	88	n/a	n/a
Honduras	28.4	101	60.5	50	14.2	110	10.3	117
Sri Lanka	27.2	102	52.9	79	10.1	118	18.6	80
Ecuador	27.0	103	48.6	93	23.7	87	8.8	120
India	26.9	104	32.3	115	9.5	119	38.8	35
El Salvador	26.8	105	41.2	108	31.3	59	8.0	121
Senegal	26.7	106	42.0	107	18.7	99	19.4	77
Egypt	26.4	107	51.7	84	12.6	113	14.8	99
Peru	25.7	108	43.3	106	21.9	91	11.8	112
Guatemala	25.2	109	44.6	102	18.8	98	12.1	110
Paraguay	24.9	110	51.2	86	18.3	100	5.3	122
Mali	24.9	111	39.3	110	14.0	112	21.3	72
Zambia	24.7	112	44.3	103	15.8	106	13.9	104
Benin	24.5	113	39.5	109	14.2	109	19.7	76
Bangladesh	24.1	114	30.2	120	12.3	114	29.9	45
Madagascar	24.1	115	38.7	111	22.5	89	11.1	114
Philippines	23.7	116	30.8	116	28.7	69	11.5	113
Uganda	22.6	117	43.9	105	10.2	117	13.9	103
Côte d'Ivoire	21.1	118	30.6	117	14.3	108	18.4	84
Burkina Faso	20.6	119	34.7	113	10.6	116	16.4	93
Nicaragua	19.4	120	35.2	112	14.1	111	8.8	119
Cambodia	18.5	121	28.6	121	15.9	105	11.0	115
Nigeria	18.3	122	30.2	119	8.8	121	15.9	95
Ethiopia	16.8	123	22.8	124	15.2	107	12.2	109
Pakistan	14.0	124	14.5	125	8.9	120	18.6	81
Sudan	11.9	125	25.3	123	6.6	123	3.8	123

Table 5c: Infrastructure pillar

	INFRASTRUCTURE		Info & comm. technologies (ICT)		Energy		General infrastructure	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank	Score	Rank
Norway	55.5	1	64.6	13	55.2	1	46.8	17
Hong Kong (SAR), China	53.9	2	70.2	4	42.2	4	49.2	8
Canada	53.1	3	66.2	11	36.9	6	56.2	5
Australia	52.3	4	70.3	3	26.9	26	59.7	3
Sweden	51.7	5	67.4	10	40.3	5	47.3	15
Korea, Rep.	48.2	6	81.0	1	22.4	54	41.2	30
Finland	48.0	7	57.1	23	34.5	8	52.3	6
New Zealand	47.8	8	64.7	12	31.6	12	47.1	16
Singapore	47.6	9	69.0	5	26.0	31	47.9	13
Bahrain	46.9	10	58.8	18	24.3	43	57.7	4
Denmark	45.9	11	68.9	6	30.7	16	38.1	49
Iceland	45.4	12	51.8	26	52.1	3	32.4	84
Japan	45.4	13	68.8	7	25.6	36	41.7	27
United States of America	44.6	14	67.4	9	30.5	18	36.0	63
Switzerland	44.5	15	57.1	22	35.0	7	41.4	29
Austria	44.0	16	58.4	21	32.6	9	41.0	32
United Kingdom	43.6	17	70.6	2	25.7	34	34.6	70
Netherlands	43.6	18	68.3	8	23.3	48	39.2	40
Spain	43.5	19	64.0	14	26.1	30	40.5	36
Luxembourg	43.3	20	62.1	16	29.8	19	37.9	52
Germany	43.2	21	63.7	15	25.2	40	40.9	33
France	43.1	22	61.9	17	25.7	32	41.6	28
Ireland	39.5	23	55.5	24	28.9	22	34.1	72
Belgium	39.5	24	58.6	19	23.1	49	36.8	56
Israel	38.4	25	54.4	25	27.7	24	32.9	83
Estonia	38.0	26	58.5	20	19.9	65	35.5	68
Slovenia	36.6	27	51.8	27	24.6	42	33.3	78
Cyprus	36.5	28	46.0	30	22.3	55	41.0	31
Paraguay	36.3	29	17.2	93	53.3	2	38.5	44
Italy	35.9	30	44.7	34	26.1	29	36.9	55
United Arab Emirates	35.8	31	45.6	32	30.5	17	31.3	90
Colombia	35.7	32	37.6	43	31.3	14	38.2	46
China	35.4	33	28.4	59	14.5	92	63.3	2
Slovak Republic	35.2	34	38.1	42	17.5	77	50.1	7
Lithuania	35.0	35	47.7	28	17.4	79	40.0	38
Portugal	34.7	36	45.1	33	25.2	39	33.9	75
Croatia	34.1	37	47.2	29	19.5	67	35.5	66
Qatar	33.9	38	34.8	46	26.3	28	40.6	35
Hungary	33.9	39	45.8	31	17.7	74	38.1	48
Latvia	33.4	40	40.5	38	23.1	50	36.7	57
Uruguay	33.4	41	34.1	47	28.5	23	37.6	54
Greece	32.6	42	44.1	36	25.4	38	28.1	107
Mongolia	32.4	43	25.5	66	5.7	114	66.0	1
Czech Republic	32.4	44	41.1	37	19.8	66	36.2	59
Brazil	32.2	45	30.4	53	23.9	45	42.3	23
Argentina	31.5	46	32.6	50	20.2	63	41.7	26
Peru	31.4	47	24.4	69	30.9	15	39.0	42
Kuwait	31.2	48	30.8	52	31.8	11	31.0	95
Guyana	31.1	49	13.3	104	n/a	n/a	48.9	9
Chile	31.1	50	37.4	44	20.4	62	35.5	67
Panama	30.7	51	28.3	60	25.7	33	38.1	50
Poland	30.4	52	39.8	39	17.5	76	33.8	76
Malaysia	30.1	53	44.2	35	12.8	98	33.3	79
Romania	30.0	54	35.5	45	16.2	81	38.5	43
Costa Rica	29.6	55	25.5	67	31.5	13	31.8	87
Viet Nam	29.3	56	22.1	79	18.0	73	47.9	12
Morocco	29.2	57	21.4	81	22.1	56	44.0	19
Tunisia	29.0	58	27.2	61	22.0	58	37.7	53
Bulgaria	28.9	59	38.5	40	14.9	89	33.2	80
Kazakhstan	28.5	60	33.8	48	9.5	108	42.1	24
Madagascar	28.2	61	8.8	118	n/a	n/a	47.5	14
Saudi Arabia	27.8	62	30.2	54	15.0	88	38.2	47
India	27.7	63	16.3	94	18.5	72	48.3	11

Table 5c: Infrastructure pillar *(continued)*

	INFRASTRUCTURE		Info & comm. technologies (ICT)		Energy		General infrastructure	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank	Score	Rank
Turkey	27.5	64	30.1	55	21.1	61	31.3	91
Mexico	27.0	65	29.0	57	15.7	85	36.4	58
Oman	26.7	66	26.3	64	11.9	101	41.9	25
Sri Lanka	26.6	67	17.3	92	32.2	10	30.3	100
Philippines	26.5	68	22.3	78	29.4	21	27.7	112
Ecuador	26.4	69	22.4	77	14.4	94	42.6	21
Albania	26.4	70	21.3	83	22.0	59	36.0	64
Bolivia	26.3	71	18.6	89	11.9	102	48.6	10
Macedonia	26.2	72	32.8	49	14.5	93	31.2	92
Russian Federation	25.8	73	31.1	51	14.1	95	32.2	85
Mauritius	25.7	74	24.0	70	n/a	n/a	27.3	114
Lebanon	25.6	75	23.1	74	11.1	104	42.7	20
Bangladesh	25.5	76	13.6	103	27.0	25	35.9	65
Nicaragua	25.3	77	18.2	91	22.6	52	34.9	69
Thailand	25.0	78	21.3	82	15.6	86	38.1	51
South Africa	24.8	79	20.3	86	15.8	84	38.2	45
Senegal	24.6	80	11.1	110	22.1	57	40.6	34
Indonesia	24.5	81	16.2	95	14.9	90	42.4	22
Guatemala	24.5	82	23.0	75	25.6	35	24.7	120
Serbia	24.5	83	26.7	63	13.7	97	33.0	82
Venezuela	24.4	84	24.8	68	12.4	100	36.0	62
Namibia	24.4	85	9.3	114	24.8	41	39.0	41
Honduras	23.9	86	18.7	88	22.6	53	30.6	96
El Salvador	23.9	87	20.6	84	26.7	27	24.4	121
Botswana	23.9	88	14.8	98	22.7	51	34.1	73
Cambodia	23.8	89	11.2	108	29.8	20	30.4	99
Algeria	23.7	90	13.6	102	11.6	103	45.7	18
Bosnia & Herzegovina	23.4	91	23.5	73	16.0	82	30.6	98
Iran	22.9	92	20.4	85	8.0	112	40.3	37
Ghana	22.9	93	11.5	107	25.5	37	31.6	88
Armenia	22.9	94	15.3	96	13.8	96	39.4	39
Zambia	22.6	95	18.3	90	21.2	60	28.4	103
Jordan	22.6	96	29.4	56	11.1	105	27.4	113
Kyrgyzstan	22.5	97	21.8	80	14.9	91	31.0	94
Sudan	22.4	98	11.7	106	23.5	46	31.9	86
Azerbaijan	22.0	99	22.8	76	12.6	99	30.6	97
Egypt	21.7	100	25.9	65	10.8	106	28.3	105
Ukraine	21.5	101	27.1	62	9.2	109	28.2	106
Kenya	21.4	102	14.3	100	20.1	64	29.7	101
Moldova, Rep.	21.3	103	24.0	71	6.0	113	34.0	74
Mali	21.2	104	11.2	109	n/a	n/a	31.1	93
Cameroon	21.0	105	10.5	111	24.3	44	28.3	104
Nigeria	21.0	106	9.2	115	17.7	75	36.1	60
Benin	20.8	107	9.7	112	16.8	80	36.1	61
Uganda	20.8	108	8.0	121	n/a	n/a	33.7	77
Jamaica	20.6	109	23.8	72	9.9	107	28.1	108
Niger	20.3	110	7.6	124	n/a	n/a	33.0	81
Georgia	20.2	111	19.5	87	18.9	71	22.3	123
Ethiopia	20.0	112	8.5	119	23.4	47	28.0	109
Côte d'Ivoire	19.7	113	15.2	97	19.0	70	24.8	119
Pakistan	19.5	114	14.7	99	15.9	83	27.7	111
Brunei Darussalam	19.4	115	38.3	41	15.5	87	4.4	125
Tajikistan	19.3	116	9.4	113	19.3	69	29.2	102
Malawi	18.9	117	6.4	125	n/a	n/a	31.4	89
Swaziland	18.5	118	11.8	105	n/a	n/a	25.2	118
Trinidad and Tobago	18.1	119	28.5	58	9.1	110	16.8	124
Tanzania	17.8	120	8.9	116	19.4	68	25.2	117
Rwanda	17.6	121	8.3	120	n/a	n/a	26.9	115
Burkina Faso	17.4	122	8.9	117	n/a	n/a	25.8	116
Syrian Arab Republic	16.9	123	14.3	101	8.4	111	27.9	110
Zimbabwe	16.4	124	7.7	123	17.4	78	23.9	122
Yemen	15.4	125	7.9	122	3.8	115	34.6	71

Table 5d: Market sophistication pillar

	MARKET SOPHISTICATION		Credit		Investment		Trade and competition	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank	Score	Rank
Hong Kong (SAR), China	87.0	1	78.0	11	91.4	1	91.7	2
Singapore	78.7	2	65.1	20	78.5	2	92.5	1
United Kingdom	74.4	3	96.4	1	74.4	5	52.6	57
United States of America	70.9	4	88.3	4	77.8	3	46.6	83
Switzerland	70.1	5	78.3	10	67.8	6	64.1	15
Ireland	65.3	6	81.7	7	39.2	32	75.1	8
Denmark	64.5	7	89.2	2	44.4	26	59.8	27
South Africa	63.9	8	64.8	21	77.3	4	49.6	68
Canada	63.4	9	68.7	16	67.0	7	54.4	45
Malaysia	62.1	10	58.0	26	52.0	13	76.3	6
Netherlands	61.8	11	71.7	14	43.4	28	70.3	11
Korea, Rep.	61.8	12	66.3	19	62.8	9	56.4	39
Belgium	60.6	13	68.3	17	40.0	31	73.5	10
Germany	59.3	14	85.0	5	34.5	44	58.5	30
Sweden	58.9	15	57.6	27	58.4	10	60.8	24
Australia	58.8	16	71.9	13	54.8	12	49.8	66
Israel	58.6	17	61.7	24	56.0	11	58.0	35
New Zealand	57.9	18	79.3	9	48.3	19	46.3	87
Japan	57.9	19	79.7	8	45.6	23	48.5	75
Iceland	57.9	20	88.3	3	26.8	68	58.4	32
Spain	57.5	21	71.7	14	51.8	14	49.2	71
Luxembourg	57.5	22	35.0	79	46.4	22	91.0	3
Austria	56.5	23	85.0	5	22.2	89	62.2	20
Bahrain	56.1	24	53.2	32	37.9	37	77.1	5
Finland	56.1	25	76.7	12	38.3	34	53.2	53
China	54.1	26	49.1	36	63.6	8	49.7	67
Norway	53.8	27	53.3	31	48.6	18	59.7	28
France	53.8	28	68.3	17	43.5	27	49.5	69
Estonia	53.4	29	59.7	25	30.8	56	69.9	13
Saudi Arabia	52.7	30	48.3	41	47.9	20	61.8	22
United Arab Emirates	52.4	31	51.1	34	31.8	52	74.3	9
Cyprus	49.5	32	56.1	30	29.7	60	62.8	18
Thailand	49.0	33	44.7	48	38.6	33	63.6	17
Czech Republic	47.7	34	46.6	46	26.1	72	70.3	12
Albania	47.5	35	44.6	49	48.7	17	49.2	70
Hungary	47.4	36	42.5	54	24.5	80	75.2	7
Chile	47.4	37	43.7	52	42.5	29	56.0	40
Latvia	47.3	38	62.9	23	25.0	75	54.1	48
Viet Nam	47.0	39	64.3	22	19.2	101	57.5	37
Kuwait	46.9	40	40.7	59	44.5	25	55.4	43
Mongolia	46.6	41	48.9	37	32.3	51	58.5	31
Portugal	46.3	42	56.7	28	31.3	54	51.1	62
Slovak Republic	45.9	43	38.4	66	14.0	116	85.4	4
Jordan	44.7	44	29.5	94	47.4	21	57.1	38
India	44.6	45	38.2	67	49.7	15	46.1	88
Brunei Darussalam	44.1	46	24.6	103	49.7	16	58.1	34
Azerbaijan	44.1	47	35.5	75	44.7	24	52.2	59
Lithuania	43.9	48	50.7	35	15.7	112	65.4	14
Nicaragua	43.5	49	48.4	40	33.3	47	48.7	73
Bulgaria	43.0	50	52.1	33	19.0	102	57.9	36
Bosnia & Herzegovina	42.9	51	48.8	38	33.3	47	46.7	81
Peru	42.7	52	48.2	42	31.0	55	49.0	72
Italy	42.7	53	56.7	28	25.9	74	45.6	89
Panama	42.5	54	47.7	43	17.2	108	62.6	19
Trinidad and Tobago	42.4	55	33.1	86	33.2	49	60.8	25
Mauritius	42.1	56	44.1	51	28.9	61	53.2	52
Cambodia	41.6	57	39.1	64	35.3	42	50.4	64
Georgia	41.1	58	48.7	39	19.9	99	54.8	44
Poland	40.9	59	39.4	62	27.7	65	55.5	42
Slovenia	40.3	60	34.7	80	22.2	90	64.0	16
Kyrgyzstan	40.2	61	45.5	47	22.3	88	52.8	56
Macedonia	40.2	62	41.4	56	20.3	95	58.8	29
Kenya	40.1	63	47.5	44	30.2	58	42.7	97

Table 5d: Market sophistication pillar *(continued)*

	MARKET SOPHISTICATION		Credit		Investment		Trade and competition	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank	Score	Rank
Ukraine	39.6	64	41.2	57	23.1	84	54.3	47
Namibia	39.4	65	39.9	61	16.1	111	62.0	21
Qatar	39.2	66	25.1	99	36.5	39	55.9	41
Lebanon	39.0	67	35.4	77	28.3	64	53.4	51
Honduras	38.9	68	44.4	50	20.0	97	52.4	58
Romania	38.9	69	39.1	63	26.0	73	51.6	61
Moldova, Rep.	38.6	70	24.0	104	33.6	46	58.3	33
Guatemala	38.4	71	40.6	60	26.7	70	48.0	76
Turkey	38.0	72	29.5	95	32.5	50	52.0	60
Mexico	37.2	73	33.4	85	27.4	66	50.8	63
Colombia	36.8	74	36.1	74	35.4	41	38.8	109
Kazakhstan	36.6	75	33.0	88	23.8	82	53.1	54
Russian Federation	36.4	76	29.1	98	36.2	40	43.8	94
Paraguay	36.2	77	38.4	65	16.7	109	53.6	50
Botswana	35.7	78	37.4	70	21.2	94	48.5	74
Croatia	35.7	79	36.3	72	16.4	110	54.4	46
Brazil	35.7	80	31.6	90	37.0	38	38.4	112
Rwanda	35.3	81	31.3	91	42.0	30	32.5	120
Oman	35.1	82	25.0	100	19.3	100	61.1	23
Egypt	35.0	83	33.0	87	29.7	59	42.3	99
Morocco	34.4	84	38.1	68	22.5	86	42.6	98
Tunisia	34.3	85	34.5	81	24.1	81	44.2	93
El Salvador	34.2	86	41.0	58	14.7	113	46.9	80
Serbia	34.2	87	43.4	53	18.4	103	40.8	103
Swaziland	33.8	88	35.4	78	13.0	119	52.9	55
Zambia	33.7	89	36.2	73	17.5	107	47.4	78
Armenia	33.7	90	42.2	55	14.4	114	44.3	92
Guyana	33.5	91	21.8	106	18.0	104	60.7	26
Algeria	33.4	92	17.2	111	35.3	42	47.7	77
Bolivia	33.4	93	47.2	45	13.2	118	39.8	106
Costa Rica	32.7	94	35.4	76	9.1	124	53.7	49
Tajikistan	32.3	95	15.8	115	38.0	35	43.1	95
Greece	32.3	96	37.7	69	13.8	117	45.4	91
Indonesia	32.2	97	23.0	105	27.0	67	46.7	82
Philippines	32.0	98	21.0	108	24.9	76	50.1	65
Ghana	31.8	99	29.3	96	26.7	69	39.3	108
Nigeria	31.1	100	21.1	107	24.8	77	47.3	79
Zimbabwe	30.5	101	15.7	117	34.0	45	41.8	101
Bangladesh	30.5	102	33.5	84	22.6	85	35.3	114
Uruguay	30.4	103	33.8	82	14.3	115	43.1	96
Ecuador	30.4	104	32.0	89	12.6	120	46.5	84
Jamaica	30.2	105	25.0	101	20.1	96	45.4	90
Sri Lanka	29.7	106	30.4	92	17.7	106	41.1	102
Malawi	28.7	107	17.1	112	30.2	57	38.7	111
Argentina	28.3	108	29.2	97	17.7	105	38.1	113
Senegal	28.3	109	18.5	110	20.0	97	46.4	85
Cameroon	28.1	110	16.0	114	28.7	62	39.5	107
Pakistan	28.1	111	29.9	93	22.5	87	31.8	121
Madagascar	28.0	112	5.9	125	38.0	35	40.0	105
Uganda	28.0	113	33.7	83	11.5	123	38.7	110
Mali	26.9	114	13.8	118	24.7	78	42.3	100
Tanzania	26.9	115	24.8	102	23.2	83	32.8	118
Ethiopia	26.8	116	20.4	109	28.7	62	31.2	123
Iran	26.1	117	36.9	71	11.5	122	29.8	125
Yemen	26.0	118	10.7	121	26.7	70	40.6	104
Syrian Arab Republic	25.1	119	10.4	123	31.3	53	33.5	116
Burkina Faso	24.4	120	15.7	116	24.7	78	32.8	117
Côte d'Ivoire	23.6	121	11.9	119	12.5	121	46.4	86
Benin	23.3	122	16.8	113	22.0	91	31.3	122
Niger	22.4	123	10.8	120	22.0	91	34.5	115
Sudan	21.0	124	10.6	122	22.0	91	30.3	124
Venezuela	15.5	125	6.8	124	7.1	125	32.5	119

Table 5e: Business sophistication pillar

	BUSINESS SOPHISTICATION		Knowledge workers		Innovation linkages		Knowledge absorption	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank	Score	Rank
Singapore	79.1	1	87.3	3	68.3	1	81.7	2
Luxembourg	74.0	2	92.8	1	67.6	2	61.7	6
Ireland	73.8	3	75.9	15	55.6	11	90.0	1
Switzerland	68.0	4	88.3	2	61.5	6	54.3	11
Hong Kong (SAR), China	66.9	5	65.6	29	59.2	8	75.8	3
Finland	63.9	6	84.0	5	57.7	9	49.9	14
Sweden	63.1	7	83.9	6	54.1	14	51.4	13
Netherlands	61.6	8	77.3	10	53.5	16	54.1	12
Malaysia	58.5	9	69.0	23	44.9	29	61.6	7
Canada	58.4	10	71.0	20	59.8	7	44.3	26
Denmark	58.1	11	82.6	7	53.3	17	38.2	46
United Kingdom	57.8	12	74.1	17	53.1	18	46.1	23
Israel	56.8	13	86.6	4	37.2	41	46.7	21
Japan	55.9	14	82.3	8	41.2	35	44.3	27
United States of America	54.8	15	76.1	12	50.5	22	37.8	47
Belgium	54.4	16	80.6	9	50.9	20	31.6	73
Australia	54.1	17	76.0	14	49.6	23	36.7	50
Czech Republic	53.0	18	76.0	13	33.5	60	49.6	16
Iceland	52.8	19	75.8	16	55.6	10	27.0	94
Germany	51.6	20	66.9	25	43.3	30	44.5	25
Estonia	51.6	21	66.9	26	41.8	32	46.0	24
Austria	51.6	22	69.3	21	46.2	25	39.1	41
Norway	50.8	23	71.4	19	45.5	28	35.6	55
France	50.3	24	72.9	18	38.3	39	39.8	40
Thailand	50.2	25	52.7	39	41.8	33	56.1	10
Korea, Rep.	49.8	26	57.7	35	33.5	61	58.4	8
Qatar	49.5	27	44.9	54	54.2	13	n/a	n/a
United Arab Emirates	49.5	28	69.3	22	63.4	5	15.6	124
China	49.3	29	64.8	30	35.8	46	47.3	19
Hungary	48.9	30	46.2	49	31.8	66	68.7	5
New Zealand	48.7	31	66.0	27	43.2	31	36.8	49
Swaziland	48.0	32	57.7	34	16.9	117	69.3	4
Costa Rica	47.9	33	49.2	42	53.8	15	40.7	37
Slovenia	47.8	34	67.2	24	35.3	50	41.0	36
Niger	45.4	35	34.2	80	66.7	3	35.2	58
Italy	45.2	36	65.6	28	34.4	55	35.4	57
Russian Federation	44.9	37	64.0	31	27.6	83	43.0	31
Oman	44.7	38	32.6	82	63.8	4	37.6	48
Lebanon	44.5	39	60.1	32	26.8	87	46.6	22
Spain	43.9	40	59.7	33	33.6	59	38.5	45
Cyprus	43.9	41	41.3	60	51.2	19	39.0	42
Bosnia & Herzegovina	43.7	42	77.1	11	31.8	67	22.2	112
South Africa	42.3	43	48.0	46	36.4	44	42.4	32
Chile	41.9	44	55.0	37	35.3	49	35.5	56
Ukraine	41.5	45	45.5	51	31.8	69	47.3	20
Brazil	41.5	46	48.3	45	34.4	56	41.8	34
Poland	41.4	47	56.3	36	25.7	92	42.3	33
Saudi Arabia	41.3	48	42.2	57	46.8	24	35.0	60
Namibia	40.8	49	39.8	62	50.6	21	32.0	71
Sudan	40.7	50	39.7	63	n/a	n/a	41.7	35
Kenya	39.5	51	44.9	55	45.8	27	27.8	88
Jamaica	38.9	52	48.7	44	31.8	68	36.2	53
Argentina	38.3	53	41.9	58	29.7	75	43.1	30
Croatia	38.2	54	45.4	53	25.2	93	43.9	28
Peru	37.9	55	46.6	48	35.4	48	31.7	72
Slovak Republic	37.8	56	49.1	43	32.4	64	32.0	70
Portugal	37.6	57	45.4	52	34.1	57	33.3	63
Viet Nam	37.5	58	26.0	98	37.7	40	48.8	18
Latvia	37.3	59	52.2	40	32.7	62	27.1	93
Kazakhstan	37.0	60	38.1	69	23.9	98	49.0	17
Philippines	36.7	61	46.7	47	30.9	70	32.4	68
Ecuador	36.3	62	42.9	56	41.7	34	24.2	108
Ghana	35.7	63	35.3	77	22.3	104	49.7	15

Table 5e: Business sophistication pillar *(continued)*

	BUSINESS SOPHISTICATION		Knowledge workers		Innovation linkages		Knowledge absorption	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank	Score	Rank
Cambodia	35.6	64	24.8	103	54.8	12	27.3	92
Bulgaria	35.4	65	41.5	59	24.1	97	40.6	38
Colombia	35.4	66	37.4	73	30.2	73	38.5	44
Guyana	35.3	67	23.5	106	25.7	91	56.7	9
Trinidad and Tobago	35.3	68	37.8	70	36.4	43	31.6	74
Venezuela	34.0	69	45.6	50	27.6	84	29.0	81
Panama	33.7	70	26.8	95	46.2	26	28.1	84
Romania	33.5	71	35.6	76	21.7	107	43.2	29
Malawi	33.5	72	54.6	38	24.3	96	21.6	114
Serbia	33.3	73	38.2	68	22.9	102	39.0	43
Lithuania	33.3	74	51.6	41	28.1	82	20.2	116
Kyrgyzstan	32.7	75	34.0	81	36.1	45	28.0	86
Greece	32.3	76	39.0	66	30.4	72	27.7	89
Jordan	32.3	77	24.0	105	38.4	38	34.4	61
Bahrain	32.2	78	37.8	71	34.6	53	24.2	107
Mongolia	32.2	79	36.9	74	24.5	95	35.2	59
Azerbaijan	31.7	80	22.7	108	32.2	65	40.1	39
Rwanda	31.2	81	28.5	91	34.8	51	30.4	77
Ethiopia	31.1	82	31.2	84	38.5	37	23.7	109
Mauritius	31.0	83	26.8	94	33.9	58	32.4	69
India	30.8	84	24.4	104	34.8	52	33.2	65
Bolivia	30.7	85	36.7	75	28.2	81	27.4	91
Egypt	30.7	86	39.3	64	25.8	90	26.9	95
Armenia	30.3	87	38.4	67	19.9	111	32.7	67
Uruguay	30.1	88	29.7	89	32.6	63	28.0	85
Mexico	29.9	89	37.4	72	26.1	88	26.1	96
Turkey	29.4	90	41.1	61	21.8	106	25.1	103
Sri Lanka	29.3	91	30.9	86	34.6	54	22.4	111
Botswana	29.2	92	32.2	83	22.0	105	33.3	64
Paraguay	28.6	93	30.6	87	27.2	86	28.0	87
Indonesia	28.2	94	7.0	124	40.8	36	36.7	51
Macedonia	27.9	95	25.7	100	22.3	103	35.7	54
Brunei Darussalam	27.4	96	27.8	93	29.2	78	25.4	102
Moldova, Rep.	27.3	97	39.0	65	14.6	121	28.1	83
Tunisia	26.9	98	16.7	119	36.6	42	27.4	90
Nicaragua	26.7	99	27.9	92	19.4	113	32.8	66
Guatemala	26.6	100	19.9	113	35.6	47	24.5	105
Nigeria	26.6	101	26.2	97	23.3	100	30.3	78
Honduras	26.5	102	28.6	90	21.7	108	29.3	80
Georgia	26.4	103	26.6	96	23.7	99	28.9	82
Cameroon	26.4	104	26.0	99	16.5	118	36.6	52
Yemen	25.9	105	30.0	88	n/a	n/a	21.8	113
Benin	25.7	106	34.6	78	18.2	115	24.4	106
Algeria	25.1	107	25.1	102	18.8	114	31.5	75
El Salvador	24.9	108	31.1	85	20.9	110	22.8	110
Pakistan	24.7	109	18.8	115	29.7	76	25.5	100
Morocco	24.1	110	20.1	111	21.7	109	30.4	76
Uganda	23.9	111	16.8	118	29.9	74	24.9	104
Zambia	23.8	112	16.9	117	28.9	79	25.4	101
Burkina Faso	23.5	113	25.1	101	28.8	80	16.6	122
Iran	23.5	114	21.3	110	29.2	77	19.8	117
Kuwait	23.1	115	23.2	107	27.5	85	18.5	120
Madagascar	22.4	116	13.9	122	19.7	112	33.7	62
Mali	22.4	117	22.2	109	26.1	89	18.8	119
Syrian Arab Republic	21.7	118	34.5	79	14.8	120	15.8	123
Zimbabwe	21.6	119	n/a	n/a	17.7	116	25.6	98
Tanzania	21.4	120	20.0	112	24.9	94	19.4	118
Bangladesh	19.9	121	12.3	123	30.6	71	16.9	121
Tajikistan	19.9	122	14.4	121	15.6	119	29.6	79
Albania	19.6	123	19.0	114	14.5	122	25.5	99
Senegal	19.4	124	14.5	120	23.1	101	20.6	115
Côte d'Ivoire	19.3	125	18.0	116	14.0	123	25.9	97

Table 5f: Scientific outputs pillar

	SCIENTIFIC OUTPUTS		Knowledge creation		Knowledge impact		Knowledge diffusion	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank	Score	Rank
Sweden	62.1	1	75.1	3	39.3	29	71.9	4
Switzerland	62.0	2	73.4	4	50.8	12	61.9	10
Finland	58.5	3	70.9	5	35.6	40	69.1	5
Israel	57.5	4	77.4	2	33.1	58	62.0	8
United States of America	57.4	5	60.4	9	52.5	11	59.4	12
Netherlands	53.8	6	55.4	11	39.7	28	66.4	6
Korea, Rep.	53.7	7	80.8	1	26.5	78	53.6	17
Iceland	53.2	8	49.8	13	72.5	2	37.2	38
China	52.7	9	54.2	12	55.1	9	48.6	21
United Kingdom	52.3	10	45.0	14	55.3	8	56.6	15
Ireland	51.2	11	33.7	25	37.7	33	82.2	1
Qatar	50.6	12	1.2	114	100.0	1	n/a	n/a
Japan	49.8	13	69.4	6	21.5	95	58.5	14
Germany	49.8	14	69.2	7	26.6	77	53.6	18
Singapore	48.9	15	32.0	31	36.8	35	78.0	2
New Zealand	47.6	16	64.6	8	54.7	10	23.5	77
Denmark	46.3	17	58.3	10	35.7	38	44.8	25
Hungary	45.5	18	22.4	37	50.3	13	63.9	7
Cyprus	43.7	19	14.9	51	69.7	3	46.3	24
Luxembourg	43.2	20	33.7	24	34.1	52	61.9	9
Canada	42.5	21	38.2	19	46.2	14	43.0	28
France	41.1	22	36.8	21	32.1	63	54.5	16
Kuwait	38.1	23	5.1	76	34.3	50	75.0	3
Hong Kong (SAR), China	38.1	24	5.1	77	57.5	4	51.7	20
Estonia	38.1	25	33.5	26	40.9	22	39.9	32
Norway	37.4	26	38.1	20	35.1	45	39.1	35
Belgium	37.0	27	33.4	28	37.2	34	40.5	30
Moldova, Rep.	36.8	28	43.4	15	40.0	25	27.0	62
Czech Republic	36.2	29	25.5	33	46.1	16	37.0	39
Austria	35.4	30	40.8	16	26.6	76	39.0	36
Guyana	34.3	31	7.9	65	n/a	n/a	60.7	11
Slovenia	33.9	32	38.9	17	34.9	47	27.9	60
Australia	33.1	33	34.3	23	39.8	27	25.3	70
Russian Federation	32.9	34	33.4	27	34.5	49	30.7	49
Lebanon	31.0	35	15.1	49	n/a	n/a	46.9	22
Romania	30.7	36	11.7	56	40.1	24	40.4	31
Georgia	30.6	37	32.9	29	36.5	36	22.3	84
Malaysia	30.4	38	8.8	63	30.4	65	52.1	19
Paraguay	30.0	39	1.0	118	n/a	n/a	59.0	13
Ukraine	29.9	40	34.9	22	24.5	84	30.2	52
Armenia	29.7	41	24.7	34	38.6	30	25.8	67
Spain	29.4	42	24.0	35	34.6	48	29.6	55
Costa Rica	28.4	43	3.2	95	40.0	26	42.1	29
Bangladesh	28.2	44	2.3	105	35.8	37	46.6	23
Iran	28.2	45	32.1	30	26.9	74	25.5	68
Italy	27.8	46	27.7	32	23.8	87	31.9	47
Tajikistan	27.7	47	13.8	53	34.2	51	35.1	43
Mongolia	27.4	48	38.7	18	n/a	n/a	16.0	119
Mauritius	27.2	49	2.0	108	57.1	5	22.7	82
Bulgaria	27.2	50	14.9	52	44.4	17	22.4	83
Slovak Republic	26.5	51	13.6	54	38.2	31	27.9	61
Macedonia	26.3	52	7.4	66	42.6	20	29.1	57
Kyrgyzstan	26.1	53	21.7	39	32.6	60	23.8	76
Bahrain	25.6	54	2.9	98	55.3	7	18.5	109
Tanzania	25.5	55	3.5	88	55.8	6	17.3	114
Viet Nam	25.3	56	2.9	97	40.2	23	32.9	46
Croatia	25.2	57	20.9	40	30.3	66	24.4	74
Brazil	25.2	58	11.4	58	30.5	64	33.6	45
Portugal	24.8	59	16.0	47	32.7	59	25.9	66
India	24.8	60	10.4	62	24.2	85	39.9	33
Ghana	24.6	61	4.9	80	29.6	68	39.3	34
Azerbaijan	24.4	62	6.2	71	41.9	21	25.1	72
Sudan	24.2	63	0.4	124	46.2	15	26.1	64

Table 5f: Scientific outputs pillar *(continued)*

	SCIENTIFIC OUTPUTS		Knowledge creation		Knowledge impact		Knowledge diffusion	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank	Score	Rank
Thailand	23.9	64	8.6	64	27.0	73	36.2	40
Serbia	23.7	65	20.1	41	15.1	109	35.9	41
Poland	23.7	66	18.6	43	23.3	91	29.1	56
Argentina	23.5	67	18.3	44	21.6	93	30.7	50
Latvia	23.3	68	22.3	38	24.1	86	23.4	79
Côte d'Ivoire	22.9	69	1.7	113	35.5	41	31.6	48
Trinidad and Tobago	22.7	70	1.9	110	44.2	18	22.0	87
Cameroon	22.5	71	5.8	74	28.1	71	33.6	44
Mali	22.5	72	1.9	109	43.8	19	21.6	89
Venezuela	22.4	73	2.4	102	29.1	69	35.8	42
Oman	22.3	74	3.3	93	35.1	46	28.7	58
Swaziland	22.3	75	1.0	119	n/a	n/a	43.6	27
Philippines	22.3	76	3.3	92	19.6	102	43.8	26
Jordan	22.1	77	22.4	36	21.1	98	22.9	81
Lithuania	21.8	78	11.3	60	33.5	54	20.7	95
South Africa	21.5	79	13.2	55	32.5	62	18.9	105
Uruguay	21.5	80	10.7	61	35.2	44	18.7	107
Kazakhstan	21.4	81	15.5	48	27.7	72	20.9	93
Sri Lanka	20.9	82	6.1	72	26.8	75	29.8	54
Greece	20.6	83	17.6	46	24.5	83	19.6	100
Syrian Arab Republic	20.5	84	4.2	83	37.9	32	19.5	102
Chile	20.4	85	11.6	57	25.6	81	23.9	75
Tunisia	19.5	86	11.3	59	23.7	89	23.5	78
Morocco	19.5	87	5.8	73	25.6	82	27.0	63
Brunei Darussalam	19.3	88	1.0	117	n/a	n/a	37.6	37
Ethiopia	19.1	89	4.8	81	33.8	53	18.7	106
Turkey	18.9	90	17.9	45	17.7	106	21.0	92
Ecuador	18.5	91	1.8	112	33.3	57	20.4	96
Uganda	18.5	92	5.1	79	33.3	56	17.1	116
Saudi Arabia	18.3	93	2.1	106	35.2	43	17.5	113
Indonesia	18.3	94	1.0	116	23.2	92	30.5	51
Yemen	18.2	95	1.0	120	33.4	55	20.2	97
Albania	18.0	96	0.9	121	35.3	42	17.8	112
Niger	17.6	97	3.4	90	30.0	67	19.4	103
Pakistan	17.5	98	3.7	85	23.4	90	25.5	69
Zambia	17.4	99	3.3	91	29.0	70	19.7	99
Egypt	17.2	100	7.1	67	23.7	88	20.8	94
Botswana	16.9	101	3.6	86	n/a	n/a	30.1	53
Mexico	16.7	102	3.9	84	18.0	105	28.2	59
Bosnia & Herzegovina	16.7	103	6.7	69	18.4	104	24.9	73
Senegal	16.0	104	4.6	82	17.3	107	26.0	65
Kenya	15.8	105	5.3	75	20.1	100	22.1	86
Madagascar	15.7	106	2.5	100	21.5	94	23.1	80
El Salvador	15.6	107	n/a	n/a	9.2	112	21.9	88
Benin	15.1	108	5.1	78	n/a	n/a	25.2	71
Cambodia	15.1	109	2.5	101	26.5	79	16.2	118
Peru	14.5	110	1.1	115	26.0	80	16.3	117
Burkina Faso	14.3	111	2.5	99	21.4	96	19.1	104
Guatemala	14.2	112	0.7	123	19.9	101	22.1	85
Namibia	14.1	113	14.9	50	n/a	n/a	13.4	120
Colombia	14.1	114	2.3	104	18.6	103	21.4	90
Bolivia	14.1	115	3.5	87	20.8	99	18.1	111
Nigeria	13.4	116	2.0	107	21.1	97	17.1	115
Malawi	13.2	117	6.7	68	32.5	61	0.3	123
Jamaica	12.8	118	6.3	70	13.6	110	18.4	110
United Arab Emirates	12.6	119	1.8	111	35.7	39	0.3	122
Nicaragua	11.4	120	3.3	94	n/a	n/a	19.6	101
Honduras	11.2	121	0.8	122	11.5	111	21.4	91
Rwanda	8.7	122	3.4	89	4.0	113	18.6	108
Panama	8.4	123	3.1	96	2.4	114	19.8	98
Zimbabwe	6.6	124	19.0	42	0.0	115	0.9	121
Algeria	6.1	125	2.3	103	15.9	108	0.0	124

Table 5g: Creative outputs pillar

	CREATIVE OUTPUTS		Creative intangibles		Creative goods and services	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank
Hong Kong (SAR), China	57.6	1	54.8	27	60.4	1
Sweden	56.7	2	59.3	14	54.1	4
Switzerland	54.4	3	63.5	9	45.3	11
Canada	54.0	4	54.2	29	53.9	5
Denmark	52.4	5	50.7	43	54.1	3
Germany	51.7	6	56.1	23	47.2	8
Norway	50.6	7	50.9	40	50.4	6
Netherlands	50.6	8	46.7	55	54.5	2
Austria	49.0	9	58.0	16	40.0	15
Jordan	48.9	10	70.1	4	27.7	47
Estonia	48.9	11	55.2	25	42.6	13
Brazil	46.9	12	56.2	22	37.5	20
Czech Republic	46.8	13	47.1	54	46.5	10
United Arab Emirates	46.6	14	71.5	3	21.7	58
New Zealand	45.6	15	60.1	13	31.1	35
Hungary	44.9	16	43.2	71	46.5	9
France	44.7	17	58.0	17	31.4	34
United Kingdom	44.3	18	49.2	48	39.4	16
Portugal	44.1	19	54.5	28	33.7	25
Slovenia	43.8	20	51.4	36	36.2	23
Nigeria	43.8	21	57.2	19	30.3	37
Argentina	43.4	22	57.8	18	29.0	42
Serbia	43.4	23	36.5	96	50.2	7
United States of America	43.2	24	48.0	51	38.3	18
Iceland	42.3	25	56.9	21	27.7	48
Belgium	42.3	26	40.6	83	44.0	12
Korea, Rep.	42.2	27	58.7	15	25.6	51
Finland	42.1	28	50.7	44	33.4	26
Turkey	41.6	29	50.7	42	32.5	29
Singapore	41.4	30	50.4	45	32.4	30
Viet Nam	41.3	31	46.0	59	36.7	21
Moldova, Rep.	41.1	32	55.4	24	26.7	49
Latvia	41.0	33	49.1	49	32.9	27
Spain	41.0	34	44.3	67	37.7	19
China	40.9	35	53.7	30	28.1	45
Australia	40.6	36	51.1	37	30.2	38
Israel	40.4	37	51.8	34	28.9	43
India	40.3	38	51.0	38	29.6	39
Thailand	39.9	39	50.7	41	29.1	41
Malaysia	39.9	40	55.2	26	24.7	54
Tunisia	39.6	41	71.7	2	7.6	93
Luxembourg	39.5	42	42.8	72	36.2	22
Italy	39.2	43	39.6	86	38.8	17
Chile	38.8	44	65.9	8	11.8	80
Costa Rica	38.8	45	69.5	5	8.0	92
Bulgaria	38.1	46	51.8	35	24.3	55
Colombia	37.7	47	49.8	47	25.6	52
Cyprus	37.4	48	42.7	73	32.1	31
Lithuania	37.2	49	48.0	52	26.4	50
Qatar	36.9	50	73.6	1	0.3	123
Croatia	36.7	51	40.9	79	32.6	28
Guatemala	36.7	52	63.3	10	10.2	87
Pakistan	36.3	53	41.0	78	31.7	33
Poland	35.8	54	36.3	97	35.3	24
Uruguay	35.8	55	66.6	7	5.0	105
Lebanon	35.7	56	42.1	76	29.2	40
Saudi Arabia	35.6	57	68.8	6	2.4	115
Ireland	34.2	58	40.1	84	28.2	44
Sri Lanka	34.1	59	47.7	53	20.5	60
Zimbabwe	33.9	60	25.7	117	42.1	14
Honduras	33.8	61	57.1	20	10.5	84
Panama	33.2	62	61.4	12	5.1	104
Slovak Republic	33.1	63	38.2	91	28.0	46

Table 5g: Creative outputs pillar *(continued)*

	CREATIVE OUTPUTS		Creative intangibles		Creative goods and services	
Country/Economy	Score (0–100)	Rank	Score	Rank	Score	Rank
Romania	33.0	64	35.4	102	30.6	36
Japan	32.8	65	42.3	74	23.3	57
Senegal	32.8	66	61.6	11	4.0	109
El Salvador	31.8	67	52.0	33	11.6	81
Ecuador	31.4	68	44.3	66	18.4	63
Greece	31.2	69	30.5	111	31.9	32
Ukraine	31.0	70	40.6	82	21.4	59
Mexico	30.1	71	44.2	68	16.0	66
Egypt	29.5	72	43.9	69	15.1	69
Nicaragua	29.4	73	46.3	56	12.6	79
Mauritius	29.1	74	39.1	89	19.0	62
Russian Federation	28.9	75	33.0	107	24.9	53
Peru	28.8	76	53.2	31	4.4	106
Venezuela	28.3	77	40.7	80	15.8	68
Armenia	28.1	78	39.5	87	16.7	65
Guyana	27.6	79	45.5	60	9.6	88
Uganda	27.3	80	50.2	46	4.3	108
Albania	27.3	81	34.6	105	19.9	61
Oman	27.2	82	52.2	32	2.2	117
Bolivia	26.9	83	40.7	81	13.1	77
Macedonia	26.8	84	39.7	85	13.9	72
South Africa	26.6	85	45.1	62	8.2	91
Côte d'Ivoire	26.6	86	50.9	39	2.3	116
Brunei Darussalam	26.0	87	36.0	98	16.0	67
Paraguay	25.8	88	44.4	65	7.2	94
Indonesia	25.7	89	48.6	50	2.8	111
Philippines	25.7	90	41.1	77	10.2	86
Ghana	25.6	91	42.2	75	9.0	89
Rwanda	25.3	92	44.7	63	5.9	98
Cameroon	25.1	93	44.7	64	5.5	100
Malawi	25.0	94	39.5	88	10.6	82
Jamaica	24.9	95	43.3	70	6.6	96
Bangladesh	24.7	96	35.8	99	13.6	73
Botswana	24.4	97	35.7	100	13.2	76
Cambodia	24.3	98	34.9	104	13.6	74
Trinidad and Tobago	24.3	99	38.1	93	10.4	85
Iran	23.6	100	32.6	109	14.7	71
Benin	23.6	101	46.3	57	0.9	121
Madagascar	23.6	102	29.9	113	17.2	64
Kuwait	23.5	103	46.2	58	0.9	120
Kyrgyzstan	23.3	104	22.4	121	24.1	56
Mali	23.2	105	45.1	61	1.4	118
Namibia	22.8	106	38.4	90	7.2	95
Bosnia & Herzegovina	22.5	107	30.3	112	14.7	70
Kenya	22.3	108	38.1	92	6.5	97
Morocco	22.1	109	35.7	101	8.6	90
Mongolia	21.6	110	38.0	94	5.2	101
Tanzania	21.1	111	29.3	115	12.8	78
Bahrain	20.2	112	37.8	95	2.6	113
Kazakhstan	20.2	113	29.8	114	10.5	83
Georgia	19.8	114	34.0	106	5.7	99
Burkina Faso	19.7	115	35.2	103	4.3	107
Syrian Arab Republic	18.7	116	24.2	119	13.2	75
Azerbaijan	17.9	117	33.0	108	2.7	112
Zambia	16.1	118	31.2	110	0.9	119
Tajikistan	15.0	119	27.0	116	3.0	110
Swaziland	13.9	120	22.7	120	5.2	102
Ethiopia	13.8	121	25.2	118	2.4	114
Yemen	10.7	122	20.8	122	0.6	122
Algeria	9.0	123	18.0	123	0.0	125
Niger	5.2	124	n/a	n/a	5.2	103
Sudan	5.1	125	9.9	124	0.2	124

Top 10

The top 10 countries in the Innovation Output Sub-Index are Sweden, Switzerland, the Netherlands, Germany, the US, Finland, Denmark, Israel, the UK, and Canada. The Output Sub-Index, like the overall GII, is dominated by Europe (seven countries), and includes two North American countries and Israel, which has a remarkable showing (8th on Output, 14th on the GII, and 1st at the regional level). The best-ranked economies within each region are Sweden (1st), the US (5th), Israel (8th), the Republic of Korea (11th), Brazil (32nd), India (44th), and Nigeria (62nd).

The **Netherlands** comes in 3rd on the Output Sub-Index, a performance driven by marks within the top 10 in international Patent Cooperation Treaty applications by residents, scientific and technical journal articles, total computer software spending, royalty and license fees' receipts, daily newspapers' circulation, and creative services exports. This excellent result allows it to be ranked 9th in the overall GII, despite its 16th place on the Input side.

Germany's position is interesting because it has an excellent performance on the Output Sub-Index (4th), but this does not compensate for its relative weaknesses on the Input side (21st). For the second consecutive year, Germany is not among the top 10 on the GII (12th). While its biggest strengths are in resident patent applications both at the national patent office and at the Patent Cooperation Treaty (ranked 4th and 9th, respectively), its score is due to a good balance, as it is well-positioned in practically all indicators included in pillars 6 (14th) and 7 (6th).

The **United States of America (US)** comes in at position 7 in the GII 2011, moving up from 11th position last year. The US's high position is very commendable when one considers that it is the only large country (in both size and population) in the top 10. The US has a high ranking among all Input pillars (11th) and an even higher ranking among all Output pillars (5th), benefiting from a high level of Innovation Efficiency (9th among high-income countries). The strongest pillars for the US include those of Market sophistication (4th) and Scientific outputs (5th). However, its performance across most of the other pillars is uniformly high (Institutions at position 15, Human capital and research at 13, Infrastructure at 14, Business sophistication at 15, and Creative outputs at 24). The US does have several weaknesses—such as a low number of graduates in engineering (ranking 73rd), a poor share of renewables in energy use (78th), and a deficit of 4.1 ha per capita on ecological footprint and biocapacity (108th). But it also has several key strengths, such as a very low rigidity of employment (1st, or least rigid worldwide), high tertiary enrolment (5th at 82.9%), a share of 2.8% of GDP spent on R&D (7th), the good quality of its scientific research institutions (4th), well-developed government online services (2nd), and positive credit and investment environments (ranked 4th and 3rd, respectively). Although measured through qualitative surveys, the US also is very robust in new innovation metrics such as ICT and new business model creation (ranking 9th) and ICT and organizational models creation (2nd).

Top performers by income group

By income group, the top-ranked countries in the Innovation Output Sub-Index are Sweden (1st), Brazil (32nd), China (14th), and Bangladesh (69th). In position 14, China again is the only non-high-income country in the top 30. Moldova, Brazil, Jordan, Malaysia, Costa Rica, Serbia, and Argentina all achieve rankings among the top 40. High-income countries with weak performances on the Output dimension are Saudi Arabia (66th), Greece (72nd), Oman (78th), Trinidad and Tobago (87th), Bahrain (92nd), and Brunei Darussalam (94th), all of them in the lower half of the rankings.

Brazil achieves a remarkable 32nd position on the Output Sub-Index, topping the Output rankings among middle-income countries. With a poor 68th position on the Input side bringing it down, it still achieves 47th place on the overall GII. Brazil's strengths on the Output side come from a good overall balance. The country attains positions within the top 30 on utility model and trademark applications by residents at the national office (24th and 23rd, respectively), growth rate of labour productivity (26th at 3.9%), computer and communications service exports (15th at 57% of total commercial service exports), and creative service exports (2nd at 20.9%), as well as on the use of ICT on business and organizational models (two survey questions, ranked 23rd and 25th). These results are commendable in the face of Brazil's weaknesses on the Input side (68th), where several indicators are nevertheless within the top 30: gross expenditure on R&D (30th at 1.1% of GDP); share of renewables in energy use (24th with 44.5%); ecological footprint

and biocapacity (7th with a reserve of 6.1 ha per capita), depth of credit information (25th), market capitalization (23rd), total value of stock traded (27th), firms offering formal training (13th, with 52.9%), and the state of cluster development (30th, a survey question). Brazil is also taking important steps towards technological catch-up and knowledge absorption, particularly in the areas of high-tech imports (19th at 15.7% of total imports) and computer and communications service imports (16th, at 49.4% of total commercial service imports).

Bangladesh, a low-income country with a GDP per capita of 1,416 in purchasing power parity (PPP) dollars, tops the Output rankings among low-income countries, reaching position 69. The country's stronger points on the Output side are its growth rate of labour productivity (29th at 3.8%) and the 71.8% share of computer and communications service exports of total commercial service exports (2nd globally). Bangladesh is yet another case of a country achieving more with less. In effect, its positions on the Input pillars are rather weak, particularly on Business sophistication (121st), Human capital and research (114th), Institutions (105th), and Market sophistication (102nd). Its relatively better ranking on Infrastructure (76th) is leveraged by its positions within the top 40 on efficiency in energy use (where it ranks 4th, at 11.2 PPP$/kg oil eq.), its share of renewables in energy use (34th), its gross capital formation (38th at 24.4% of GDP), and its ecological footprint and biocapacity (39th, although with a deficit of 0.2 ha per capita).

The Innovation Input Sub-Index

The Innovation Input Sub-Index variables provide information on indicators that measure elements that must be in place to foster innovation in an economy.

Top 10

The top 10 economies on the Innovation Input Sub-Index are Singapore, Hong Kong (SAR, China), Switzerland, Ireland, Sweden, Finland, Denmark, Canada, Luxembourg, and the UK. Regional leaders are Singapore (1st), Switzerland (3rd), Canada (8th), Israel (20th), Chile (36th), South Africa (40th), and India (87th).

Hong Kong (SAR, China) is ranked 4th on the GII and is the runner-up after another Asian economy, Singapore (discussed above), on the Input Sub-Index. Hong Kong (SAR, China) has an interesting profile, as it has positions within the top 5 in all pillars except the two that are traditionally linked to innovation: Human capital and research (30th) and Scientific outputs (24th). These two relatively low rankings mean that it reaches only 27th place on Efficiency among high-income countries (66th overall).

The result for **Ireland** is driven by its excellent marks in Institutions and Market and Business sophistication (ranked 6th, 6th, and 3rd, respectively). Although Ireland presents a good environment and potential for innovation, it lags behind on the Output Sub-Index where it ranks 22nd, attaining positions 13 on the GII, and 83 on the Efficiency Index (32nd among high-income countries).

Finland comes next, placing 6th on Input, 6th on Output, 5th on the GII, and 35th on Efficiency (12th on the latter among high-income countries). Finland comes in 1st on three indicators: freedom of the press, rule of law; and exports of computer and communications services. Finland is also ranked among the top 10 on five pillars: Institutions, Human capital and research, Infrastructure, Business sophistication, and Scientific outputs; the country's relative weaknesses are on the Market sophistication and Creative outputs pillars.

Top performers by income group

By income group, in descending order of income, the best performers present no surprises. With the exception of Singapore (1st), they are the same countries leading in the overall GII rankings by income group: Malaysia (27th), China (43rd), and Ghana (65th). Among non-high-income countries, Malaysia is the only country in the top 30, followed by Chile, Lithuania, and South Africa in the top 40. Brunei Darussalam, ranked 70th, is the only high-income country in the lower half of the rankings.

Although scores on the Input and Output Sub-Indices might differ by substantial amounts leading to important shifts in rankings from one Sub-Index to the other for particular countries, there is a positive relationship between the two, as shown by Figure 2. The data seem to confirm that efforts made on enabling environments are rewarded with increased innovation outputs.

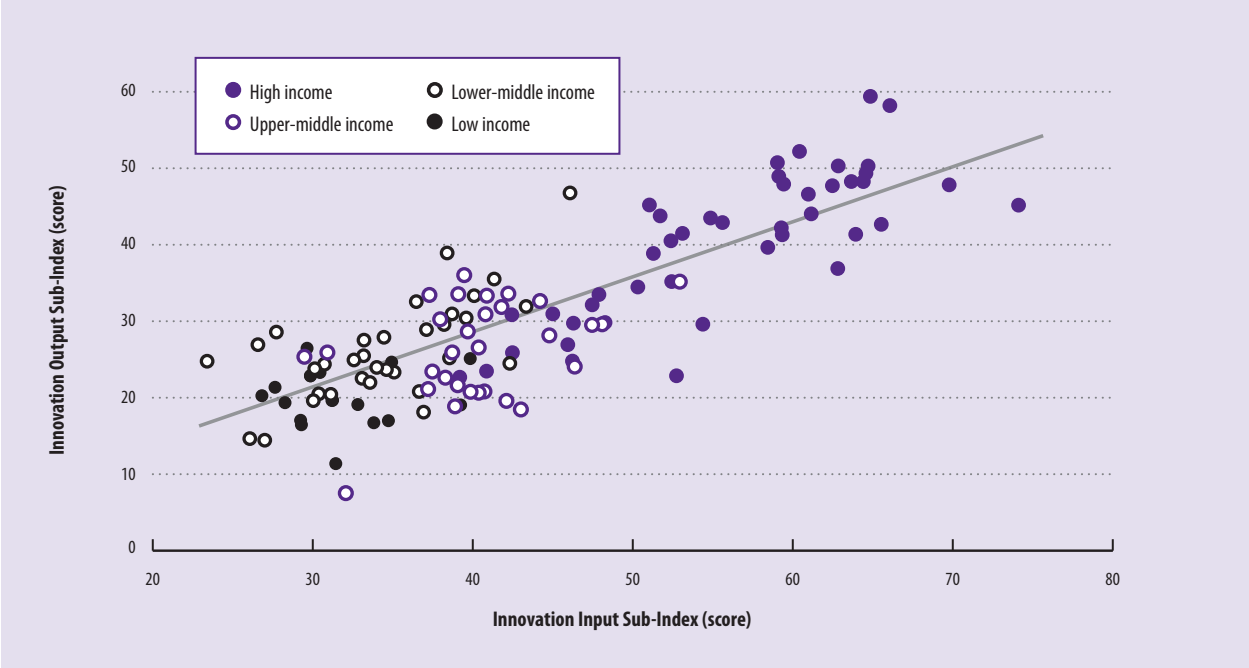
The Innovation Efficiency Index

While the GII is calculated as the average of the Input and Output

Table 6: Innovation Efficiency Index rankings: Top 10

Rank	Country/Economy	Efficiency Rank	Input Rank	Output Rank	Income	Income Rank	Region	Regional Rank	Population (US$ millions)	GDP per capita (current PPP$)
1	Côte d'Ivoire	1.06	125	79	LM	**1**	SSF	**1**	21.6	1,701.2
2	Nigeria	1.03	119	62	LM	2	SSF	2	158.3	2,203.3
3	China	1.02	43	14	LM	3	EAS	**1**	1,354.1	6,828.0
4	Pakistan	1.01	123	67	LM	4	SAS	**1**	184.8	2,608.6
5	Moldova, Rep.	1.01	77	29	LM	5	ECS	**1**	3.6	2,854.3
6	Sweden	0.92	5	**1**	HI	**1**	ECS	2	9.3	37,904.6
7	Brazil	0.91	68	32	UM	**1**	LCN	**1**	195.4	10,412.1
8	Argentina	0.90	82	40	UM	2	LCN	2	40.7	14,538.3
9	India	0.89	87	44	LM	6	SAS	2	1,214.5	3,270.1
10	Bangladesh	0.89	114	69	LI	**1**	SAS	3	164.4	1,416.3

Figure 2: Innovation Output Sub-Index vs. Innovation Input Sub-Index



Note: Countries/economies are classified according to the World Bank Income Group Classification (January 2011).

Sub-Indices, the Innovation Efficiency Index is calculated as the ratio of the Output over the Input Sub-Indices. Although this index is designed to be neutral to the countries' stages of development, as shown by the discussion of the top performers below, valuable insights can be obtained from an analysis among countries with similar levels of development. Table 6 provides Innovation Efficiency Index rankings for the top 10; Tables 7a through 7d give the scores per income group, together with the Input, Output, and Efficiency rankings. To facilitate comparisons, these tables also include the difference between the Input and Output rankings, as well as the population and GDP per capita of each country/economy.

Top 10

The top 10 countries in the Innovation Efficiency Index are

Table 7a: Innovation Efficiency Index rankings (high-income countries/economies)

Rank	Country/Economy	Efficiency Score	Efficiency Rank	Input Rank	Output Rank	Difference	Population (US$ millions)	GDP per capita (current PPP$)
1	Sweden	0.92	6	5	**1**	4	9.3	37,904.6
2	Hungary	0.89	11	33	16	17	10.0	19,764.3
3	Switzerland	0.88	12	3	2	**1**	7.6	45,116.9
4	Netherlands	0.86	13	16	3	13	16.7	40,714.7
5	Germany	0.86	15	21	4	17	82.1	36,267.4
6	Qatar	0.85	18	31	19	12	1.5	91,378.7
7	Israel	0.83	22	20	8	12	7.3	27,759.2
8	Korea, Rep.	0.81	25	17	11	6	48.5	27,168.5
9	United States of America	0.80	26	11	5	6	317.6	45,989.2
10	Estonia	0.79	31	24	20	4	1.3	19,451.4
11	Czech Republic	0.78	33	26	24	2	10.4	25,232.0
12	Finland	0.78	35	6	6	0	5.3	34,719.7
13	Cyprus	0.77	37	30	27	3	0.9	30,223.4
14	France	0.77	39	23	21	2	62.6	33,655.5
15	New Zealand	0.76	46	15	15	0	4.3	29,072.2
16	Denmark	0.76	47	7	7	0	5.5	36,761.7
17	Iceland	0.76	48	13	13	0	0.3	37,595.1
18	United Kingdom	0.76	50	10	9	**1**	61.9	36,495.8
19	Slovenia	0.76	51	32	30	2	2.0	27,004.4
20	Canada	0.75	54	8	10	–2	33.9	37,945.6
21	Kuwait	0.73	57	51	51	0	3.1	48,631.3
22	Norway	0.72	59	14	18	–4	4.9	55,672.1
23	Austria	0.71	60	19	23	–4	8.4	38,363.1
24	Italy	0.70	63	37	39	–2	60.1	31,908.6
25	Japan	0.70	64	18	26	–8	127.0	32,452.8
26	Croatia	0.69	65	45	48	–3	4.4	19,805.4
27	Hong Kong (SAR), China	0.69	66	2	12	–10	7.1	44,303.8
28	Portugal	0.69	67	34	36	–2	10.7	24,569.4
29	Belgium	0.68	71	22	28	–6	10.7	36,249.0
30	Latvia	0.68	72	38	45	–7	2.2	15,412.8
31	Spain	0.67	74	29	34	–5	45.3	32,544.8
32	Ireland	0.65	83	4	22	–18	4.6	41,278.2
33	Luxembourg	0.65	84	9	25	–16	0.5	83,758.8
34	Poland	0.64	85	41	55	–14	38.0	19,058.7
35	Slovak Republic	0.62	92	35	54	–19	5.4	22,356.3
36	Singapore	0.61	94	**1**	17	–16	4.8	50,632.8
37	Greece	0.61	95	50	72	–22	11.2	29,663.4
38	Australia	0.59	97	12	31	–19	21.5	39,230.7
39	Saudi Arabia	0.59	98	44	66	–22	26.2	23,395.4
40	Brunei Darussalam	0.58	101	70	94	–24	0.4	51,204.6
41	Trinidad and Tobago	0.57	103	58	87	–29	1.3	25,571.7
42	United Arab Emirates	0.54	109	25	56	–31	4.7	57,743.7
43	Oman	0.54	110	42	78	–36	2.9	25,462.1
44	Bahrain	0.43	122	28	92	–64	0.8	35,174.1

Côte d'Ivoire, Nigeria, China, Pakistan, Moldova, Sweden, Brazil, Argentina, India, and Bangladesh. This list includes some of the most densely inhabited countries in the world: China, India, Brazil, Pakistan, Bangladesh, and Nigeria are all among the 10 most populous countries in this year's sample, and three of them place 1st on Efficiency in their regions.

Three BRIC countries (Brazil, India, and China) are in this select list, with the fourth, the Russian Federation, coming in only at 52nd place. By region, the best performers are Côte d'Ivoire (1st), China (3rd), Pakistan (4th), Moldova (5th), Brazil (7th), Jordan (16th), and the US (26th). By income group, in descending order of income, leaders are Sweden (6th), Brazil (7th), Côte d'Ivoire (1st), and Bangladesh (10th).

Although **Côte d'Ivoire** tops the Efficiency rankings, its overall performance is poor. The country

Table 7b: Innovation Efficiency Index rankings (upper-middle-income countries/economies)

Rank	Country/Economy	Efficiency Score	Efficiency Rank	Input Rank	Output Rank	Difference	Population (US$ millions)	GDP per capita (current PPP$)
1	Brazil	0.91	7	68	32	36	195.4	10,412.1
2	Argentina	0.90	8	82	40	42	40.7	14,538.3
3	Venezuela	0.86	14	115	74	41	29.0	12,322.9
4	Serbia	0.86	17	71	38	33	9.9	11,719.2
5	Iran	0.84	19	106	71	35	75.1	11,558.4
6	Lebanon	0.82	23	57	41	16	4.3	13,069.7
7	Turkey	0.80	28	80	53	27	75.7	13,885.0
8	Costa Rica	0.80	29	53	37	16	4.6	11,105.7
9	Romania	0.76	49	55	47	8	21.2	14,278.0
10	Russian Federation	0.76	52	59	50	9	140.4	18,962.6
11	Bulgaria	0.74	55	47	43	4	7.5	13,332.7
12	Uruguay	0.72	58	66	61	5	3.4	13,189.1
13	Colombia	0.67	75	74	70	4	46.3	8,959.2
14	Malaysia	0.66	77	27	35	–8	27.9	14,012.0
15	Macedonia	0.66	78	61	68	–7	2.0	10,822.7
16	Mauritius	0.63	89	46	63	–17	1.3	12,838.4
17	Mexico	0.62	90	81	88	–7	110.6	14,335.1
18	Lithuania	0.62	91	39	59	–20	3.3	16,747.1
19	Chile	0.62	93	36	57	–21	17.1	14,330.7
20	Albania	0.59	96	78	95	–17	3.2	8,373.0
21	Azerbaijan	0.57	104	83	100	–17	8.9	9,638.2
22	Peru	0.55	108	72	98	–26	29.5	8,629.5
23	Kazakhstan	0.52	112	64	103	–39	15.8	11,509.9
24	South Africa	0.52	113	40	83	–43	50.5	10,277.8
25	Botswana	0.51	114	62	104	–42	2.0	13,384.5
26	Panama	0.51	115	60	101	–41	3.5	13,057.1
27	Jamaica	0.49	120	73	115	–42	2.7	7,632.6
28	Bosnia & Herzegovina	0.47	121	54	111	–57	3.8	8,490.6
29	Namibia	0.43	123	49	116	–67	2.2	6,410.1
30	Algeria	0.23	125	101	125	–24	35.4	8,172.5

is ranked a low 117th on the overall GII and last on the Input Sub-Index, coming in among the bottom 25 on all five Input pillars. With such feeble and fragile conditions, its top rank on Efficiency and its 79th position on the Output Sub-Index are praiseworthy indeed. On closer inspection, Côte d'Ivoire presents rather mixed results on the Output side. The main leverage on the Output score comes from some progress on two indicators marked by a year of global crisis: with a small growth in labour productivity of 0.1%, it achieves position 77; and with zero FDI net outflows, it is ranked 91st—a result that is possible because it is compared with countries with high levels of divestment, such as Belgium (–16.7%) and Bahrain (–8.7%), at the bottom of this indicator. The real strength of Côte d'Ivoire comes from the share of 57.4% of its commercial service exports in computer, communications, and other services (ranked 13th globally). This is coherent with the relatively good scores assigned by the World Economic Forum survey on the use of ICT on business and organizational models, where it ranks 64th and 91th, respectively.

Top performers by income group

Among high-income economies, the Europe and Central Asia region dominates the top 5, while Middle Eastern and North African countries present mixed results: Qatar and Israel are ranked 6th and 7th, respectively, and Oman and Bahrain are at the bottom. Apart from the Republic of Korea, ranked 8th, East Asian and Pacific countries have rather weak performances (New Zealand comes next at position 15). Twenty-four out of 44 high-income countries have lower rankings on Outputs than on Inputs.

Among upper-middle-income countries, some show a capacity to achieve more innovation outputs from less favourable conditions: Brazil, Argentina, Serbia, and Costa Rica make it to the top 40 on the Output Sub-Index, surmounting lower positions on the Input

Table 7c: Innovation Efficiency Index rankings (lower-middle-income countries/economies)

Rank	Country/Economy	Efficiency Score	Efficiency Rank	Input Rank	Output Rank	Difference	Population (US$ millions)	GDP per capita (current PPP$)
1	Côte d'Ivoire	1.06	1	125	79	46	21.6	1,701.2
2	Nigeria	1.03	2	119	62	57	158.3	2,203.3
3	China	1.02	3	43	14	29	1,354.1	6,828.0
4	Pakistan	1.01	4	123	67	56	184.8	2,608.6
5	Moldova, Rep.	1.01	5	77	29	48	3.6	2,854.3
6	India	0.89	9	87	44	43	1,214.5	3,270.1
7	Jordan	0.86	16	56	33	23	6.5	5,597.0
8	Viet Nam	0.83	20	63	42	21	89.0	2,953.1
9	Sri Lanka	0.83	21	96	65	31	20.4	4,771.6
10	Paraguay	0.81	24	92	64	28	6.5	4,522.5
11	Guyana	0.80	27	75	49	26	0.8	3,088.2
12	Senegal	0.79	30	107	82	25	12.9	1,816.6
13	Cameroon	0.79	32	110	85	25	20.0	2,204.9
14	Armenia	0.78	34	84	60	24	3.1	5,278.9
15	Tunisia	0.77	36	79	58	21	10.4	8,272.5
16	Ukraine	0.77	40	67	52	15	45.4	6,317.8
17	Guatemala	0.77	41	97	73	24	14.4	4,719.5
18	Ecuador	0.77	44	100	77	23	13.8	8,267.7
19	Thailand	0.74	56	48	46	2	68.1	7,995.1
20	Philippines	0.70	62	93	84	9	93.6	3,541.7
21	El Salvador	0.68	69	91	86	5	6.2	6,629.3
22	Honduras	0.68	70	98	96	2	7.6	3,841.6
23	Bolivia	0.68	73	109	105	4	10.0	4,419.3
24	Egypt	0.67	76	88	89	–1	84.5	5,672.6
25	Nicaragua	0.66	79	105	106	–1	5.8	2,641.3
26	Indonesia	0.65	80	95	97	–2	232.5	4,198.8
27	Georgia	0.65	81	76	75	1	4.2	4,774.1
28	Syrian Arab Republic	0.65	82	111	110	1	22.5	4,730.0
29	Mongolia	0.58	102	52	81	–29	2.7	3,522.3
30	Morocco	0.57	105	86	102	–16	32.4	4,494.4
31	Sudan	0.56	107	124	122	2	43.2	2,209.7
32	Yemen	0.53	111	121	123	–2	24.3	2,469.6
33	Swaziland	0.49	117	85	117	–32	1.2	4,998.4

side. While other countries with excellent Input scores lag behind—Malaysia (position 27 on Input, and 35 on Output), Chile (36 and 57), Lithuania (39 and 59), and South Africa (40 and 83)—all four countries in the top 40 on the Input Sub-Index have lower (worse) ranks on Output. In this income group, 17 countries out of 30 have lower rankings on Output than on Input.

The same analysis among lower-middle-income countries leads to encouraging results. Six of the top 10 countries in the Efficiency Index come from this income group: Côte d'Ivoire, Nigeria, China, Pakistan, and Moldova take the first five spots, and India comes in 9th. China, Moldova, and Jordan are among the top 40 on the Output Sub-Index (at positions 14, 29, and 33, respectively). Within this income group, only 7 out of 33 countries have lower Output rankings than Input rankings, with the major differences found in Swaziland (which shows a drop of 32 positions from Input to Output scores), Mongolia (a difference of 29 positions), and Morocco (16 positions). The other four countries have their Input and Output ranks less than two positions apart.

Among low-income countries, Bangladesh takes the lead (and makes it to the top 10 on Efficiency for all countries combined), at positions 114 and 69 on the Input and Output Sub-Indices, respectively. Ten out of 18 countries within this income group have Output scores lower than their Input scores, all of them Sub-Saharan African countries except for Cambodia (–5 positions). The bigger negative differences are found in Kenya (–45), Rwanda (–29), Zambia (–26), Niger (–22), Malawi (–14), and Ghana (–11).

Table 7d: Innovation Efficiency Index rankings (low-income countries/economies)

Rank	Country/Economy	Efficiency Score	Efficiency Rank	Input Rank	Output Rank	Difference	Population (US$ millions)	GDP per capita (current PPP$)
1	Bangladesh	0.89	10	114	69	45	164.4	1,416.3
2	Tajikistan	0.77	38	120	99	21	7.1	1,972.1
3	Uganda	0.77	42	112	91	21	33.8	1,217.2
4	Mali	0.77	43	113	93	20	13.3	1,185.5
5	Tanzania	0.77	45	108	90	18	45.0	1,355.7
6	Zimbabwe	0.76	53	122	107	15	12.6	500.0
7	Kyrgyzstan	0.71	61	89	80	9	5.6	2,283.3
8	Benin	0.68	68	118	112	6	9.2	1,507.9
9	Ghana	0.63	86	65	76	–11	24.3	1,552.4
10	Cambodia	0.63	87	103	108	–5	15.1	1,915.0
11	Madagascar	0.63	88	104	109	–5	20.1	1,048.6
12	Burkina Faso	0.58	99	117	118	–1	16.3	1,186.9
13	Malawi	0.58	100	99	113	–14	15.7	858.2
14	Ethiopia	0.56	106	116	121	–5	85.0	934.4
15	Zambia	0.49	116	94	120	–26	13.3	1,428.6
16	Rwanda	0.49	118	90	119	–29	10.3	1,069.7
17	Kenya	0.49	119	69	114	–45	40.9	1,572.6
18	Niger	0.36	124	102	124	–22	15.9	674.6

Figure 3 plots the GII scores against the Innovation Efficiency Index scores. The relationship is positive, as expected, implying that more efficient countries achieve better GII scores. The figure also shows that efficiency is not related to the level of economic development of nations, and illustrates the relatively better performance of lower-middle-income countries, which are for the most part located in the bottom quadrant to the right.

Regional rankings

The next section considers the rankings of countries by region. For reasons of space each country profile and region cannot be discussed in detail, but snapshots are provided for the main countries that lead in the rankings.

To put the discussion of rankings in perspective, Figure 4 presents in a bar graph average pillar scores by region. Regions are discussed following the sum of average scores, in descendent order. Regional trends are further discussed in the concluding remarks to this chapter.

North America

Only two North American countries are represented. The US, in 7th position, is discussed among the leaders of the Output Sub-Index.

Canada is in a commendable 8th position in the GII, 10th in Output and 8th in Input. It places among the top 10 on the Institutions, Infrastructure, Market and Business sophistication, and Creative outputs pillars. Canada also has lower, but still performing, positions on the sectors traditionally linked to innovation—Human capital and research (19th) and Scientific outputs (21st).

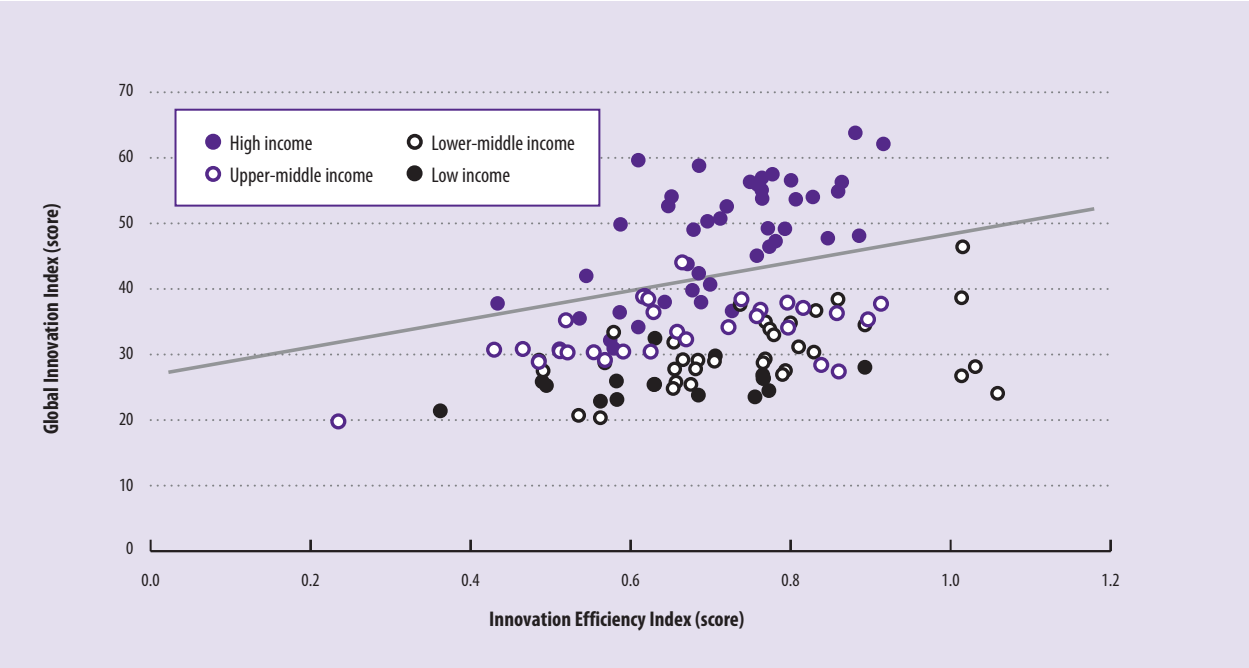
Europe and Central Asia

A total of 44 economies from Europe and Central Asia are represented in this year's GII report, 19 of which are within the top 30 and only one—Tajikistan—among the bottom 25. The top positions, not surprisingly, correspond to high-income countries (down to regional position 24). Among high-income countries, Poland (43rd), Croatia (44th), and Greece (63rd) lag behind.

The five **Nordic countries** Sweden (2nd), Finland (5th), Denmark (6th), Iceland (11th), and Norway (18th) have very strong performances globally as well as regionally, where they are within regional top 10 positions on the overall GII, Output, and Input Indices (except Norway, which is ranked 11th on the GII regional rankings).

Within the European Union (EU), among the **15 original EU countries (EU15)**,[28] the Netherlands and the UK are in the top 10, followed

Figure 3: Global Innovation Index vs. Innovation Efficiency Index



Note: Countries/economies are classified according to the World Bank Income Group Classification (January 2011).

by Germany (12th), Ireland (13th), Luxembourg (17th), Austria (19th), and France (22nd). The rest of the EU15 countries—Belgium (24th) and the four Mediterranean countries, Spain (32nd), Portugal (33rd), Italy (35th), and Greece (63rd)—have lost key positions to some of the 12 countries that recently acceded to the EU (the EU12 group).[29]

The **EU12 group** is led by high-income countries Estonia (23rd), Hungary (25th), the Czech Republic (27th), Cyprus (28th), and Slovenia (30th); in the same high-income group, Latvia (36th), the Slovak Republic (37th), and Poland (43rd) have relatively low scores. Upper-middle-income countries are all in the second quintile: Lithuania (40th), Bulgaria (42nd), and Romania (50th).

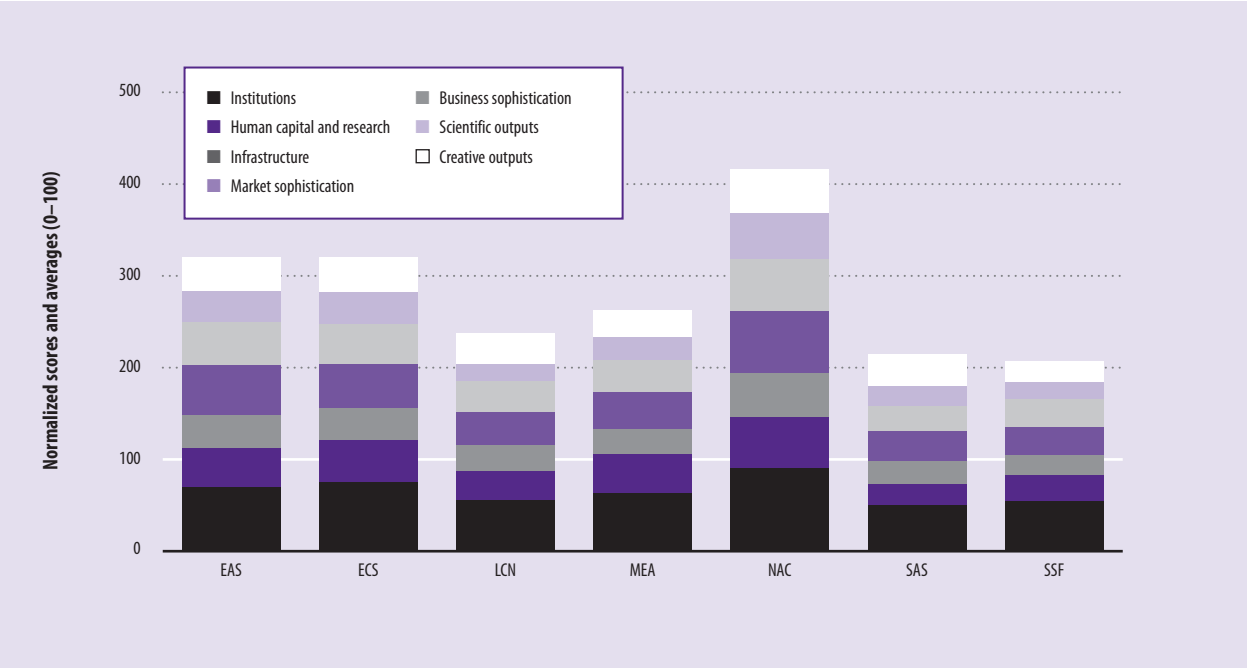
Among **non-EU countries** in the region, lower-middle-income Moldova (39th) leads, ahead of high-income Croatia (44th) and upper-middle-income countries Serbia (55th), Russian Federation (56th), Turkey (65th), and Macedonia (67th) in the third quintile; and Bosnia and Herzegovina (76th), Albania (80th), Kazakhstan (84th), and Azerbaijan (88th) in the fourth quintile. The remaining lower-middle-income countries show relatively good performances as well, and all are ranked in the third quintile: Ukraine (60th), Armenia (69th), and Georgia (73rd).

The score of lower-middle-income **Moldova** is admirable. The country places 25th in the regional rankings and 39th in the overall GII, surpassing 14 wealthier countries in its region. Moldova's main assets are on the Output side: it has a dynamic intellectual property system (placing among the top 10 positions on indicators for resident patent, utility model, and trademark applications at the national office, and on trademark applications through the Madrid System); it also exhibits strong growth in labour productivity (4th at 8.1%) and creative goods exports (5th at 5.8%). With Sweden, it is the only European country ranked in the top 10 on Efficiency (5th).

East Asia and the Pacific

Fifteen economies are represented from East Asia and the Pacific, seven of which are within the top 30 and only one of which—Cambodia—is in the bottom 25. In addition, the top ranked countries among upper- and lower-middle-income countries—Malaysia and China, respectively—come from this region. Figure 4 shows that this region is almost tied with Europe and Central Asia in its innovation performance, even though in terms

Figure 4: Average scores for selected country groups



Note: Countries/economies are classified according to the World Bank Regional Classification (January 2011): EAS = East Asia & Pacific; ECS = Europe & Central Asia; LCN = Latin America & Caribbean; MEA = Middle East & North Africa; NAC = North America; SAS = South Asia; and SSF = Sub-Saharan Africa.

of economic development it is much closer to the Middle East and North Africa.[30]

In East Asia and the Pacific, the first two economies are in the global top 10: Singapore (3rd) and Hong Kong (SAR, China, 4th). Five more are in the top 30: New Zealand (15th), the Republic of Korea (16th), Japan (20th), and Australia (21st)—all high-income countries—and China (29th). Among the regional high-income countries, only Brunei Darussalam lags behind at position 75.

The **Republic of Korea** tops the regional rankings on the Output Sub-Index, where it places 11th worldwide. Like the Netherlands, Germany, and the US, the Republic of Korea is among the most efficient innovators among high-income countries. The country has important strengths across the board, with commendable 1st place positions on two sub-pillars, ICT and knowledge creation, and on five indicators—gross tertiary enrolment (with an impressive 98.1%), the government's online service index, e-participation, the depth of credit information, and resident patent applications at the national office. It also places among the top 10 positions on three other sub-pillars: tertiary education, investment, and knowledge absorption, achieving positions among the top 10 on an impressive further 14 indicators (refer to the country profile for details). Four of these are Output indicators: resident patent applications through the Patent Cooperation Treaty (6th), resident utility model applications at the national office (5th), high-tech exports (4th with 28.43% of total exports), and the survey question on the use of ICT on business models.

Malaysia, the only upper-middle-income country in the region, tops the income group globally on both the GII and the Input Sub-Index (it is overtaken by Brazil on the Output Sub-Index). Further details about Malaysia can be found in the discussion of the country under the GII section. Among the lower-middle-income group are some of the world's most efficient global innovators: China (1st on all three main indices within its income group, and in the general rankings it is 29th on the GII and 14th on Output), Thailand (48th and 46th), and Viet Nam (51st and 42nd). Mongolia (68th) the Philippines (91st), and Indonesia (99th) lag behind. Within their income groups, middle-income countries Malaysia and China reach the regional top 10 on all four indices.

The Middle East & North Africa

The GII includes 16 countries from the Middle East and North Africa, of which only two—Israel (14th) and Qatar (26th)—are ranked among the top 30; they are also both high-income countries. Three countries from the region are within the bottom 25: the Syrian Arab Republic, Yemen, and Algeria. This region does not have any low-income countries represented in this year's GII rankings. The other five high-income economies in the region have rather disappointing positions: United Arab Emirates is at 34th place, Bahrain is at 46th, Kuwait at 52nd, Saudi Arabia at 54th, and Oman at 57th.

Israel tops the regional rankings on all three main indices, and is ranked 4th on Efficiency at the regional level. Israel's strength comes from pillar 6, Scientific outputs (where it ranks 4th), with good showings in knowledge creation and knowledge diffusion (2nd and 8th). The country also scores within the top 10 on resident filings at the Patent Cooperation Treaty (4th), scientific and technical journal articles (1st), high-tech exports (8th), and computer and communications service exports (5th).

Three upper-middle-income countries come from this region. Lebanon (regional 6th, overall 49th) and Iran (regional 13th and overall 95th) present weaknesses in both Input and Output indicators, and neither country is among the top 40 overall, while Algeria scores lowest among the countries with sufficient data to be included in the sample.

Jordan, an exception, ranks 4th in the region and 41st overall. The other lower-middle-income countries—Tunisia (regional 10th, overall 66th), Egypt (11th and 87th), and Morocco (12th and 94th)—are all in the lower half of the GII rankings, with Syrian Arab Republic (14th and 115th) and Yemen (15th and 123rd) among the bottom 25.

The position of **Jordan,** at 41st overall, is notable because it is more than 25 positions ahead of its closest competitor in the same region and income group, Tunisia. Although Jordan is only 8th in the region on Input, it is 3rd on Output. Jordan's leverage comes from Creative outputs, with a strong dynamism at the level of residents' trademark registrations at the national level (where it reached 1st place) and a relatively high level of creative goods exports. Jordan is also among the top 20 on the percentage of graduates in science (ranked 9th overall) and on tertiary inbound mobility (14th), two statistics that reflect the regional attractiveness of its higher education and its relevance to innovation. The country is among the top 20 on market capitalization and stocks traded as well, both of which demonstrate the dynamism of its economy.

Latin America & Caribbean

Twenty countries from Latin America and the Caribbean are included in this year's GII. None of them reach the top 30 on any of the three main indices (GII, Input, and Output), and three are ranked among the bottom 25: Venezuela, Nicaragua, and Bolivia. Trinidad and Tobago, the only high-income country in the region in the sample, is ranked a disappointing 72nd on the GII, 58th on the Input Sub-Index, and 87th on the Output Sub-Index.

Three upper-middle-income countries from Latin America and the Caribbean hold top positions within their income group: Chile (38th overall and 2nd in its income group after Malaysia), Costa Rica (45th and 5th), and Brazil (47th and 6th). Argentina (58th), Uruguay (64th), and Colombia (71st) are in the third quintile; Panama (77th), Mexico (81st), Peru (83rd), and Jamaica (92nd) in the fourth; and Venezuela (102nd) is down among the bottom 25. Costa Rica is the only country in the region to be among the regional top 3 on the GII and the Input and Output Sub-Indices (ranked 45th, 53rd, and 37th, respectively).

Chile, the second country in terms of GDP per capita after Argentina (with a per capita GDP of PPP $14,331) at 38th place is the only country in the region to have reached a position within the top 40. Chile's strengths are in its Input Sub-Index (36th), with a commendable 18th position on the Institutions pillar and positions in the top 20 on six indicators: regulatory quality (18th), tax rate (19th at 25% of profits), government online service (18th), market capitalization (11th at 128% of GDP), applied tariff rate (7th at 1.0%), and FDI net inflows (18th at 7.8% of GDP). On the Output side, its strengths are in FDI net outflows (12th at 4.88% of GDP), and resident trademark registrations at the national office (6th). Chile's standing is supported by a good overall balance, because its major weaknesses are precisely in the two areas traditionally linked to innovation: Human capital and research (71st, with a disappointing 89th ranking on elementary education) and Scientific outputs (85th). Chile presents one of the lowest Efficiency Index scores of all the countries in the sample, where it is ranked 16th regionally and 93rd globally.

Among lower-middle-income countries in this region, Guyana (61st) and Paraguay (74th) are in the third quintile; Guatemala (86th), El Salvador (90th), Ecuador (93rd), and Honduras (98th) are in the fourth; and Nicaragua (110th) and Bolivia (112nd) are among the bottom 25.

South Asia

The four countries from South Asia in the sample show mixed results. India, ranked 62 overall, tops the regional rankings; it is followed by Sri Lanka (82), Bangladesh (97), and Pakistan (105). These four countries, however, have their strengths on the Output side. In effect, although they rank between 87th (India) and 123rd (Pakistan) on the Input Sub-Index, they rank between 44th (India) and 69th (Bangladesh) on the Output Sub-Index, with Efficiency Index rankings ranging between 4th (Pakistan) and 21st (Sri Lanka).

After China, **India** is the second most densely populated country, with 1.2 billion inhabitants; it is also eleventh in GDP, with US$1,310 billion. A lower-middle-income country, it comes second after Sri Lanka in GDP per capita in PPP dollars in the region. India is ranked 62nd on the GII, 1st in its region, and 8th in its income group—after China, Moldova, Jordan, Thailand, Viet Nam, Ukraine, and Guyana. India comes in at 44th on the Output Sub-Index, within the top 30 on labour productivity growth (21st with 4.5%) and computer and communications services exports (4th globally, with 70.0% of total commercial service exports). It also has positions within the top 40 on two knowledge diffusion indicators: high-tech exports (32nd, at 6.34% of GDP) and FDI net outflows (38th, at 1.08% of GDP). On pillar 6, Creative outputs, it ranks 39th on national feature films produced, 22nd on daily newspapers, 9th on creative goods exports, and 29th on creative services exports. India's position, however, is dragged down by its poor performance on the Input side (ranked 87th): India is in the last quintile on sub-pillars business environment, elementary education, tertiary education, and knowledge workers. But the country has high marks—within the top 40—on R&D (35th); general infrastructure (11th), a result driven by its 9th position on gross capital formation (at 35% of GDP); and investment (15th), a result driven by a deep and dynamic stock market.

Sub-Saharan Africa

A total of 24 countries from Sub-Saharan Africa are included in the rankings, none of which made it to the top 30, and 17 of which are ranked within the bottom 25. In this year's edition, not a single country from this region is classified as high-income. Regional leaders on the GII and the Output, Input, and Efficiency measures are Mauritius, Nigeria, South Africa, and Côte d'Ivoire. While only Mauritius and Nigeria achieve positions within the top 70 on the Output Sub-Index, it is noteworthy that six countries achieve this threshold on the Input Sub-Index: South Africa (40th), Mauritius (46th), Namibia (49th), Botswana (62nd), Ghana (65th), and Kenya (69th).

Among upper-middle-income countries, Mauritius (53rd overall) achieves the top regional spot on the GII, while South Africa (59th overall) is the runner-up, followed far behind by Namibia (regional 4th and overall 78th) and Botswana (5th and 79th).

With a population of 1.2 million and a GDP per capita of 12,838 in PPP dollars, **Mauritius** is the second least populous country in the region (after Swaziland) and the second wealthiest (after Botswana), so its top regional position is not entirely surprising. This island in the Indian Ocean is the only country in the region with rankings in the upper half of all three main indices. On the Input side, its major strength comes from the Institutions pillar (ranked 25th), where it achieves 8th position on the business environment sub-pillar. Its 6th position on tertiary education is driven essentially by excellent rates of outbound mobility and gross enrolment of tertiary students (ranked 9th and 4th, with 29.6% and 7.7%, respectively). On the Output side, its score is leveraged primarily by its new business density score (11th globally) and by its relatively high level of creative goods exports (ranked 16th, at 3.4% of total goods exports).

South Africa tops the regional Input Sub-Index (40th globally). It is not only the most heavily populated country within this income group, by far, with 50.5 million inhabitants (the other three countries have less than 5.5 million combined), but it is also ranked 3rd in per capita income, with 10,277.8 PPP dollars per capita (Namibia, at PPP$6,410.1, is the least developed of the four). The Input score is driven essentially by an 8th position (in the global ranking) on market sophistication, the result of positions within the top 10 on the strength of legal rights for credit (7th), depth of credit information (1st), and domestic credit to private sector (9th, with 145.1%). Its achievement on investment protection is one of the stronger globally (10th), leading to deep stock

Figure 5: Average scores by income group and by pillar (0–100)

Note: Countries/economies are classified according to the World Bank Income Group Classification (January 2011).

markets (2nd on market capitalization and 8th on stocks traded, with 246.5% and 119.8% of GDP, respectively). Ranked 83rd on the Output Sub-Index, South Africa achieves placement among the top 40 in only two areas: resident patent applications through the Patent Cooperation Treaty and computer software spending (8th globally, with 0.9% of GDP).

Lower-middle-income countries all have poor performances. These are Nigeria (7th regionally and 96th overall), Senegal (8th and 100th), Swaziland (9th and 101st), Cameroon (10th and 103rd), Côte d'Ivoire (18th and 117th), and Sudan (24th and 124th).

Although **Nigeria**'s position on the GII is rather low (96th), this country obtained the top regional position on the Output Sub-Index, where it is ranked 62nd, and has the second-best Efficiency Index score globally. Nigeria's leverage is similar to 1st place on the Efficiency raking, Côte d'Ivoire. This is again a country with positions in the bottom 25 on all Input rankings, although it attains relatively high scores on two measures extremely sensitive to the economic recession: growth in labour productivity (41st at 2.55%), and FDI net outflows (80th, but at a level of 0.1%, Nigeria scored 47.5 over 100). Nigeria also obtained relatively high scores from the survey questions on ICT use on business and organizational models (45th and 70th). On the Creative outputs pillar, Nigeria is ranked 5th globally on the production of national feature films per million population.

Low-income countries Ghana (70th) and Kenya (89th) get relatively high scores—both within the regional top 10—reaching 1st and 3rd place, respectively, in the overall low-income group. Of the remaining countries in this income group and region, Tanzania, Uganda, and Mali fare relatively better on the Output Sub-Index (their ranks are in the fourth quintile); while the scores of Rwanda, Zambia, and Malawi are driven by somewhat better scores on the Input Sub-Index (ranking in the fourth quintile). Of the rest, Madagascar, Côte d'Ivoire, Benin, Zimbabwe, Burkina Faso, Ethiopia, and Niger are in the last quintile (the bottom 25) in all three main indices (GII, Input, and Output).

Conclusions

Innovation is a global phenomenon: it is not only OECD countries that

Table 8: Heatmap for GII top 10 economies and regional and income group averages (1–100)

Country/Economy	GII	Institutions	Human capital and research	Infrastructure	Market sophistication	Business sophistication	Input	Scientific outputs	Creative outputs	Output	Efficiency
Switzerland	63.82	92.60	55.14	44.50	70.08	68.02	66.07	62.00	54.40	58.20	0.88
Sweden	62.12	87.27	63.27	51.66	58.93	63.13	64.85	62.07	56.73	59.40	0.92
Singapore	59.64	90.42	74.69	47.62	78.71	79.11	74.11	48.93	41.42	45.18	0.61
Hong Kong (SAR), China	58.80	92.77	48.38	53.87	87.00	66.85	69.77	38.11	57.56	47.83	0.69
Finland	57.50	89.17	66.46	47.98	56.06	63.87	64.71	58.53	42.06	50.29	0.78
Denmark	56.96	94.24	60.17	45.90	64.48	58.07	64.57	46.26	52.43	49.34	0.76
United States of America	56.57	86.48	57.37	44.63	70.91	54.82	62.84	57.44	43.17	50.30	0.80
Canada	56.33	93.28	53.89	53.13	63.36	58.40	64.41	42.51	54.01	48.26	0.75
Netherlands	56.31	87.46	47.60	43.60	61.81	61.64	60.42	53.81	50.59	52.20	0.86
United Kingdom	55.96	86.37	56.06	43.65	74.44	57.78	63.66	52.27	44.27	48.27	0.76
Regions											
East Asia & Pacific	42.50	69.78	42.44	36.10	54.06	47.34	49.95	33.73	36.37	35.05	0.70
Europe & Central Asia	42.96	74.64	46.47	34.40	47.93	44.06	49.50	34.65	38.04	36.34	0.73
Latin America & Caribbean	31.43	55.69	31.61	28.17	35.51	34.02	37.00	19.00	32.73	25.86	0.70
Middle East & North Africa	34.45	62.81	42.53	27.98	39.93	34.48	41.54	25.46	29.27	27.36	0.65
North America	56.45	89.88	55.63	48.88	67.14	56.61	63.63	49.97	48.59	49.28	0.77
South Asia	29.92	50.08	23.06	24.82	33.23	26.16	31.47	22.88	33.87	28.37	0.91
Sub-Saharan Africa	26.92	54.32	28.65	21.34	31.40	30.19	33.18	18.15	23.16	20.66	0.64
Income levels											
High income	48.08	82.12	52.32	39.98	54.58	50.42	55.88	39.10	41.29	40.19	0.71
Upper-middle income	33.36	63.09	36.60	27.55	38.89	35.80	40.39	21.51	31.16	26.33	0.65
Lower-middle income	30.42	51.96	29.82	24.62	35.45	30.96	34.56	22.66	29.89	26.27	0.76
Low income	25.91	50.70	27.83	20.97	30.63	28.29	31.68	18.43	21.83	20.13	0.64

Note: Darker shadings indicate better performances. Countries/economies are classified according to the World Bank Income Group and Regional Classifications (January 2011).

innovate—innovation leaders are found across the world, as is evident from the presence of European, Asian, and North American economies in the GII top 10. All regions show potential as leading hubs of innovation (all are represented in the upper half of the GII rankings). The worldwide relevance of innovation capabilities and results highlights the need for a global perspective in understanding the concept and underlying premises of innovation, which this Report has adopted.

Identifying the underlying conditions of a country and comparing performances among peers is the key to a good understanding of the implications of a country's ranking in the GII. This Report attempts to abide by this underlying principle in the analysis of the GII rankings. Countries evaluate their performances by comparing themselves with other countries within their region, but in this Report we assess results on the basis of the development stages of countries (captured by the World Bank income classifications).

In this light, the results of the GII confirm that average rankings decrease monotonically with income levels. On average, high-income countries outpace developing

countries by a wide margin across the board (Figure 5). But some gaps are closing among developing countries in some areas, notably in the Institutions and Creative outputs pillars. In the Scientific outputs pillar, lower-middle-income countries outperform upper-middle-income countries (albeit by only a small margin of 1.15 points).

Upon closer inspection, the heatmap in Table 8 shows that, without exception, *all* regional and income group averages are greater than 30 on the Innovation Input Sub-Index (scores range between 0 and 100), which is remarkable. In this group of enablers to innovation, the biggest progress has been made on the Institutions pillar, where income group and regional averages are all above 50; and on the Market sophistication pillar (where all averages are above 30). This shows that significant efforts have been made worldwide towards increasing political stability; improving regulatory environments; and enhancing environments for business, credit, investment, trade and competition—all crucial elements for the fulfilment of innovation capabilities. These findings also imply that some laggards are driving the scores down.

Governments have recognized that, in order to take advantage of the potential that innovation represents for their citizens, they must adopt policies that are friendlier towards technological catch-up and the absorption of knowledge so that firms can build global innovation networks and so that they can foster trans-border flows of knowledge and intellectual property. And, in turn, firms in the private sector are doing their part by participating more fully in the financing and execution of R&D projects; by making venture capital available; and through joint venture and strategic alliance deals, dynamic stock markets, increased employment in knowledge intensive services, and so on.

The main cause for alarm comes from the Human capital and research pillar. Although the average score across countries is 39.1 (this score comes in 3rd, after the Institutions and Market sophistication pillar scores), the gap between high-income and low-income countries is the second biggest (after Institutions), at 24.9 points. A skilled workforce is essential to economic development, and deficits in this area have longstanding effects surmountable only after consistent improvements have been made, and then only after two or three generations. The only indicator from the Human capital and research pillar on which low-income countries do better on average is the percentage of graduates in science.

Looking more closely at the indicators, we see that in 41 of the 80 indicators selected for this year's GII, average scores for high-, upper-middle-, lower-middle, and low-income countries are monotonically decreasing. This decrease implies that, on average, the wealthier the country, the higher its score. But in 39 occasions this is not the pattern (see Tables 1a through 1g, included in the description of each pillar). For three indicators— 2.2.5 Tertiary outbound mobility, 4.1.4 Microfinance gross loans, and 6.2.1 Growth rate of GDP PPP$ per worker (labour productivity)—the order is reversed, which is encouraging. In two other instances, the higher average score corresponds to low-income countries: these are indicators 3.2.4 Share of renewables in energy use, and 5.2.3 R&D financed by abroad. In six other cases, the mean of the average scores of low- and middle-income countries is higher than the mean of the average of scores for upper- and middle-income countries: 2.1.2 Public expenditure/pupil over GDP per capita, 2.2.2 Graduates in science, 3.3.2 Gross capital formation, 3.3.3 Ecological footprint and biocapacity, 6.1.3 Domestic resident utility model applications, and 7.1.1 Domestic resident trademark applications (the last two are scaled by GDP PPP dollars).

It could be argued that some aspects dictated by the nature of available statistics —the prominence of trade-related variables over output variables and the importance of count variables over value variables— may have the effect of making low- or lower-middle-income economies stand out or of biasing results in favour of small countries. However, among the indicators listed above, these concerns affect only the count statistics on utility model and trademarks applications.

At the regional level, North America is not truly comparable to the other regions. The sample includes only two countries from this region, Canada and the US, which lead on all categories except on Innovation Efficiency. Otherwise, regional comparisons show that Europe and Central Asia lead on the GII, the Output Sub-Index, and four pillars (Institutions, Human capital and Research, and Scientific and Creative outputs), while East Asia and the Pacific lead on the Input Sub-Index and the three remaining pillars (Infrastructure and both Market and Business sophistication).

East Asia and the Pacific and the Middle East and North Africa have similar income structures in

the GII sample, and yet, on average, the latter region prevails only on Human capital and research, and by an insignificant margin (less than a point). On all the other pillars, differences range between 7.0 and 14.1 points. Latin America and the Caribbean comes next in terms of overall development, followed by South Asia and Sub-Saharan Africa. Although these regions achieve average positions in the GII in accordance with their level of development, the leverage for countries from Latina America and the Caribbean comes from the Input Sub-Index and each of its pillars, while South Asian countries prevail on the Output Sub-Index and pillars.

At the regional level, perhaps lacking in this global picture is Sub-Saharan Africa —a continent where more progress could be made. In the rankings, Sub-Saharan African countries overtake South Asian ones on the Input Sub-Index and on the Institutions, Human capital and research, and Business sophistication pillars. But they lag behind on everything else.

It cannot be claimed, however, that the GII model captures all dimensions of innovation across continents. Analytical chapters included in this year's Report illustrate the richness of innovation, which is difficult to define, much less to encapsulate in a particular metric. Chapter 2 provides insights on innovations taking place across sectors, products, processes, and business models in Latin America.

In this respect, BRIC countries are improving their innovation capacity dramatically: China, Brazil, and, to a minor extent, India have achieved encouraging results, especially on the Output side. Chapter 3 illustrates how these countries are confronting their development challenges with the example of India, which has pioneered promising avenues in the areas of frugal and inclusive innovation. The rise of innovation in the BRIC countries will change the face not only of innovation but also of global dynamics. Large populations in countries such as India and China will demand a different type of products and services— those that provide high quality but at affordable costs while respecting sustainability in the environment. Companies will have to adjust to these new models of innovation. Western firms are setting up centres of innovation and learning in these BRIC countries now. Other nations will need to learn to treat BRIC countries as sources of knowledge and learning for innovation, not just as threats to domestic industries or competitiveness.

Innovation efficiency is important, and it varies across regions. Some countries demonstrate a strong input performance but fail to translate their enabling capabilities into tangible results and vice versa. Lower-middle-income countries now fare better than those of all other income groups in efficiency. China, Moldova, Jordan, Viet Nam, India, Thailand, Guyana, Ukraine, Tunisia, Armenia, and Nigeria all achieved rankings in the upper half of the Output Sub-Index. Innovation capabilities might progress in waves, with uneven and non-linear improvements, leading to multiple equilibria. High-income countries, which come next in terms of innovation efficiency (and which are far ahead in Output scores) enjoy a stable position at the top of all main rankings. Upper-middle-income countries seem to be finding difficulties in translating their favourable enabling conditions into innovation results. Low-income countries frequently have difficulties in taking off, and although their institutional environments show improvements, they lack the human resources and capabilities to achieve their full potential, even taking into consideration that they are beginning at a low level.

More formal analysis, beyond the scope of this Report, is required to explore in depth the linkages and dynamics between development stages and innovation phenomena in depth. However, this Report provides a snapshot from which particular insights can be obtained.

Innovation is a multi-stakeholder effort, with many different roles for the different actors. Governments have a role in setting the right environment and policies. Firms have to improve their innovation readiness and innovation results—they must protect and leverage their intellectual property, increase their investment in R&D, and make better use—through international trade, linkages, and the adoption of ICT—of innovations developed elsewhere. Societies and individual citizens also have to look at different aspects that help them create a broader capacity for innovation. All of these stakeholders must collaborate in order to foster and sustain innovation.

It is important—and feasible—to take action. Multiple success stories and best practices worldwide demonstrate ways to advance innovation, and they can be studied and drawn upon. Since innovation can potentially expand everywhere, what can be done to help it thrive in developing economies? This Report offers important avenues for action in this regard. Some 'weak pillars' need strengthening: In more than

one country, a relatively poor performance on pillar 2, Human capital and research, goes hand in hand with low levels of Scientific outputs. Indeed, the highest correlation of the Scientific outputs pillar is with pillar 2. A more formal analysis is required to infer causality, however.

Despite our best efforts, we have only scratched the surface of the challenge of fully capturing the richness of innovation. The first task was to adopt a definition and a model of innovation that would capture innovation efforts in emerging economies. What worked in theory implied a real challenge in terms of data collection, particularly in the areas of innovation linkages and creative outputs.

The computation methodology adopted for the Report, which has the advantages of being transparent and replicable, is extremely sensitive to modelling choices and missing data points. To ensure the credibility of the whole exercise, results were submitted to a statistical audit by the Joint Research Centre of the European Commission. The Audit assessed the robustness of the GII rankings to other modelling techniques, and involved the imputation of missing data, computation of geometric averages instead of arithmetic averages (the former are less compensatory), random weights, and principle components analysis, among others (refer to the appendix to this chapter). Based on the Audit, five countries with unreliable rankings were dropped from the Report. In addition, the Audit confirmed that, for the top 40 and the bottom 14 countries, median ranks after 4,000 runs of Monte Carlo simulations differed from the GII ranks by less than five positions.

We hope that this work makes a small but concrete step forward in improving our understanding of the broad phenomenon of innovation in firms, governments, and societies. It is linked to some of the ongoing work by other organizations such as the OECD. We intend to work collaboratively with them to enhance our common understanding of this phenomenon.

The challenge is to translate these intellectual ideas and ranks into action—to connect with different stakeholders engaged in innovation in different economies and become part of the evolving dialogue around innovation. Readers are invited to contact INSEAD if they wish to explore some of the insights of the Report in more detail within their own contexts.

Notes

1 Archibugi and Coco, 2004.

2 The Confederation of Indian Industry has been a partner of the GII since 2009.

3 OECD/EC, 2005, also known as the *Oslo Manual 2005*.

4 OECD/EC *Oslo Manual*, 1992, 1997, and 2005 editions.

5 OECD/EC, 2005, p. 146.

6 OECD, 2010b.

7 OECD/EC, 2005, p. 58. Disruptive innovations can, for example, change the structure of the market, create new markets or render existing products obsolete; see Christensen, 1997.

8 UNCTAD, 2007, p. 6.

9 OECD 2010b, Box 1, p. 1.

10 Gibbons et al., 1994.

11 Etzkowitz and Leydesdorff, 2000; and Leydesdorff and Etzkowitz, 1996. These authors point out that mode 2 'is the original format of science before its academic institutionalization in the 19th century […] Mode 2 represents the material base of science, how it actually operates. Mode 1 is a construct, built upon that base in order to justify autonomy for science, especially in an earlier era when it was still a fragile institution and needed all the help it could get'.

12 OECD/EC, 2005, p. 38.

13 Silberglitt et al., 2006.

14 BCG /NAM, 2009.

15 EC, 2011.

16 EIU, 2009.

17 World Economic Forum, 2010.

18 The first two are part of the six World Governance Indicators (GII) constructed by the World Bank. The Press Freedom Index is calculated by Reporters Without Borders (it is included in the Voice and Accountability Index, one of the two World Governance Indicators of the World Bank not included in the GII).

19 Ease of Doing Business Data Notes, http://www.doingbusiness.org.

20 ITU, 2010. The 2010 edition is based on 2008 statistics. We are grateful to Susan Teltscher, Head, and Esperanza Magpantay, Statistician, Market Information and Statistics Division, Telecommunication Development Bureau, International Telecommunication Union for sharing their data and for a useful discussion regarding the GII framework.

21 We are grateful to Karen Treanton, Head of Energy Balances, Prices and Emissions Section of the Energy Statistics Division of the International Energy Agency (IEA) for sharing the IEA series. Electricity consumption per capita is typically used as a proxy for infrastructure, as the power grid, to reach consumers, needs to come directly into houses and businesses. While the International Energy Agency reports on the level of energy use as a percentage of GDP in constant PPP terms, we followed the World Bank in taking the reciprocal of that variable to provide a measure of efficiency in the use of energy resources.

22 The surveys are conducted by the World Bank in partnership with academic and international institutions and private companies and individuals engaged in international logistics (refer to Appendix XX Sources and definitions for details). Hard statistics on infrastructure are available, but it was deemed that for the purposes of assessing infrastructure as an enabler to innovation, expert opinion was more appropriate.

23 In last year's edition of the GII, the measure included was the average loan balance per borrower (over GNI per capita). It was replaced by the gross loan portfolio over GDP to complement indicator 4.2.1.

24 We are grateful to Cornelius Bubenzer, Financial Markets Executive, Thomson Reuters, for his assistance with the extraction of data from the Thomson One database.